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The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis

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Tittle: The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis

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Tittle: The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis

Abstract

Introduction The breakout of the COVID-19 poses a significant threat to the TB patients. However, the prevalence status, treatment, and possible outcomes of the coinfection are poorly characterised. We aimed to systematically review the evidence on this topic and provide comprehensive information to guide the control and treatment of TB and COVID-19 (TB-COVID) coinfection in future research.

Methods An extensive screening was conducted using 6 electronic databases to search eligible studies from November 1, 2019 to March 19, 2021. Prevalence rate, treatment, and outcomes of TB-COVID coinfection in different countries were extracted. Random effects models were used to calculate mean fatality rates of coinfection with 95% confidence intervals (CI). The risks of bias were assessed with JBI Critical Appraisal Checklist for Study Reporting Prevalence Data and JBI Critical Appraisal Checklist for Case Report. A meta-analysis was conducted for subgroups on in-hospital fatality rate.

Results Total 42 of 3109 studies were included into the analysis (35 case reports and 7 retrospective cohort studies). Nineteen countries reported coinfected patients, including high and low TB prevalence countries. The only study revealing prevalence rate came from West Cape Province, South Africa (people aged above 20, 0.04% till June 1, 2020 and 0.06% till to June 9, 2020).The treatment regimes for coinfected patients were of

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great heterogeneity. The mean overall and in-hospital fatality rates of coinfection were 13.9% (95% CI: 1.6% to 26.2%) and 17.5% (95% CI: 8.9% to 26.0%). The mean in-hospital fatality rate for high-income country (Italy and Argentina) and low- and middle-income countries (LIMICs) (India, Philippines, South Africa) were 6.5% (95% CI: -0.8% ~ 13.9%) and 22.5% (95% CI: 19.0% ~ 26.0%).

Conclusion TB-COVID coinfection is common globally and the coinfected patients are suffering from higher fatality risk than normal COVID-19 patients. Outcomes shared significant differences between high-income countries and LIMICs.

Registration PROSPERO CRD42021253660

Keywords: COVID-19, tuberculosis, coinfection, prevalence, health outcome, treatment, systematic review

Strengths and limitations of this study

- This systematic review and meta-analysis gathered all the available evidence from selected databases and provided comprehensive information on prevalence status, treatment plans and possible outcomes of TB-COVIDcoinfection.
- We emphasized on the evidence from case reports and thedetailed information about patients' basic information, treatment and outcomes.
- Sub-group analysis was conducted to explore the possible reason for different fatality rates in different areas.
- We did not include comorbidity as a keyword and MeSH term in the searching

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process, which might keep us from some relevant studies.

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1. Introduction

Individuals with chronic respiratory infections, including tuberculosis (TB) are first to experience the adverse effects of a pneumotropic pandemic[1]. The burden and impact of TB control and treatment is widely acknowledged with the ongoing of coronavirus disease 2019 (COVID-19) pandemic. Patients with TB have an increased risk of mortality during a coinfection with COVID-19[2]. On the contrary, the COVID-19 outbreak has put on the TB patients and entire health care system in an unprecedented crisis. The number of people who fell ill with TB in 2019 is deemed 10 million[3]. Just like COVID-19, the prevalence of TB varies geographically. The top 7 countries account for two-thirds of the total global cases in 2018, and most of high prevalence rate countries locate in South-East Asia and Africa[4]. Recently many experts have recognized the effects of the COVID-19 pandemic on TB treatment and control, and some researches identified TB as a risk factor to severe COVID-19[5]. As warned by the investigator in the Global Tuberculosis Network Group, COVID-19 may boost TB given infection and mortality[6]. In low- and middle-income countries (LIMICs), the situation is even worse. As estimated by Kissler, the transmission of SARS-CoV-2 is likely to enter into regular circulation[7], which would be a deadly threat to people in LIMICs, especially when most of them cannot stand the chance of getting COVID-19 vaccine[8]. Regrettably, people in these countries are also the main victim suffering from tuberculosis[3].

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Cases of coinfection and studies about prevalence status have been reported from several countries[9]. However, much of the research up to now has been descriptive in nature and focused on the observation of several coinfected patients, which can only provide weak evidence to guide the treatment for TB and COVID-19 (TB-COVID) coinfection. Furthermore, few scholar has conducted a systematic review in this field, and there is no general agreement about the prevalence, treatment and outcome of the coinfection worldwide.

Although few health alert and guidelines for people with increased risk for adverse outcomes in COVID-19 have been issued by World Health Organization (WHO) and Center for Disease Control and Prevention (CDC)[10], there is a dearth of information regarding the impact of COVID-19 in patients with TB and the actual impact of tuberculosis on occurrence and clinical outcomes of COVID-19 is not clear. This study aims to systematically analyze the current evidence on global prevalence status, treatment protocols, and possible outcomes of TB and COVID-19 coinfection.

2. Methods

This systematic review protocol followed the International Prospective Register of Systematic Reviews (PROSPERO) guideline and has been registered in PROSPERO's database (registration number: CRD42021253660). The study reporting followed the Preferred Reporting Items in Systematic Reviews and Meta-Analyses (PRISMA)

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guideline[11] (Table S1 in appendix 1).

2.1 Eligibility Criteria of Included Studies

Studies on patients who had been confirmed with both TB and COVID-19 were included. We excluded studies that only mentioned the fatality and the number of coinfected patients less than 10 to avoid large deviation in fatality rate, which would affect the results of Meta-Analysis (MA).

This systematic review included any kind of treatment for TB or COVID-19, including supportive therapy, directed therapy, antiretroviral treatment, ICU care, etc.

We sought prevalence rate data of any region and figured out all countries that reported coinfected patients. The primary outcome of interest was in-hospital survival or mortality and secondary outcomes were disease progression, time to discharge from the ICU and hospital (for survivors), and time to death (for the dyed).

Case reports, cross-sectional studies, and cohort studies were included to explore the global prevalence status. Case reports, cross-sectional studies, cohort studies, and quasi-experimental studies that revealed the treatment and outcome of TB-COVID coinfection were targeted. We excluded all reviews, editorials or commentaries, and

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clinical guidelines.

2.2 Search Strategy

An extensive search of literature was conducted using PubMed, Web of Science, Scopus, ProQuest, Embase, and Cochrane library databases. Manual reference screens from included studies were performed to ensure the inclusion of all relevant studies. The study's publication date was between November 1, 2019 and March 19, 2021. With the librarian's (JZ) help, we used building blocks searching strategy to ensure all the relevant studies were screened. The searching process utilized Medical Subject Heading (MeSH) terms and Tittle/Abstract words around two areas: 1) COVID-19, including COVID-19, SARS-COV-2, novel virus, etc., 2) tuberculosis, including tuberculosis, TB, Mycobacterium tuberculosis, etc. (see Table S2 in appendix 2). The citation manager EndNote 20 (Thomson Corporation, Canada) removed duplicate studies, and we spent special efforts on the same patients or cohorts among different papers.

2.3 Data Extraction

Two reviewers (QW and SG) independently conducted the study selection (review of titles and abstracts, review of full texts, and final decision), data abstraction, and quality assessment. Covidence was utilized to help us finish all the above processes. All non-English and non-Chinese articles were translated to English by Baidu Translation

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(Baidu Co., Ltd, China). During the review process, we recorded exclusion reasons for all excluded studies in the full-text screen step. The conflicts between 2 reviewers were discussed and eventually resolved with another author's help (NX).

Since this systematic review got 2 research objectives, 2 categories of information were extracted: 1) data about the prevalence, 2) data about treatment and outcome. For the first category, we stressed locations (nation or province, especially for case reports) of studies and the following information: age, gender, sample size, time, prevalence rate, fatality rate. As for the second category, treatment for TB and COVID-19 were recorded separately, as well as the primary and secondary outcomes (e.g., disease progression, time to discharge from the ICU and hospital, time to death). Besides, we also stressed patients' basic information in both parts, like TB site, TB type, drug resistant status, comorbidity, BCG vaccination, etc.

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2.4 Quality Assessment

We used the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Study Reporting Prevalence Data for the prevalence part as the tool[12]. Nine questions in this checklist were set to assess the quality of studies, including sample frame, sample size, data analysis, etc. All included studies were case reports for the treatment and outcome review, so we applied JBI Critical Appraisal Checklist for Case Report[13] as the tool which consists of 8 questions about patients' condition, test and assessment,

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intervention, etc.

2.5 Data Synthesis

A narrative synthesis method was introduced for listing prevalence status and key findings such as the countries and regions of the case reports to probe the potential prevalence of TB-COVID coinfection. We condensed the information about treatment into 3 categories: treatment before hospitalization, treatment in hospital, and treatment after discharge. We also sorted out the treatment in the hospital into 4 kinds: supportive treatment, antibiotics, antiviral drugs, and other treatment. The narrative synthesis in this systematic review followed Cochrane's Consumers and Communication Review Group (CC&CRG) reviews protocol[14]. We conducted MA about the fatality rate of coinfection patients using the inverse variance statistical method to pool the data and estimate its 95% confidence interval (CI). Due to the highly heterogeneous treatment method for patients as well as patients themselves, we chose the random-effects model in MA and the heterogeneity was evaluated with I^2 statistic. The funnel plot was created to test the bias.

2.6 Subgroup Set

Studies were grouped into the following subgroups for further analysis: high-income country and low- and middle-income country groups, by doing which we would like to explore the possible difference in fatality rate among different types of country.

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2.7 Patient and Public Involvement

Patients were not involved in this systematic review.

3. Results

We searched 565 studies from PubMed, 682 from ProQuest, 839 from Scopus, 644 from Web of Science, 367 from Embase, and 1 from the Cochrane database. After removing the duplicate studies, we got 1231 studies and imported them into Covidence. With the help of Covidence, we excluded 1049 studies in the title and abstract screen step. Next, we carefully reviewed the full text of the remaining studies, and found another 11 possibly relevant studies. Finally, we excluded 140 studies, in which we didn't find outcomes of interest (n=80) or quantitative data (n=39). The entire process is shown in Figure 1. Among the 42 included studies, 35 were case reports, and 7 were retrospective cohort studies. No experimental study was identified.

<Insert Figure 1 here>

3.1. Global Prevalence Status

Till to Mar. 19 2021, there were 35 case reports from 17 countries, including India[15-19], Indonesia[20], China[21-25], Saudi Arabia[26 27], Qatar[28 29], Singapore[30], Turkey[31 32], Haiti[33], Argentina[34], U.S.[35-38], Brazil[39 40], Panama[41], Nigeria[42 43], South Africa[44 45], Morocco[46],

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Italy[47 48], and France[49]. Totally, 55 cases were described in these 35 reports. China, India and U.S. were the top 3 countries with the most studies, and 29 (82.9%) studies were published in 2020. Table 1 shows the study characteristics of these case reports. Besides, we also identified coinfected patients from Belgium and Switzerland in Motta's paper[50].

Table 1 Study characteristics of case reports (n=35)

Study characteristic	Studies (%)	Cases (%)
Publication year		
2020	29 (82.9)	49 (89.1)
2021	6 (17.1)	6 (10.9)
Country		
India	5 (14.3)	5 (9.1)
Indonesian	1 (2.9)	1 (1.8)
China	5 (14.3)	11(20.0)
Saudi Arabia	2 (5.7)	2 (3.6)
Qatar	2 (5.7)	7 (12.7)
Singapore	1 (2.9)	4 (7.3)
Turkey	2 (5.7)	3 (5.5)
Haiti	1 (2.9)	1 (1.8)
Argentina	1 (2.9)	3 (5.5)
U.S.	4 (11.4)	4 (7.3)
Brazil	2 (5.7)	3 (5.5)
Panama	1 (2.9)	2 (3.6)
Nigeria	2 (5.7)	3 (5.5)
South Africa	2 (5.7)	2 (3.6)
Morocco	1 (2.9)	1 (1.8)
Italy	2 (5.7)	2 (3.6)
France	1 (2.9)	1 (1.8)

The only large-scale study about the prevalence rate came from West Cape Province, South Africa[51]. The author used the data from Western Cape Provincial Health Data Centre (WCPHDC), which included 3,460,932 individuals aged 20 or above, about half

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of the population in West Cape Province. Until June 1, 2020, when testing criteria changed with public sector tests being limited to patients >55 years of age or with comorbidity, 1,489 coinfected patients (including previous TB and current TB) were recorded, and the prevalence rate of coinfection was about 0.04%. Till June 9, 2020, 2128 coinfected patients (including previous TB and current TB) were recorded, and the prevalence rate was about 0.06%.

3.2. Treatment and Outcome

Till the end of search, no cohort study, cross-sectional study, or experimental study was identified to provide detailed information on TB and COVID-19 coinfection treatment. Table S3 and S4 in appendix 3 presented the results of the included 35 studies which reported 55 coinfected patients in total. As for the basic information, 12 patients were female, and 47 patients were aged 20 to 60 years. Up to 50 cases were reported as active TB patients, 12 of which got TB before COVID-19, and we could not identify infection orders about the rest 38 cases. Besides, 4 cases were reported as post TB[21], and 1 was reported as latent TB[24]. Although most cases were diagnosed with pulmonary TB, there were still some more complex conditions, like central nervous system involvement[18 35], tuberculous meningitis[25], tuberculous pleuritis[23 24 28], and peritoneal TB[38]. With regard to drug resistance, 37 patients' information was missing, 15, 2 and 1 of the remaining 18 cases were reported as negative[19 20 26 27 34 36 39-41 43 45 48], positive[24 33], and likely positive[15]. All the cases were symptomatic COVID-19 patients.

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Only 5 patients received treatment before hospitalization. Among them, 3 received Anti-TB Treatment (ATT)[19 31 41], 1 received antibiotic (amoxicillin)[35], and 1 took hemodialysis due to chronic kidney failure[32]. During the hospitalization, 7 patients were transferred to the Intensive Care Unit (ICU)[15 27 31 32 36 41 49]. Up to 50 cases' detailed treatment information was reported. Various kinds of oxygen support were the most common supportive treatment (21 cases), and 1 patient in China received extracorporeal membrane oxygenation[24]. Except for several cases that did not provide any detailed ATT regime (12 cases) or patients did not receive ATT (11 cases), 1 or several kinds of first-line Anti-TB drugs, including rifampicin, isoniazid, ethambutol, and pyrazinamide, were utilized in most remaining cases (27 cases). For the drug resistance TB patients, the doctors would adjust the regime for them[24 33]. Other kinds of antibiotics were also utilized, including azithromycin (21 cases), ceftriaxone (13 cases), moxifloxacin (7 cases), linezolid (4 cases), etc. However, we could not identify the purpose of these extra antibiotics due to the lack of information. We guessed it could be 1 or more of the following situations: 1) supplement to ATT regime, 2) prevention against possible bacterial infections, and 3) treatment for the existing bacterial infection. Among these 50 cases, 13 cases did not mention any kind of antiviral drug and 2 cases did not provide antiviral drugs' names. Within the remaining cases, hydroxychloroquine (HCQ) was the most utilized antiviral drug (21 cases), as well as lopinavir (8 cases), ritonavir (8 cases) and arbidol (7 cases). We also found that all the 20 Chinese coinfected patients did not take HCQ completely but took arbidol most.

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Besides, drugs like steroids (methylprednisolone, dexamethasone, prednisone, etc.), anti-coagulant drugs, vitamins (vitamin C, vitamin B6, etc.) were also prescribed for some patients. We identified information concerning treatment after discharge for 10 patients; 8 of them received ATT, and 2 received treatment for existing comorbidity. Although there was no doubt that the regimes for TB and COVID-19 coinfected patients were highly heterogeneous, we still could identify 4 main paradigms about treatment during hospitalization: 1) simultaneous antiviral treatment and ATT[16-18 20-23 25 27 28 31 32 36-38 40-42 46 48], 2) only ATT[19 23 29 30 32 33 43-45], 3) only antiviral treatment[15 21 23 24 35 41 47], 4) no antiviral treatment and ATT[34]. Check Table S5 in appendix 3 for detailed information about the treatment.

A minimal number of studies discussed about pros and cons of the treatment and the possible effects of the regimes. Faqihi held that the first-line Anti-TB drug with antiviral treatment for COVID-19 might theoretically increase the risk of side-effects[27], and Yousaf stressed the importance of adjustment to the management of coinfected patients[28]. The antagonism from lopinavir/ritonavir against rifampicin had been noticed[17 22], and Cao suggested umifenovir could be the substitute of lopinavir/ritonavir[22]. Kumar warned additive hepatotoxicity from remdisivir due to simultaneous use of anti-tubercular drugs[17]. Faqihi gave isoniazid along with pyridoxal phosphate to avoid peripheral neuropathy[27], and Farias did not start antiretroviral therapy at first for the 2 HIV, TB and COVID-19 coinfected patients to avoid complications with ATT[40]. Confronted with liver function impairment during the

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treatment, Musso replaced isoniazid and pirazinamid with amikacin and moxifloxacin to modify the ATT regimen[48]. Liu, He, Cao, and Singh discussed the impact of corticosteroids for treating lung inflammation[16 21 22 24]. Liu's recommendation was not to administer corticosteroids for COVID-19 and TB coinfected patients. Besides, since COVID-19 and TB disease might both cause immunosuppression, Liu recommended using immunomodulatory therapies such as thymosin in patients with progressively lower lymphocyte counts[24].

About the outcome of 55 reported cases, 35 were resolved, 6 were deceased, 3 were still on treatment in hospital, 1 returned to hospitalization due to positive recurrence of SARS-CoV-2 RNA, and 10 were missing. For patients discharged from the hospital, the length of stay ranged from 7 days to 2 months. Besides, in Vanzetti's study, 3 patients were described they had a longer hospital stay than average[34]. For deceased patients, the length of stay ranged from 6 to 26 days. Unfortunately, we could only find minimal information about coinfected patients' physical condition when they got discharged. Most studies only indicated that the test results for SARS-CoV-2 were negative or patients got recovered. Although Agada described the patient with poor prognostic indices, detailed physical conditions were also missing[43]. He believed the treated or untreated mycobacterium TB was a risk factor of a worse prognosis of COVID-19 patients[21]. In addition, long-term outcome or follow-up outcome for the TB and COVID-19 coinfected patient was absent.

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3.3. Fatality Rate of TB and COVID-19 Coinfection

We identified 7 retrospective cohort studies that revealed the fatality rate of TB and COVID-19 coinfection (Table S6 in appendix 3 for detailed information). These studies came from India[52], Philippines[53], Italy[54], South Africa[51], Argentina[55], Russia[56] and 1 was based on data from multiple countries[50]. Motta's study provided 2 cohorts[50], 1 of which was same as Stochino's study[54]. Thusly, we removed the overlapped cohort. Except for Gubkina's study whose targeted population was children and no one deceased[56], the overall and in-hospital fatality rate fell within the range of 5.3% ~ 23.6% and 5.0% ~ 27.3%. Table 2 presents the results of the included 7 studies.

Table 2 Fatality rate of included studies (n=7)

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First author (year)	Country	Time span	Sample size	In-hospital	age	gender	Comorbidity	Died	Fatality rate (overall or in-hospital)
Gupta 2020	India	1 February 2020 to 14 June 2020	22	yes	19-67	20 male (90.9%) 2 female (9.1%)	Hypertension 4 (18.2%) Diabetes 3 (13.6%) Seizure disorder 2 (9.1%) Hypothyroidism 1 (4.5%)	6	27.3 (in-hospital)
Motta 2020	Belgium, Brazil, France, Italy, Russia, Singapore, Spain, and Switzerland	12 March, 2020 to 5 May, 2020	49	no	27-70	/	/	7	14.3% (overall)
Sy 2020	Philippines	May 17, 2020 to June 15, 2020	106	no	/	73 male (68.9%) 33 female (31.1%)	Hypertension 22 (20.8%) Diabetes 14 (13.2%) Cancer 1 (0.9%) Renal cancer 4 (3.8%) Cardiac disease 8 (7.5%) Asthma 4 (3.8%) COPD 3 (2.8%)	25	23.6% (overall)
			66	yes	/		Hypertension 16 (24.2%) Diabetes 10 (15.2%) Cancer 1 (1.5%) Renal cancer 4 (6.1%) Cardiac disease 7 (10.6%) Asthma 1 (1.5%) COPD 1 (1.5%)	18	27.3% (in hospital)

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Stochino 2020	Italy	/	20	yes	27-47	12 male (60%) 8 female (40%)	/	1	5% (in-hospital)
Davies 2020	South Africa	Till to 9 June, 2020	2128	no	20 or above	/	/	113	5.3% (overall)
			469	yes	20 or above	/	/	102	21.7% (in-hospital)
Domingo 2020	Argentina	1 March to 30 June, 2020	23	yes	5-82	18 male (78.3%) 5 female (21.7%)	Smoking and other addictions 16 (69.6%) HIV 4 (17.4%) Psychosis 1 (4.3%) Pulmonary thromboembolism 1 (4.3%) Arterial hypertension 1 (4.3%)	2	8.7% (in-hospital)
Gubkina 2020	Russia	March to June, 2020	24	yes	/	/	/	0	0% (in-hospital)

/: No information provided

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3.4. Quality of Included Studies

We utilized JBI Critical Appraisal Checklist for Study Reporting Prevalence Data as the tool to assess 7 included studies that revealed prevalence rate or fatality rate. The grade (yes, no, unclear or N/A) was shown in Figure S1 and Table S7 in appendix4. The results suggested that insufficient sample size and inappropriate sample frame were the main risks of bias of these studies.

We utilized JBI Critical Appraisal Checklist for Case Report[13] as the tool to assess 35 included case reports, as shown in Figure S2 and Table S8 in appendix4. All the studies described patients' demographic characteristics clearly, and we could identify basic information about patients' history (28 studies), clinical condition (31 studies), diagnostic tests or assessment methods and the results (34 studies) for most studies. Whereas only 23, 20, and 22 studies provided clear messages about treatment procedures, post-intervention clinical condition, and takeaway lessons, respectively. The most poorly reported part was adverse or unanticipated events, and only 11 studies gave a clear description, which could be a high risk of bias.

3.5. Meta-Analysis on Fatality Rate

We included 6 qualified studies (8 cohorts) to estimate the fatality rate of TB and COVID-19 coinfection, 3 studies for overall fatality rate[50 51 53] and 5 studies for in-hospital fatality rate[51-55]. Gubkina's study was excluded because its targeted population

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was children, which might lead high heterogeneity to the MA. Although Davies's study sample was people aged 20 or above, we still included it into the MA.

As shown in Figure 2A, the estimated overall fatality was 13.9% (95% CI: 1.6% ~ 26.2%). Significant heterogeneity was observed between the studies (I^2=91.07%, p-value for Q statistics <0.001). The estimated in-hospital fatality rate was 17.5% (95% CI: 8.9% to 26.0%), and heterogeneity was detected among these studies too (I^2=75.47%, p-value for Q statistics <0.01) (Figure 2B). We utilized funnel plots to explore the existence of publication bias, and the results were both positive (Figure 3A, 3B).

To further explore the source of heterogeneity, we grouped these studies into the high-income country and LIMICs subgroups. However, due to the lack of enough data, we could only conduct subgroup MA on in-hospital fatality rate. The mean in-hospital fatality rate for high-income country (Italy and Argentina) and LIMICs (India, Philippines, South Africa) were 6.5% (95% CI: -0.8% ~ 13.9%) and 22.5% (95% CI: 19.0% ~ 26.0%). No heterogeneity was observed between studies for both subgroups (Figure 2C, 2D). Similarly, 2 funnel plots were created to explore the existence of publication bias, and the results were both negative (Figure 3C, 3D).

<Insert Figure 2 and Figure 3 here>

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4. Discussion

This systematic review identified the evidence on the prevalence status, treatment and possible outcome of TB and COVID-19 coinfection. Although studies about coinfection were limited and quantitative data was even rarer, we still identified 42 studies, of which 35 were case reports and 7 were retrospective cohort studies.

In terms of global prevalence status, 19 countries reported coinfected patients, including low TB prevalence countries like Belgium and Switzerland, as well as high TB prevalence countries like China and India. Therefore, we assumed the coinfection was common worldwide, which every country and region should beware of. The only large-scale study about the prevalence rate came from West Cape Province, South Africa. The rate was 0.04% on June 1, 2020 and 0.06% on June 9, 2020, increasing by 50% within only 8 days, not to mention the public sector tests were limited to patients >55 years of age or with comorbidity. From the perspective of hindsight, though the rate was low, considering it was at the beginning of the pandemic, the situation was still worrying. However, we cannot infer the global prevalence rate based on this very limited information, and we believe more studies are urgently needed, especially for countries with high COVID-19 cases and high TB prevalence.

Although case report is not a very sound study design for us to evaluate treatment and outcome information, we didn't find any other kinds of study design about this topic.

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The analysis of case reports showed the treatment for coinfection was of great heterogeneity, while we still identified 4 main paradigms: 1) simultaneous antiviral treatment and ATT, 2) only ATT, 3) only antiviral treatment, 4) no antiviral treatment and ATT. Some studies implied the doctor just followed the local clinical guidelines; however, very few studies mentioned regimen adjust due to characteristics of TB and COVID-19 coinfected patients. Therefore, we assumed the current treatment plans were more like a mix of the suggested TB regime and the COVID-19 regime. Besides, we did not detect any specific treatment protocol for patients with inactive TB, namely the previous TB or latent TB patients. Furthermore, HCQ, the most commonly used antiviral drug in these cases, was not taken by any patient from China's study. Traditional Chinese Medicine (TCM), which was widely used for COVID-19 patients in China[57], was also seldomly mentioned in these China's reports. The reason might be that these cases were reported at the beginning of the pandemic, and the doctors lacked the experience to deal with the situation. We tried multiple ways to explore the relationship between treatments and outcomes, but we did not find meaningful results. Therefore, we could not find evidence on which one is the best clinical practice due to a lack of sound investigation or experimental study, which suggested the urgency of further research.

The prolonged hospital stay for coinfected posed a significant challenge to the health care system, especially for those low-resources countries. Besides, the vaguely described health status at discharge and absence of long-term outcomes are rather

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alarming. Moreover, for discharged patients who were still on ATT, the adverse impact from pandemic to delivery of TB care services which had been confirmed in multiple countries[58-60], might impede their accessibility to services.

We tried to conduct MA on overall and in-hospital fatality rates, and the estimated results were 13.9% and 17.5%. According to Meyerowitz-Katz, the fatality rate of patients with single COVID-19 was 0.68% (95%CI: 0.053% ~ 0.82%)[61], which was relatively smaller. Therefore, we believe the coinfected patients were facing more challenges and calling for the need to refine medical equipment and interventions. In addition, significant heterogeneity was detected in both MAs on overall and in-hospital fatality rates. In contrast, no heterogeneity was detected in both sub-groups' MAs on in-hospital fatality rate. We did not detect any pattern about treatment in both high-income countries and LIMICs, based on case reports. Considering patinets in high-income countries are likely to receive care of better quality than patients in LIMICs, we believed the different results of MAs' heterogeneity hinted that the quality of care was highly associated with outcomes. In other words, the patients in high-income countries shared a better condition than those in LIMICs and were more likely to recover from TB and COVID-19 coinfection.

Regarding the quality of included studies, insufficient sample size and unappreciated sample frame were leading risk of bias for prevalence studies. In addition, the lack of

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adverse and unanticipated events, intervention and treatment procedure, and post-intervention condition were 3 main defects of the included 35 case reports.

Publication bias was found in overall and in-hospital fatality rate of the TB and COVID-19 coinfection, which suggested the current results stranded on selected studies. Thus, more studies are urgently needed to evaluate the prevalence, treatment, and long-term outcome of TB-COVID coinfection.

One main limitation of this systematic review was that we did not include comorbidity as a keyword and MeSH term in the searching process, which might keep us from some relevant studies. Besides, the diagnosis and testing criteria of COVID-19 and TB differs from countries and regions, and we were not able to transfer the data under a universal standard.

In conclusion, the results of our systematic review show coinfection was not rare worldwide, and outcomes shared significant differences between high-income countries and LIMICs. Given that some countries have higher TB and COVID-19 prevalence simultaneously, global efforts are needed to work collaboratively on TB-COVID coinfection, especially in response to those with limited medical resources.

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Ethical Statement N/A

Data availability statement Data are from the published study. Extracted data are available on request from the corresponding author.

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Conflict of Interest: The authors declare no conflict of interest.

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Contributor and Guarantor Information： Quan Wang put up with the idea and designed this study; Jie Zhao was responsible for the literature search; Quan Wang and Shasha Guo conducted the study selection (review of titles and abstracts, review of full texts, and final decision), data abstraction, and quality assessment independently; Ning Xu, as a medical doctor, provided necessary information about treatment and drugs; Quan Wang finished the manuscript; Hui Li, Qiang Sun, Xiaolin Wei and Quanfang Dong refined the manuscript; Qiang Sun was responsible for this study. The corresponding author, Qiang Sun, attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

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Dissemination to participants and related patient and public communities It is not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

Figure legends

Figure 1 A flow diagram outlines the study selection process.

Figure 2 Forest plots of fatality rate accorting to hospitalization, subgroups according to economic status of regions. A: overall fatality rate; B: in-hospital fatality rate; C: in-hospital fatality rate (high-income country subgroup); D: in-hospital fatality rate (low- and middle-income country subgroup)

Figure 3 Funnel plots of fatality rate accorting to hospitalization, subgroups according to economic status of regions. A: overall fatality rate; B: in-hospital fatality rate; C: in-hospital fatality rate (high-income country subgroup); D: in-hospital fatality rate (low- and middle-income country subgroup)

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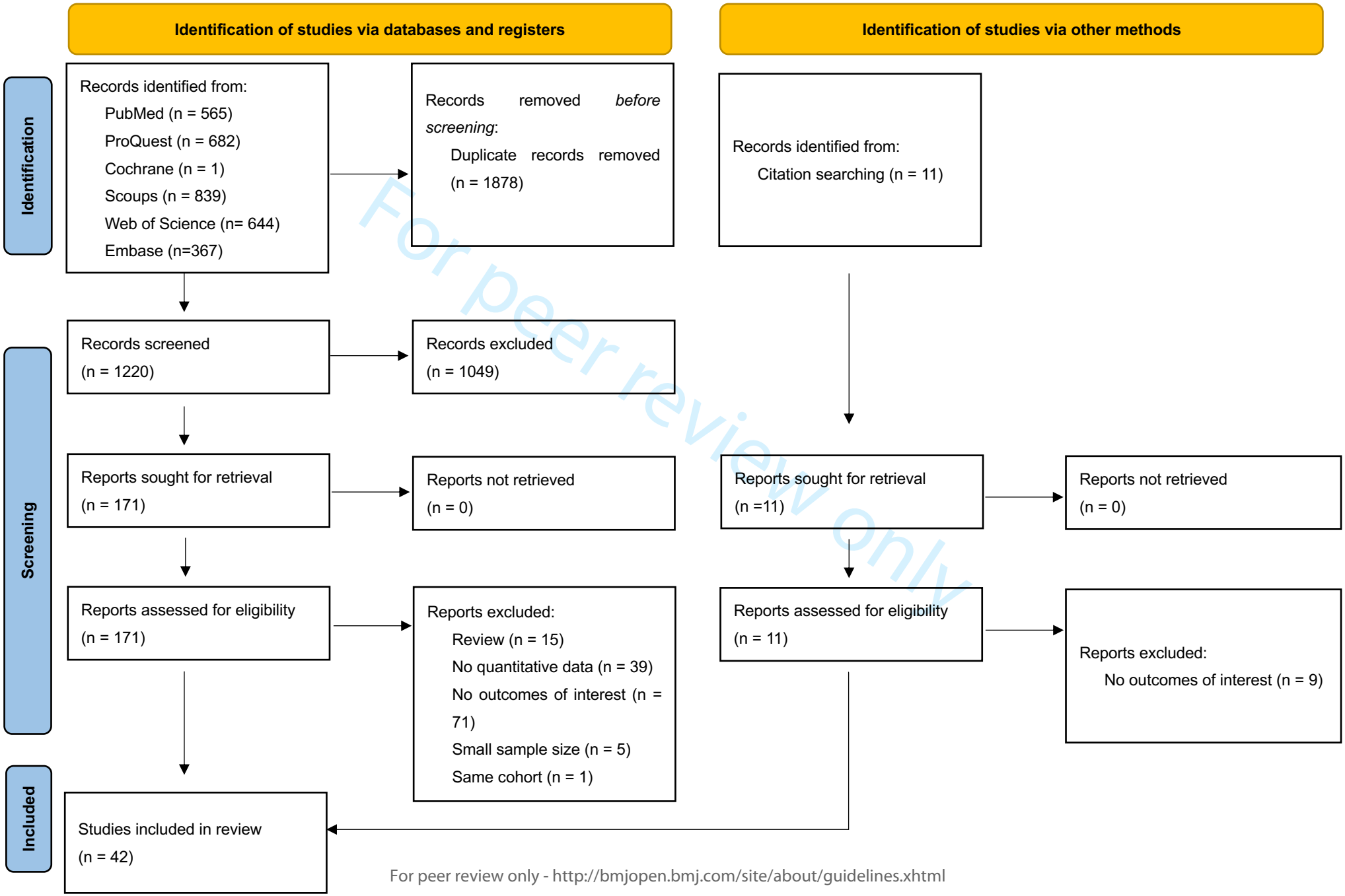
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Figure 1 Flow chart of literature search



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Figure 2 Forest plot of included studies

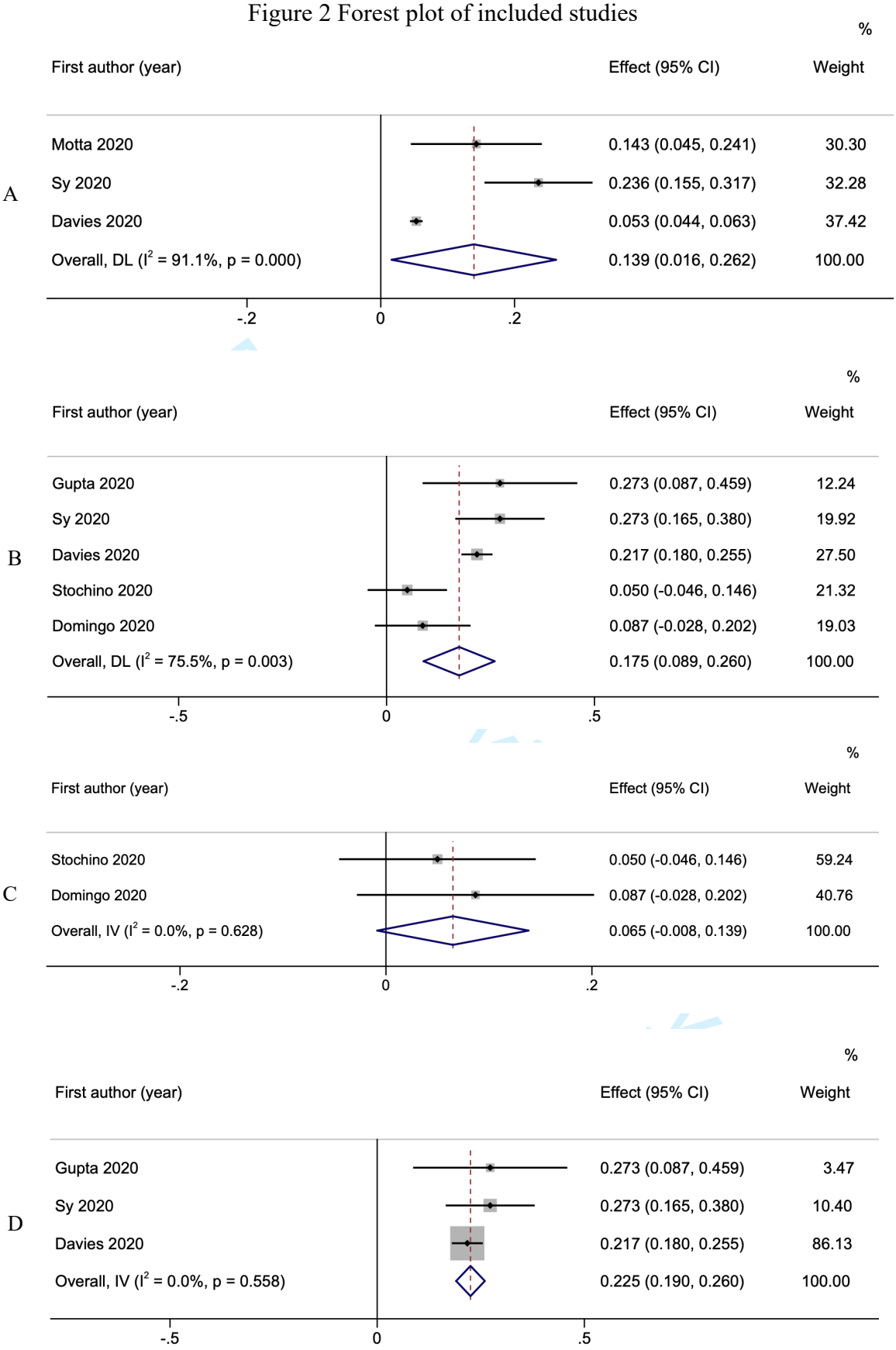
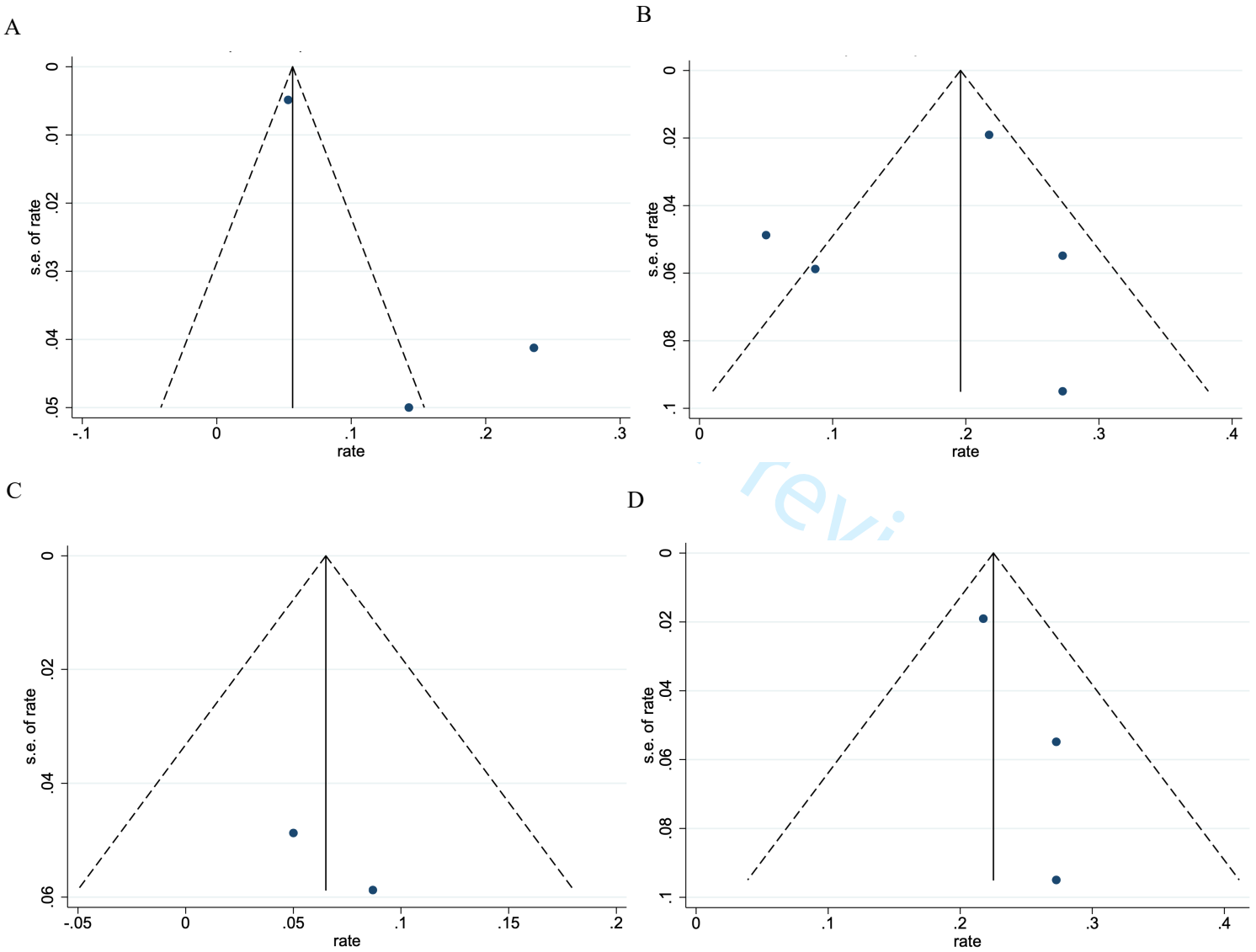


Figure 3 Funnel plot of included studies



Appendix 1

Table S1 The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P1-P2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P4-P5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P6-P7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	P7-P8, appendix 2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the	P8-P9

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Section and Topic	Item #	Checklist item	Location where item is reported
		process.	
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P8-P9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P8-P9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P9
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	P8-p9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P8-P9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P8-P10
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P8-P10
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P8-P10
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	P9-P10

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Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P10
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P10-P12
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	P10-P13
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	P19 appendix 4
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	P10-P18, appendix 3
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P11-P20
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	P19-P20
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P19
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	P19
Certainty of	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P15-P20

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Section and Topic	Item #	Checklist item	Location where item is reported
evidence			
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P21
	23b	Discuss any limitations of the evidence included in the review.	P24
	23c	Discuss any limitations of the review processes used.	N/A
	23d	Discuss implications of the results for practice, policy, and future research.	P22-P24
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P2, P5-6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P5-6
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	P5-6
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P25
Competing interests	26	Declare any competing interests of review authors.	P25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	P24

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Appendix 2

Table S2 Search strategies to identify studies reporting the prevalence status, treatment and outcomes of tuberculosis and COVID-19

Database searched: PubMed

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	("covid-19"[MeSH Terms] OR "sars-cov-2"[MeSH Terms]) AND "tuberculosis"[MeSH Terms]	244
2	((covid-19[Title/Abstract]) OR (Sars-cov-2[Title/Abstract])) AND((TB[Title/Abstract]) OR (tuberculos*[Title/Abstract]))	523
3	#1 OR #2	565

Database searched: Web of Science

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	((TS=(covid-19) OR TS=(sars-cov-2) OR AB=(covid-19) OR AB=(sars-cov-2)) OR (TS=(covid-19)) OR (TS=(sars-cov-2)) OR (AB=(covid-19))) AND (TS=(tuberculosis)) OR (TS=(Mycobacterium tuberculosis)) OR (TS=(Mycobacterium tuberculosis)) OR (AB=(tuberculo*)) OR (AB=(TB)) OR (AB=(Mycobacterium tuberculosis)) OR (AB=(M.tuberculosis)) OR (AB=(tubercle bacillus)))	644

Database searched: ProQuest

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	su(COVID-19 OR SARS-COV-2) AND su(TB OR tuberculo* OR (Mycobacterium tuberculosis) OR (M.tuberculosis))	682

Database searched: Scopus

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Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	(TITLE-ABS-KEY (covid-19 OR sars-cov-2) AND TITLE-ABS-KEY (tb OR tuberculos* OR (mycobacterium AND tuberculosis) OR (m.tuberculosis)))	839

Database searched: Cochrane database

Date: Mars 19, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	MeSH descriptor: [COVID-19] explode all trees	257
2	MeSH descriptor: [SARS-COV-2] explode all trees	204
3	MeSH descriptor: [tuberculosis] explode all trees	2,029
4	(#1 OR #2) AND 3	1

Database searched: Embase

Date: Mars 19, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	('covid 19':ab,ti OR 'sars cov 2':ab,ti) AND [01-11-2019]/sd NOT [20-3-2021]/sd	81,923
2	(tuberculosis:ab,ti OR 'mycobacterium tuberculosis':ab,ti OR tuberculo*:ab,ti OR 'tb':ab,ti OR 'm.tuberculosis':ab,ti OR 'tubercle bacillus':ab,ti) AND [01-11-2019]/sd NOT [20-3-2021]/sd	19,510
3	#1 AND #2	367

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Appendix 3

Table S3 Characteristics of reported TB and COVID-19 coinfection patients (n=55)

Basic information	Number (%)
Gender	
female	12 (21.8)
male	43 (78.2)
Age	
< 20	3 (5.5)
20-39	23 (41.8)
40-59	22 (40.0)
≥60	7 (12.7)
Comorbidity	
diabetes mellitus	8 (14.5)
hypertension	5 (9.1)
HIV	4 (7.3)
Type of TB	
active TB	50 (90.9)
post TB	4 (7.3)
latent TB	1 (1.8)
TB site	
pulmonary TB	47 (85.5)
central nervous system involvement	2 (3.6)
tuberculous meningitis	1 (1.8)

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tuberculous pleuritis	1 (1.8)
peritoneal TB	1 (1.8)
not reported	6 (10.9)
BCG Vaccination	
yes	2 (3.6)
not reported	53 (96.4)
Drug resistance	
no	15 (27.3)
yes	2 (3.6)
likely	1 (1.8)
not reported	37 (67.3)
Infection order	
TB before COVID-19	17
COVID-19 before TB	0
not reported	38
Symptomatic COVID-19	
yes	55 (100.0)
no	0 (0.0)

Table S4 Detailed basic information of included case reports (n=35)

Author (Year)	Country	Cases number	Age	Gender	Comorbidity	BCG Vaccination	TB site	Drug resistance	Active TB	TB before COVID-19	Symptomatic COVID-19

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Sarma 2020[1]	India	1	53	female	chronic kidney disease (CKD) stage 4, type 2 diabetes mellitus	/	pulmonary TB	likely	yes	yes	yes
Singh 2020[2]	India	1	76	female	/	/	pulmonary TB	/	yes	/	yes
Kumar 2020[3]	India	1	38	male	/	/	pulmonary TB	/	yes	/	yes
Ata 2020[4]	India	1	28	male	no HIV	/	pulmonary TB central nervous system involvement	/	yes	/	yes
Yadav 2020[5]	India	1	43	male	non-diabetes	/	pulmonary TB	no	yes	/	yes
Baskara 2021[6]	Indonesian	1	42	male	diabetes mellitus	/	pulmonary TB	no	yes	yes	yes
Wang 2020[7]	China	1	45	male	/	/	tuberculous meningitis	/	yes	/	yes
He 2020[8]	China	3	67	male	hypertension	/	/	/	no	yes	yes
			26	male	/	/	/	/	no	yes	yes
			76	male	exfoliative dermatitis, Arthrolithiasis	/	/	/	no	yes	yes
Cao 2020[9]	China	1	47	female	/	/	pulmonary TB	/	yes	yes	yes

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			50	male	anemia, hypoproteinemia	/	pulmonary TB	/	yes	yes	yes
Yao 2020[10]	China	3	44	male	diabetes, pleural effusion	/	pulmonary TB, tuberculous pleuritis	/	yes	/	yes
			57	male	diabetes	/	pulmonary TB	/	yes	yes	yes
			48	male	/	/	/	/	no	/	yes
			26	male	/	/	pulmonary TB	yes	yes	yes	yes
Liu 2020[11]	China	3	46	male	/	/	pulmonary TB, tuberculous pleuritis	/	no	yes	yes
Khayat 2021[12]	Saudi Arabia	1	40	female	no HIV and hepatitis	/	pulmonary TB	no	yes	/	yes
Faqihi 2020[13]	Saudi Arabia	1	60	male	hypertensive and diabetic	/	pulmonary TB	no	yes	/	yes
			34	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
Yousaf 2020[14]	Qatar	6	32	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			50	male	no HIV, hepatitis B, and hepatitis C	/	pulmonary TB	/	yes	/	yes

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					infection; have diabetes						
			35	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			27	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			35	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB, tuberculous pleuritis	/	yes	/	yes
AlKhateeb 2020[15]	Qatar	1	28	male	/	/	pulmonary TB	/	yes	/	yes
Tham 2020[16]	Singapore	4	32	male	/	/	pulmonary TB	/	yes	/	yes
			33	male	/	/	pulmonary TB	/	yes	/	yes
			22	male	/	/	pulmonary TB	/	yes	/	yes
			40	male	/	/	pulmonary TB	/	yes	/	yes
Cinar 2020[17]	Turkey	1	55	male	/	/	/	/	yes	/	yes
Sarinoglu 2020[18]	Turkey	2	77	female	diabetes mellitus, hypertension, chronic obstructive lung disease, chronic kidney failure	/	pulmonary TB	/	yes	/	yes
			39	female	/	/	pulmonary TB	/	yes	/	yes
Lopinto	France	1	58	male	pulmonary	/	pulmonary TB	/	yes	yes	yes

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2020[19]					embolism						
Musso 2021[20]	Italy	1	45	male	No HIV	/	pulmonary TB	no	yes	yes	yes
Luciani 2020[21]	Italy	1	32	female	No HIV	yes	pulmonary TB	/	yes	/	yes
Bouaré 2020[22]	Morocco	1	32	female	/	/	pulmonary TB	/	yes	/	yes
Essajee 2020[23]	South Africa	1	2	female	/	yes	pulmonary TB	no	yes	/	yes
Pillay 2021[24]	South Africa	1	44	female	HIV, pneumocystis iirovecii pneumonia, community-acquired pneumonia, pulmonary embolism	/	pulmonary TB	/	yes	/	yes
Agada 2021[25]	Nigeria	1	50	male	No HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	no	yes	/	yes
Gbenga 2020[26]	Nigeria	2	33	male	/	/	pulmonary TB	/	yes	/	yes
			30	male	/	/	pulmonary TB	/	yes	/	yes
Rivas 2020[27]	Panama	2	29	male	pleural empyema	/	pulmonary TB	no	yes	/	yes
			53	male	HIV	/	pulmonary TB	/	yes	yes	yes
Farias 2020[28]	Brazil	2	39	male	HIV, hepatitis B	/	pulmonary TB	no	yes	yes	yes
			43	male	HIV, anxiety	/	pulmonary TB	no	yes	/	yes

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Pinheiro 2020[29]	Brazil	1	68	male	disorder diabetes, hypertension, chronic liver disease secondary to schistosomiasis	/	pulmonary TB	no	yes	/	yes
Gerstein 2021[30]	USA	1	49	male	cirrhosis, portal hypertension	/	pulmonary TB, peritoneal TB	/	yes	/	yes
Cutler 2020[31]	USA	1	61	male	Parkinson's disease	/	pulmonary TB	/	yes	/	yes
Sasson 2020[32]	USA	1	44	male	no HIV; hypertension, type II diabetes mellitus, a previous stroke secondary to a left atrial thrombus	/	/	no	yes	/	yes
Freij 2020[33]	USA	1	5	female	no HIV	/	central nervous system involvement	/	yes	/	yes
Vanzetti 2020[34]	Argentina	3	33	female	/	/	pulmonary TB	no	yes	/	yes
			34	male	/	/	pulmonary TB	no	yes	yes	yes
			17	male	/	/	pulmonary TB	no	yes	/	yes
Vilbrun 2020[35]	Haiti	1	26	male	no HIV	/	pulmonary TB	yes	yes	yes	yes

/: No information provided

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Table S5 Detailed treatment information of included cases (n=55)

Author (Year)	Cases number	ICU	Treatment before hospitalization	Treatment in hospital	Treatment after discharge	Post-intervention clinical condition	Outcome	Days to die or discharge
Sarma 2020[1]	1	yes	/	**Supportive treatment:** oxygen support (intubated) **Antibiotic:** ampicillin, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** vitamin C, midazolam, fentanyl	/	/	resolved	/
Singh 2020[2]	1	no	/	**Supportive treatment:** / **Antibiotic:** ceftriaxone, azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** N-acetyl cysteine, methylprednisolone, vitamin C	/	/	on treatment	/

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Kumar 2020[3]	1	no	/	**Supportive treatment:** oxygen support, mechanical ventilation **Antibiotic:** ATT, other antibiotic (no detail) **Antiviral drug:** hydroxychloroquine sulphate **Other treatment:** /	/	/	died	/
Ata 2020[4]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** steroids	rifampicin, ethambutol, and pyrazinamide	/	resolved	28

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Yadav 2020[5]	1	no	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** / **Other treatment:** /	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, ethambutol, and pyrazinamide) **Antiviral drug:** / **Other treatment:** /	/	/	/	/
Baskara 2021[6]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), azithromycin, ceftriaxone, ceftazidime **Antiviral drug:** hydroxychloroquine, oseltamivir **Other treatment:** anti-coagulant drugs	treatment for diabetes mellitus	/	resolved	13

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Wang 2020[7]	1	no	/	**Supportive treatment:** / **Antibiotic:** rifampicin, amikacin, moxifloxacin **Antiviral drug:** arbidol **Other treatment:** tolvaptan, aspirin, mannitol, potassium supplement, sodium supplement	/	/	died	14
He 2020[8]	3	no	/	**Supportive treatment:** noninvasive ventilation support **Antibiotic:** antibiotic (no details and not ATT) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** methylprednisolone, probiotics, traditional Chinese medicine, immunoglobulin	/	/	resolved	37

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no	/	**Supportive treatment:** / **Antibiotic:** antibiotic (no details and not ATT) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** probiotics, traditional Chinese medicine	/	skin rashes with pruritus	ninth day after discharge, returned to the hospital because of positive recurrence of SARS-CoV-2 RNA	28
no	/	**Supportive treatment:** nasal cannula oxygen support **Antibiotic:** ATT, antibiotic (no details) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** methylprednisolone, glucocorticoid, probiotics, traditional Chinese medicine, immunoglobulin	/	/	resolved	24

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Cao 2020[9]	1	no	/	**Supportive treatment:** noninvasive ventilation support **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), moxifloxacin hydrochloride **Antiviral drug:** lopinavir and ritonavir, umifenovir **Other treatment:** polyene phosphati-dylcholine, reduced glutathione	/	Hepatic injury during the treatment	resolved	More than 14 days
Yao 2020[10]	3	no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** ATT **Antiviral drug:** / **Other treatment:** corticosteroid	/	/	died	22

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		no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** antiviral drugs (no details) **Other treatment:** acarbose, metformin	ATT	/	resolved	26
		no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** antibiotics (no details) **Antiviral drug:** antiviral drugs (no details) **Other treatment:** corticosteroids, immunoglobulin	/	/	resolved	27
Liu 2020[11]	3	no	/	**Supportive treatment:** extracorporeal membrane oxygenation, haemoperfusion **Antibiotic:** moxifloxacin, linezolid, no ATT **Antiviral drug:** arbidol **Other treatment:**	/	/	resolved	14

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		caspofungin, thymopentin					
no	/	**Supportive treatment:** nasal high-flow oxygen therapy **Antibiotic:** moxifloxacin, linezolid, cycloserine, clofazimine, pyrazinamide **Antiviral drug:** arbidol **Other treatment:** /	ATT	/	resolved	14
no	/	**Supportive treatment:** nasal high-flow oxygen therapy **Antibiotic:** moxifloxacin **Antiviral drug:** arbidol **Other treatment:** /	/	/	resolved	9

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Khayat 2021[12]	1	no	/	/	/	/	/	/
Faqihi 2020[13]	1	yes	/	**Supportive treatment:** respiratory support **Antibiotic:** moxifloxacin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir, ritonavir, ribavirin **Other treatment:** dexamethasone, prophylactic anticoagulation	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), isoniazid and rifampicin (later)	/	resolved	27
Yousaf 2020[14]	6	no	/	**Supportive treatment:** / **Antibiotic:** ceftriaxone, azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/

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AlKhateeb 2020[15]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details) **Antiviral drug:** / **Other treatment:** /	/	/	resolved	1 month
Tham 2020[16]	4	no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details) **Antiviral drug:** / **Other treatment:** /	/	/	/	/
		no	/		/	/	/	/
		no	/		/	/	/	/
		no	/		/	/	/	/
Cinar 2020[17]	1	yes	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), meropenem **Antiviral drug:** favipiravir **Other treatment:**	/	myelodysplastic syndrome (MDS/RAEB1 FAB subtype) during the treatment	/	/

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				convalescent plasma product, tocilizumab					
Sarinoglu 2020[18]	2	yes	hemodialysis	**Supportive treatment:** / **Antibiotic:** ATT (no details), piperacillin/tazobactam **Antiviral drug:** favipiravir, hydroxychloroquine **Other treatment:** /	hemodialysis	/	/	/	
		no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details), azithromycin, plaquanil, ciprofloxacine, tazocilli **Antiviral drug:** / **Other treatment:**	/	/	/	/	

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				/				
Lopinto 2020[19]	1	yes	/	/	/	/	resolved	14
Musso 2021[20]	1	no	/	**Supportive treatment:** chest tube drainage **Antibiotic:** amikacin, moxifloxacin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hidroxicloroquine **Other treatment:** corticosteroids	/	/	died	14

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Luciani 2020[21]	1	no	/	**Supportive treatment:** / **Antibiotic:** amoxicillin/clavulanic acid, clarithromycin, piperacillin/tazobactam, linezolid **Antiviral drug:** hydroxychloroquine, lopinavir, ritonavir **Other treatment:** paracetamol	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	/	resolved	> 12
Bouaré 2020[22]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), azithromycin **Antiviral drug:** chloroquine **Other treatment:** /	/	/	resolved	10

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Essajee 2020[23]	1	no	/	**Supportive treatment:** high flow nasal oxygen **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), **Antiviral drug:** / **Other treatment:** prednisone, aspirin, dexamethasone	/	/	on treatment	/
Pillay 2021[24]	1	no	/	**Supportive treatment:** rebreather mask **Antibiotic:** ATT, amoxicillin/clavulanic acid, azithromycin, trimethoprim, sulfamethoxazole **Antiviral drug:** / **Other treatment:** pyridoxine, prednisone, prophylactic clexane, dexamethasone, vitamin D, zinc, thiamine, ascorbic acid, therapeutic anticoagulation	/	/	on treatment	/

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Agada 2021[25]	1	no	/	**Supportive treatment:** supplemental oxygen **Antibiotic:** ceftriaxone, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** no **Other treatment:** dexamethasone, acetaminophen, pyridoxine phosphate, zinc-sulphate, Vitamin C	ATT	/	resolved	14
Gbenga 2020[26]	2	no	/	**Supportive treatment:** / **Antibiotic:** azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir, ritonavir **Other treatment:** Vitamin C, Zinc sulfate, oral prednisolone, tabs pyridoxine	/	/	resolved	17

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		no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir/ritonavir, **Other treatment:** vitamin C, zinc sulfate, tabs pyridoxine, IV hydrocortisone, clexane	/	/	died	6
Rivas 2020[27]	2	yes	/	**Supportive treatment:** invasive mechanical ventilatory support **Antibiotic:** ATT (isoniazid, rifampin, ethambutol, and pyrazinamide), levofloxacin, azithromycin, **Antiviral drug:** hydroxychloroquine, tenofovir, lamivudine, and dolutegravir **Other treatment:** low molecular weight heparin	/	nosocomial pneumonia	resolved	2 months

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		no	rifampicin and isoniazid	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ceftriaxone, azithromycin, **Antiviral drug:** hydroxychloroquine **Other treatment:** enoxapari	/	/	resolved	7
Farias 2020[28]	2	no	/	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ATT (isoniazid, rifampin, ethambutol, and pyrazinamide), ceftriaxone, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	21

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		no	/	**Supportive treatment:** / **Antibiotic:** azithromycin, ceftriaxone, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	7
Pinheiro 2020[29]	1	no	/	/	/	/	/	/
Gerstein 2021[30]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (levofloxacin, rifampicin, isoniazid, ethambutol) **Antiviral drug:** hydroxychloroquine, **Other treatment:** convalescent plasma	ATT (rifampicin, isoniazid)	/	resolved	42

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Cutler 2020[31]	1	no	/	**Supportive treatment:** nasal canula oxygen supplementation **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	/
Sasson 2020[32]	1	yes	intubated for hypoxia	**Supportive treatment:** intubated for hypoxia **Antibiotic:** azithromycin, ceftriaxone, cefepime, vancomycin, piperacillin/tazobactam, meropenem, caspofungin, metronidazole, voriconazole, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** dexamethasone, methylprednisolone, vitamin B6, metformin, alogliptin, long-acting insulin and rapid-acting insulin	/	/	resolved	> 35

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Freij 2020[33]	1	no	amoxicillin	**Supportive treatment:** / **Antibiotic:** azithromycin **Antiviral drug:** hydroxychloroquine, remdesivir **Other treatment:** dexamethasone, antihypertensive medications	/	/	died	26
		no	/	/	/	/	resolved	
		no	/	/	/	/	resolved	
Vanzetti 2020[34]	3	no	/	**Supportive treatment:** oxygen support **Antibiotic:** / **Antiviral drug:** / **Other treatment:** corticosteroids	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	/	resolved	longer hospital stay than average

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Vilbrun 2020[35]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (bedaquiline, levofloxacin, linezolid, clofazimine, pyrazinimide) **Antiviral drug:** / **Other treatment:** /	/	/	/	resolved	3 weeks from the diagnosis

/: No information provided

Table S6 Detailed treatment information of included retrospective cohort studies (n=7)

First author (year)	Country	Time	single center	Sample size	In-hospital	age	gender	Comorbidity	Died	Fatality rate (in-hospital or total)
Gupta 2020[36]	India	1 February 2020 to 14 June 2020	yes	22	yes	19-67	20 male (90.9%) 2 female (9.1%)	Hypertension 4 (18.2%) Diabetes 3 (13.6%) Seizure disorder 2 (9.1) Hypothyroidism 1 (4.5%)	6	27.3 (in-hospital)
Motta 2020[37]	Belgium, Brazil, France, Italy, Russia, Singapore,	12 March, 2020 to 5 May, 2020	no	49	no	27-70	/	/	7	14.3% (total)

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	Spain, and Switzerland									
Sy 2020[38]	Philippines	May 17, 2020 to June 15, 2020	no	106	no	/	73 male (68.9%) 33 female (31.1%)	Hypertension 22 (20.8) Diabetes 14 (13.2) Cancer 1 (0.9) Renal cancer 4 (3.8) Cardiac disease 8 (7.5) Asthma 4 (3.8) COPD 3 (2.8)	25	23.6% (total)
				66	yes	/		Hypertension 16 (24.2) Diabetes 10 (15.2) Cancer 1 (1.5) Renal cancer 4 (6.1) Cardiac disease 7 (10.6) Asthma 1 (1.5) COPD 1 (1.5)	18	27.3% (in hospital)
Stochino 2020[39]	Italy	/	yes	20	yes	27-47	12 male (60%) 8 female (40%)	/	1	5% (in-hospital)
Davies 2020[40]	South Africa	Till to 9 June, 2020	no	2128	no	20 or above	/	/	113	5.3% (total)
			no	469	yes	20 or	/	/	102	21.7%

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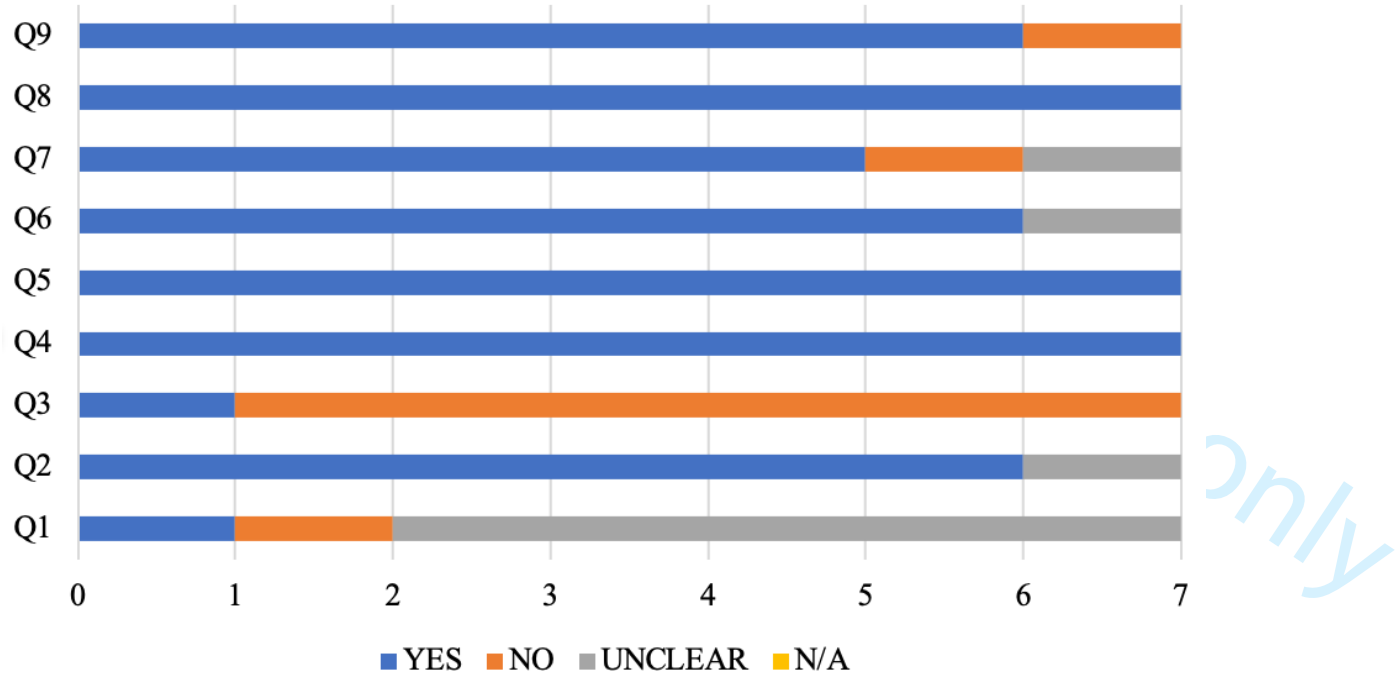
						above				(in-hospital)
Domingo 2020[41]	Argentina	1 March to 30 June, 2020	yes	23	yes	5-82	18 male (78.3%) 5 female (21.7%)	Smoking and other addictions 16(69.6) HIV 4(17.4) Psychosis 1(4.3) Pulmonary thromboembolism 1(4.3) Arterial hypertension 1(4.3)	2	8.7% (in-hospital)
Gubkina 2020[42]	Russia	March to June, 2020	24	yes	/	/	/		0	0% (in-hospital)

/: No information provided

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Appendix 4

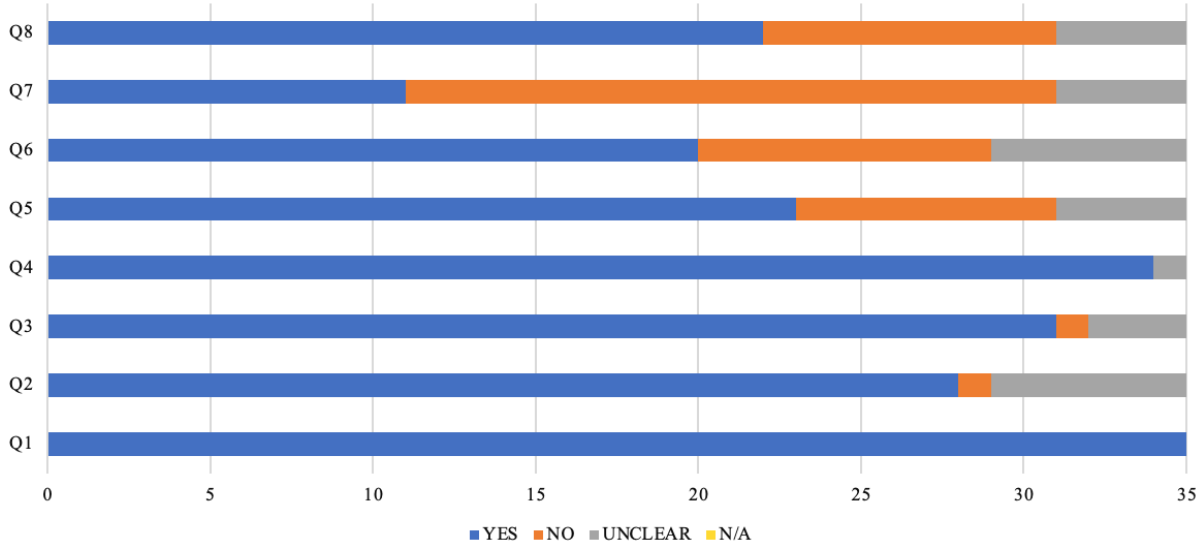
Figure S1 The results of quality assessment for prevalence study



Note: Four options for each question: yes, no, unclear and not appliable. Q1: Was the sample frame appropriate to address the target population?

Q2: Were study participants sampled in an appropriate way? Q3: Was the sample size adequate? Q4: Were the study subjects and the setting described in detail? Q5: Was the data analysis conducted with sufficient coverage of the identified sample? Q6: Were valid methods used for the identification of the condition? Q7: Was the condition measured in a standard, reliable way for all participants? Q8: Was there appropriate statistical analysis? Q9: Was the response rate adequate, and if not, was the low response rate managed appropriately?

Figure S2 The results of quality assessment for case report



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Note: Four options for each question: yes, no, unclear and not appliable. Q1: Were patient's demographic characteristics clearly described? Q2: Was the patient's history clearly described and presented as a timeline? Q3: Was the current clinical condition of the patient on presentation clearly described? Q4: Were diagnostic tests or assessment methods and the results clearly described? Q5: Was the intervention(s) or treatment procedure(s) clearly described? Q6: Was the post-intervention clinical condition clearly described? Q7: Were adverse events (harms) or unanticipated events identified and described? Q8: Does the case report provide takeaway lessons?

Table S7 Quality assessment of included case reports (n=35)

Study (Year) \ Results	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Agada 2021	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Yao 2020	Yes	Yes	Yes	Unclear	Unclear	No	No	No
Yadav 2020	Yes	Yes	Yes	Yes	No	No	No	No
Wang 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Vilbrun 2020	Yes	Yes	Yes	Yes	No	Unclear	No	Yes
Tham 2020	Yes	Yes	Yes	Yes	No	No	No	No
Singh 2020	Yes	Yes	Yes	Yes	Unclear	No	Yes	Yes
Sasson 2020	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear
Sarma 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Sarinoglu 2020	Yes	Yes	Yes	Yes	Unclear	Yes	No	Yes
Rivas 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No

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Pinheiro 2020	Yes	Yes	Unclear	Yes	No	No	No	No
Pillay 2021	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Musso 2021	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Luciani 2020	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Lopinto 2020	Yes	Yes	Yes	Yes	No	Unclear	Yes	Yes
Liu 2020	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Kumar 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Khayat 2021	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Gbenga 2020	Yes	Yes	Yes	Yes	Unclear	Unclear	Unclear	No
Freij 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cutler 2020	Yes	Yes	Unclear	Yes	Yes	Yes	No	Yes
Cinar 2020	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes
Cao 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farias 2020	Yes	Yes	Yes	Yes	Yes	Unclear	No	Yes
Bouare 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Baskara 2021	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
AlKhateeb 2020	Yes	Yes	Yes	Yes	No	Yes	No	Unclear

Note: Four options for each question: yes, no, unclear and not appliable. Q1: Were patient's demographic characteristics clearly described? Q2: Was the patient's history clearly described and presented as a timeline? Q3: Was the current clinical condition of the patient on presentation clearly described? Q4: Were diagnostic tests or assessment methods and the results clearly described? Q5: Was the intervention(s) or treatment procedure(s) clearly described? Q6: Was the post-intervention clinical condition clearly described? Q7: Were adverse events (harms) or unanticipated events identified and described? Q8: Does the case report provide takeaway lessons?

Table S8 Quality assessment of included retrospective cohort studies (n=7)

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Results / Study (Years)	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
Gupta 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Motta 2020	Unclear	Unclear	No	Yes	Yes	Yes	Yes	Yes	Yes
Sy 2020	Yes	Yes	No	Yes	Yes	Unclear	Unclear	Yes	No
Stochino 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Davies 2020	Unclear	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Domingo 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Gubkina 2020	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes

Note: Four options for each question: yes, no, unclear and not appliable. Q1: Was the sample frame appropriate to address the target population? Q2: Were study participants sampled in an appropriate way? Q3: Was the sample size adequate? Q4: Were the study subjects and the setting described in detail? Q5: Was the data analysis conducted with sufficient coverage of the identified sample? Q6: Were valid methods used for the identification of the condition? Q7: Was the condition measured in a standard, reliable way for all participants? Q8: Was there appropriate statistical analysis? Q9: Was the response rate adequate, and if not, was the low response rate managed appropriately?

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The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis (Till to March 2021)

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Primary Subject Heading:	Global health
Secondary Subject Heading:	Public health, Infectious diseases
Keywords:	COVID-19, Tuberculosis < INFECTIOUS DISEASES, PUBLIC HEALTH

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The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis (Till to March 2021)

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The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis (Till to March 2021)

Abstract

Introduction The breakout of COVID-19 poses a significant threat to the TB patients. TB and COVID-19 (TB-COVID) coinfection means the disease caused by both mycobacterium tuberculosis and SARS-COV-2 infection. Currently, the prevalence status, treatment, and outcomes of the coinfection are poorly characterised. We aimed to systematically review the evidence on this topic and provide comprehensive information to guide the control and treatment of TB-COVID coinfection.

Methods An extensive screening was conducted using 6 electronic databases to search eligible studies till to March 19, 2021. Prevalence rate, treatment, and outcomes of TB-COVID coinfection were extracted. Random effects models were used to calculate mean fatality rates of coinfection with 95% confidence intervals (CI). The risks of bias were assessed with JBI Critical Appraisal Checklist for Study Reporting Prevalence Data and JBI Critical Appraisal Checklist for Case Report. A meta-analysis was conducted for subgroups on in-hospital fatality rate.

Results Total 42 of 3109 studies were included into the analysis (35 case reports and 7 retrospective cohort studies). Nineteen countries reported coinfected patients, including high and low TB prevalence countries. The only study revealing prevalence rate came from West Cape Province, South Africa (people aged above 20, 0.04% till June 1, 2020

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and 0.06% till to June 9, 2020).The treatment regimes for coinfected patients were highly heterogeneous. The mean overall and in-hospital fatality rates of coinfection were 13.9% (95% CI: 1.6% to 26.2%) and 17.5% (95% CI: 8.9% to 26.0%). The mean in-hospital fatality rate for high-income country (Italy and Argentina) and low- and middle-income countries (LMICs) (India, Philippines, South Africa) were 6.5% (95% CI: -0.8% ~ 13.9%) and 22.5% (95% CI: 19.0% ~ 26.0%).

Conclusion TB-COVID coinfection is common globally and the coinfected patients are suffering from higher fatality risk than normal COVID-19 patients. Outcomes shared significant differences between high-income countries and LMICs.

Registration PROSPERO CRD42021253660

Keywords: COVID-19, tuberculosis, coinfection, prevalence, health outcome, treatment, systematic review

Strengths and limitations of this study

- This systematic review and meta-analysis provided comprehensive information on prevalence status, treatment plans and possible outcomes of TB-COVID coinfection.
- We collected evidence from case reports regarding information about patients' basic information, treatment regimens and outcomes.
- Sub-group analysis was conducted to explore the possible reason for different fatality rates in different areas.

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- We did not include comorbidity as a keyword and MeSH term in the searching process, which might keep us from some relevant studies.

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1. Introduction

Individuals with chronic respiratory infections, including tuberculosis (TB) had been disproportionally hit by the coronavirus disease 2019 (COVID-19) pandemic[1]. The burden and impact of TB control and treatment is widely acknowledged with the ongoing of pandemic. Patients with TB have an increased risk of mortality during a coinfection with COVID-19[2]. On the other side, the COVID-19 outbreak has put on the TB patients and entire health care system in an unprecedented crisis. The number of people who fell ill with TB in 2019 is setimated to 10 million[3]. The top seven countries account for two-thirds of the total global cases in 2018, while most of high prevalence rate countries locate in South-East Asia and Africa[4]. Recently many experts have recognized the effects of the COVID-19 pandemic on TB treatment and control, and some researches identified TB as a risk factor to severe COVID-19[5]. As warned by Global Tuberculosis Network Group, COVID-19 has fueled TB infections and mortality[6]. There have been some studies suspected that the COVID-19 could reactivate or worsen TB[6 7]. From the perspective of pathology, the disorders induced by SARS-COS-2 and Mycobacterium tuberculosis to the immunomodulation tend to induce an unbalanced inflammatory response, which can promote the progression and worsening of both diseases[8]. Besides, the use of immunosuppressive drugs on severe COVID-19 patients may lead to increased likelihood of active TB caused by reactivation or new infection[9]. In low- and middle-income countries (LMICs), the situation is even worse. As estimated by Kissler, the transmission of SARS-CoV-2 is likely to enter into regular circulation[10], which would be a deadly threat to people in LMICs, especially when most

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of them cannot stand the chance of getting COVID-19 vaccine[11]. Regrettably, people in these countries are also the main victim suffering from tuberculosis[3].

Although cases of coinfection and studies about prevalence status have been reported from several countries[12], we did not find a common definition about the TB and COVID-19 (TB-COVID) coinfection. In this systematic, we preliminarily described the TB-COVID coinfection as the disease caused by both mycobacterium tuberculosis and SARS-COV-2 infection. However, much of the research up to now has been descriptive in nature and focused on the observation of several coinfected patients, which can only provide weak evidence to guide the treatment for TB-COVID coinfection. Furthermore, few scholar has conducted a systematic review in this field, and there is no general agreement about the prevalence, treatment and outcome of the coinfection worldwide.

Although few health alert and guidelines for people with increased risk for adverse outcomes in COVID-19 have been issued by World Health Organization (WHO) and Center for Disease Control and Prevention (CDC)[13], there is a dearth of information regarding the impact of COVID-19 in patients with TB and the actual impact of tuberculosis on occurrence and clinical outcomes of COVID-19 is not clear. This study aims to systematically analyze the current evidence on global prevalence status, treatment protocols, and possible outcomes of TB and COVID-19 coinfection.

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2. Methods

This systematic review protocol followed the International Prospective Register of Systematic Reviews (PROSPERO) guideline and has been registered in PROSPERO's database (registration number: CRD42021253660). The study reporting followed the Preferred Reporting Items in Systematic Reviews and Meta-Analyses (PRISMA) guideline[14] (Appendix Table 1).

2.1 Eligibility Criteria of Included Studies

Studies on patients who had been confirmed with both TB and COVID-19 were included. We excluded studies that only mentioned the fatality and the number of coinfected patients less than 10 to avoid large deviation in fatality rate, which would affect the results of Meta-Analysis (MA).

This systematic review included any kind of treatment for TB or COVID-19, including supportive therapy, directed therapy, antiretroviral treatment, ICU care, etc.

We sought prevalence rate data of any region and figured out all countries that reported coinfected patients. The primary outcome of interest was in-hospital survival or mortality and secondary outcomes were disease progression, time to discharge from the

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ICU and hospital (for survivors), and time to death (for the dyed).

Case reports, cross-sectional studies, and cohort studies were included to explore the global prevalence status. Case reports, cross-sectional studies, cohort studies, and quasi-experimental studies that revealed the treatment and outcome of TB-COVID coinfection were targeted. We excluded all reviews, editorials or commentaries, and clinical guidelines.

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2.2 Search Strategy

An extensive search of literature was conducted using PubMed, Web of Science, Scopus, ProQuest, Embase, and Cochrane library databases. Manual reference screens from included studies were performed to ensure the inclusion of all relevant studies. The study's publication date was between November 1, 2019 and March 19, 2021. With the librarian's (JZ) help, we used building blocks searching strategy to ensure all the relevant studies were screened. The searching process utilized Medical Subject Heading (MeSH) terms and Tittle/Abstract words around two areas: 1) COVID-19, including COVID-19, SARS-COV-2, 2) tuberculosis, including tuberculosis, TB, Mycobacterium tuberculosis, M.tuberculosis (Appendix Table 2). The citation manager EndNote 20 (Thomson Corporation, Canada) removed duplicate studies, and we spent special efforts on the same patients or cohorts among different papers.

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2.3 Data Extraction

Two reviewers (QW and SG) independently conducted the study selection (review of titles and abstracts, review of full texts, and final decision), data abstraction, and quality assessment. Covidence (Covidence Company, Australia) was utilized to help us finish all the above processes. All non- English and non-Chinese articles were translated to English by Baidu Translation (Baidu Co., Ltd, China). During the review process, we recorded exclusion reasons for all excluded studies in the full-text screen step. The conflicts between 2 reviewers were discussed and eventually resolved with another author's help (NX).

Since this systematic review got 2 research objectives, 2 categories of information were extracted: 1) data about the prevalence, 2) data about treatment and outcome. For the first category, we stressed locations (nation or province, especially for case reports) of studies and the following information: age, gender, sample size, time, prevalence rate, fatality rate. As for the second category, treatment for TB and COVID-19 were recorded separately, as well as the primary and secondary outcomes (e.g., disease progression, time to discharge from the ICU and hospital, time to death). Besides, we also stressed patients' basic information in both parts, like TB site, TB type, drug resistant status, comorbidity, BCG vaccination, etc.

2.4 Quality Assessment

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We used the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Study Reporting Prevalence Data for the prevalence part as the tool[15]. Nine questions in this checklist were set to assess the quality of studies, including sample frame, sample size, data analysis, etc. All included studies were case reports for the treatment and outcome review, so we applied JBI Critical Appraisal Checklist for Case Report[16] as the tool which consists of 8 questions about patients' condition, test and assessment, intervention, etc.

2.5 Data Synthesis

A narrative synthesis method was introduced for listing prevalence status and key findings such as the countries and regions of the case reports to probe the potential prevalence of TB-COVID coinfection. We condensed the information about treatment into 3 categories: treatment before hospitalization, treatment in hospital, and treatment after discharge. We also sorted out the treatment in the hospital into 4 kinds: supportive treatment, antibiotics, antiviral drugs, and other treatment. The narrative synthesis in this systematic review followed Cochrane's Consumers and Communication Review Group (CC&CRG) reviews protocol[17]. We conducted MA about the fatality rate of coinfection patients using the inverse variance statistical method to pool the data and estimate its 95% confidence interval (CI). Due to the highly heterogeneous treatment method for patients as well as patients themselves, we chose the random-effects model in MA and the heterogeneity was evaluated with I^2 statistic. The funnel plot was created

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to test the bias.

2.6 Subgroup Set

Studies were grouped into the following subgroups for further analysis: high-income country and low- and middle-income country groups, by doing which we would like to explore the possible difference in fatality rate among different types of country.

2.7 Patient and Public Involvement

Patients were not involved in this systematic review.

3. Results

We searched 565 studies from PubMed, 682 from ProQuest, 839 from Scopus, 644 from Web of Science, 367 from Embase, and 1 from the Cochrane database. After removing 1878 duplicate studies, we got 1220 studies and imported them into Covidence. With the help of Covidence, we excluded 1049 studies in the title and abstract screen step. Next, we carefully reviewed the full text of the remaining 171 studies and then we excluded 131 studies from them. We also found another 11 possibly relevant studies during the full text reviewing process, 9 of which were excluded. The most common reasons for exclusion were no outcomes of interest (n = 80) and no quantitative data (n=39). The entire process is shown in Figure 1. Among the 42

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included studies, 35 were case reports, and 7 were retrospective cohort studies. No experimental study was identified.

<*Insert Figure 1 here*>

3.1. Global Prevalence Status

Till to Mar. 19 2021, there were 35 case reports from 17 countries, including India[18-22], Indonesia[23], China[24-28], Saudi Arabia[29 30], Qatar[31 32], Singapore[33], Turkey[34 35], Haiti[36], Argentina[37], U.S.[38-41], Brazil[42 43], Panama[44], Nigeria[45 46], South Africa[47 48], Morocco[49], Italy[50 51], and France[52]. Totally, 55 cases were described in these 35 reports. China, India and U.S. were the top 3 countries with the most studies, and 29 (82.9%) studies were published in 2020. Table 1 shows the study characteristics of these case reports. Besides, we also identified coinfected patients from Belgium and Switzerland in Motta's paper[53].

Table 1 Study characteristics of case reports (n=35)

Study characteristic	Studies (%)	Cases (%)
Publication year		
2020	29 (82.9)	49 (89.1)
2021	6 (17.1)	6 (10.9)
Country		
India	5 (14.3)	5 (9.1)
Indonesian	1 (2.9)	1 (1.8)
China	5 (14.3)	11(20.0)
Saudi Arabia	2 (5.7)	2 (3.6)
Qatar	2 (5.7)	7 (12.7)
Singapore	1 (2.9)	4 (7.3)
Turkey	2 (5.7)	3 (5.5)
Haiti	1 (2.9)	1 (1.8)

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Argentina	1 (2.9)	3 (5.5)
U.S.	4 (11.4)	4 (7.3)
Brazil	2 (5.7)	3 (5.5)
Panama	1 (2.9)	2 (3.6)
Nigeria	2 (5.7)	3 (5.5)
South Africa	2 (5.7)	2 (3.6)
Morocco	1 (2.9)	1 (1.8)
Italy	2 (5.7)	2 (3.6)
France	1 (2.9)	1 (1.8)

The only large-scale study about the prevalence rate came from West Cape Province, South Africa[54]. The author used the data from Western Cape Provincial Health Data Centre (WCPHDC), which included 3,460,932 individuals aged 20 or above, about half of the population in West Cape Province. Until June 1, 2020, when testing criteria changed with public sector tests being limited to patients >55 years of age or with comorbidity, 1,489 coinfected patients (including previous TB and current TB) were recorded, and the prevalence rate of coinfection was about 0.04%. Till June 9, 2020, 2128 coinfected patients (including previous TB and current TB) were recorded, and the prevalence rate was about 0.06%.

3.2. Treatment and Outcome

Till the end of search, no cohort study, cross-sectional study, or experimental study was identified to provide detailed information on TB and COVID-19 coinfection treatment. presented the results of the included 35 studies which reported 55 coinfected patients in total. As for the basic information, 12 patients were female, and 47 patients were

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aged 20 to 60 years. Up to 50 cases were reported as active TB patients, 12 of which got TB before COVID-19, and we could not identify infection orders about the rest 38 cases. Besides, 4 cases were reported as post TB[24], and 1 was reported as latent TB[27]. Although most cases were diagnosed with pulmonary TB, there were still some more complex conditions, like central nervous system involvement[21 38], tuberculous meningitis[28], tuberculous pleuritis[26 27 31], and peritoneal TB[41]. With regard to drug resistance, 37 patients' information was missing, 15, 2 and 1 of the remaining 18 cases were reported as negative[22 23 29 30 37 39 42-44 46 48 51], positive[27 36], and likely positive[18]. All the cases were symptomatic COVID-19 patients. Check Appendix Table 3 for detailed information.

Only 5 patients received treatment before hospitalization. Among them, 3 received Anti-TB Treatment (ATT)[22 34 44], 1 received antibiotic (amoxicillin)[38], and 1 took hemodialysis due to chronic kidney failure[35]. During the hospitalization, 7 patients were transferred to the Intensive Care Unit (ICU)[18 30 34 35 39 44 52]. Up to 50 cases' detailed treatment information was reported. Various kinds of oxygen support were the most common supportive treatment (21 cases), and 1 patient in China received extracorporeal membrane oxygenation[27]. Except for several cases that did not provide any detailed ATT regime (12 cases) or patients did not receive ATT (11 cases), 1 or several kinds of first-line Anti-TB drugs, including rifampicin, isoniazid, ethambutol, and pyrazinamide, were utilized in most remaining cases (27 cases). For the drug resistance TB patients, the doctors would adjust the regime for them[27 36]. Other kinds of antibiotics

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were also utilized, including azithromycin (21 cases), ceftriaxone (13 cases), moxifloxacin (7 cases), linezolid (4 cases), etc. However, we could not identify the purpose of these extra antibiotics due to the lack of information. We guessed it could be 1 or more of the following situations: 1) supplement to ATT regime, 2) prevention against possible bacterial infections, and 3) treatment for the existing bacterial infection. Among these 50 cases, 13 cases did not mention any kind of antiviral drug and 2 cases did not provide antiviral drugs' names. Within the remaining cases, hydroxychloroquine (HCQ) was the most utilized antiviral drug (21 cases), as well as lopinavir (8 cases), ritonavir (8 cases) and arbidol (7 cases). We also found that all the 20 Chinese coinfected patients did not take HCQ completely but took arbidol most. Besides, drugs like steroids (methylprednisolone, dexamethasone, prednisone, etc.), anti-coagulant drugs, vitamins (vitamin C, vitamin B6, etc.) were also prescribed for some patients. We identified information concerning treatment after discharge for 10 patients; 8 of them received ATT, and 2 received treatment for existing comorbidity. Although there was no doubt that the regimes for TB and COVID-19 coinfected patients were highly heterogeneous, we still could identify 4 main paradigms about treatment during hospitalization: 1) simultaneous antiviral treatment and ATT[19-21 23-26 28 30 31 34 35 39-41 43-45 49 51], 2) only ATT[22 26 32 33 35 36 46-48], 3) only antiviral treatment[18 24 26 27 38 44 50], 4) no antiviral treatment and ATT[37]. Check Appendix Table 4 and Appendix Table 5 for detailed information about the treatment.

A minimal number of studies discussed about pros and cons of the treatment and the

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possible effects of the regimes. Faqihi held that the first-line Anti-TB drug with antiviral treatment for COVID-19 might theoretically increase the risk of side-effects[30], and Yousaf stressed the importance of adjustment to the management of coinfected patients[31]. The antagonism from lopinavir/ritonavir against rifampicin had been noticed[20 25], and Cao suggested umifenovir could be the substitute of lopinavir/ritonavir[25]. Kumar warned additive hepatotoxicity from remdisivir due to simultaneous use of anti-tubercular drugs[20]. Faqihi gave isoniazid along with pyridoxal phosphate to avoid peripheral neuropathy[30], and Farias did not start antiretroviral therapy at first for the 2 HIV, TB and COVID-19 coinfected patients to avoid complications with ATT[43]. Confronted with liver function impairment during the treatment, Musso replaced isoniazid and pyrazinamide with amikacin and moxifloxacin to modify the ATT regimen[51]. Liu, He, Cao, and Singh discussed the impact of corticosteroids for treating lung inflammation[19 24 25 27]. Liu's recommendation was not to administer corticosteroids for COVID-19 and TB coinfected patients. Besides, since COVID-19 and TB disease might both cause immunosuppression, Liu recommended using immunomodulatory therapies such as thymosin in patients with progressively lower lymphocyte counts[27].

About the outcome of 55 reported cases, 35 were resolved, 6 were deceased, 3 were still on treatment in hospital, 1 returned to hospitalization due to positive recurrence of SARS-CoV-2 RNA, and 10 were missing. For patients discharged from the hospital, the length of stay ranged from 7 days to 2 months. Besides, in Vanzetti's study, 3

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patients were described they had a longer hospital stay than average[37]. For deceased patients, the length of stay ranged from 6 to 26 days. Unfortunately, we could only find minimal information about coinfected patients' physical condition when they got discharged. Most studies only indicated that the test results for SARS-CoV-2 were negative or patients got recovered. Although Agada described the patient with poor prognostic indices, detailed physical conditions were also missing[46]. He believed the treated or untreated mycobacterium TB was a risk factor of a worse prognosis of COVID-19 patients[24]. In addition, long-term outcome or follow-up outcome for the TB and COVID-19 coinfected patient was absent.

3.3. Fatality Rate of TB and COVID-19 Coinfection

We identified 7 retrospective cohort studies that revealed the fatality rate of TB and COVID-19 coinfection (Appendix Table 6 for detailed information). These studies came from India[55], Philippines[56], Italy[57], South Africa[54], Argentina[58], Russia[59] and 1 was based on data from multiple countries[53]. Motta's study provided 2 cohorts[53], 1 of which was same as Stochino's study[57]. Thusly, we removed the overlapped cohort. Except for Gubkina's study whose targeted population was children and no one deceased[59], the overall and in-hospital fatality rate fell within the range of 5.3% ~ 23.6% and 5.0% ~ 27.3%. Table 2 presents the results of the included 7 studies.

Table 2 Fatality rate of included studies (n=7)

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First author (year)	Country	Time span	Sample size	In-hospital	age	gender	Comorbidity	Died	Fatality rate (overall or in-hospital)
Gupta 2020	India	1 February 2020 to 14 June 2020	22	yes	19-67	20 male (90.9%) 2 female (9.1%)	Hypertension 4 (18.2%) Diabetes 3 (13.6%) Seizure disorder 2 (9.1%) Hypothyroidism 1 (4.5%)	6	27.3 (in-hospital)
Motta 2020	Belgium, Brazil, France, Italy, Russia, Singapore, Spain, and Switzerland	12 March, 2020 to 5 May, 2020	49	no	27-70	/	/	7	14.3% (overall)
Sy 2020	Philippines	May 17, 2020 to June 15, 2020	106	no	/	73 male (68.9%) 33 female (31.1%)	Hypertension 22 (20.8%) Diabetes 14 (13.2%) Cancer 1 (0.9%) Renal cancer 4 (3.8%) Cardiac disease 8 (7.5%) Asthma 4 (3.8%) COPD 3 (2.8%)	25	23.6% (overall)
			66	yes	/		Hypertension 16 (24.2%) Diabetes 10 (15.2%) Cancer 1 (1.5%) Renal cancer 4 (6.1%) Cardiac disease 7 (10.6%) Asthma 1 (1.5%) COPD 1 (1.5%)	18	27.3% (in hospital)

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Stochino 2020	Italy	/	20	yes	27-47	12 male (60%) 8 female (40%)	/	1	5% (in-hospital)
Davies 2020	South Africa	Till to 9 June, 2020	2128	no	20 or above	/	/	113	5.3% (overall)
			469	yes	20 or above	/	/	102	21.7% (in-hospital)
Domingo 2020	Argentina	1 March to 30 June, 2020	23	yes	5-82	18 male (78.3%) 5 female (21.7%)	Smoking and other addictions 16 (69.6%) HIV 4 (17.4%) Psychosis 1 (4.3%) Pulmonary thromboembolism 1 (4.3%) Arterial hypertension 1 (4.3%)	2	8.7% (in-hospital)
Gubkina 2020	Russia	March to June, 2020	24	yes	/	/	/	0	0% (in-hospital)

/: No information provided

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3.4. Quality of Included Studies

We utilized JBI Critical Appraisal Checklist for Study Reporting Prevalence Data as the tool to assess 7 included studies that revealed prevalence rate or fatality rate. The grade (yes, no, unclear or N/A) was shown in Appendix Figure 1 and Appendix Table 7. The results suggested that insufficient sample size and inappropriate sample frame were the main risks of bias of these studies.

We utilized JBI Critical Appraisal Checklist for Case Report[16] as the tool to assess 35 included case reports, as shown in Appendix Figure 2 and Appendix Table 8. All the studies described patients' demographic characteristics clearly, and we could identify basic information about patients' history (28 studies), clinical condition (31 studies), diagnostic tests or assessment methods and the results (34 studies) for most studies. Whereas only 23, 20, and 22 studies provided clear messages about treatment procedures, post-intervention clinical condition, and takeaway lessons, respectively. The most poorly reported part was adverse or unanticipated events, and only 11 studies gave a clear description, which could be a high risk of bias.

3.5. Meta-Analysis on Fatality Rate

We included 6 qualified studies (8 cohorts) to estimate the fatality rate of TB and COVID-19 coinfection, 3 studies for overall fatality rate[53 54 56] and 5 studies for in-hospital fatality rate[54-58]. Gubkina's study was excluded because its targeted population

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was children, which might lead high heterogeneity to the MA. Although Davies's study sample was people aged 20 or above, we still included it into the MA.

As shown in Figure 2A, the estimated overall fatality was 13.9% (95% CI: 1.6% ~ 26.2%). Significant heterogeneity was observed between the studies (I^2=91.07%, p-value for Q statistics <0.001). The estimated in-hospital fatality rate was 17.5% (95% CI: 8.9% to 26.0%), and heterogeneity was detected among these studies too (I^2=75.47%, p-value for Q statistics <0.01) (Figure 2B). We utilized funnel plots to explore the existence of publication bias, and the results were both positive (Figure 3A, 3B).

To further explore the source of heterogeneity, we grouped these studies into the high-income country and LMICs subgroups. However, due to the lack of enough data, we could only conduct subgroup MA on in-hospital fatality rate. The mean in-hospital fatality rate for high-income country (Italy and Argentina) and LMICs (India, Philippines, South Africa) were 6.5% (95% CI: -0.8% ~ 13.9%) and 22.5% (95% CI: 19.0% ~ 26.0%). No heterogeneity was observed between studies for both subgroups (Figure 2C, 2D). Similarly, 2 funnel plots were created to explore the existence of publication bias, and the results were both negative (Figure 3C, 3D).

<Insert Figure 2 and Figure 3 here>

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4. Discussion

This systematic review identified the evidence on the prevalence status, treatment and possible outcome of TB and COVID-19 coinfection. Although studies about coinfection were limited and quantitative data was even rarer, we still identified 42 studies, of which 35 were case reports and 7 were retrospective cohort studies.

Regarding the quality of included studies, we found that the sample size were insufficient, and the smapling frames were not specified, which had been the leading risk of bias for prevalence studies. In addition, the lack of reporting of adverse and unanticipated events, intervention and treatment procedure, and post-intervention condition were noticible defects of the case reports. Publication bias was found in overall and in-hospital fatality rate of the TB and COVID-19 coinfection, which suggested the current results standed on selected studies. Till to the searching date, the number of relevant studies were limited, cohort studies even more rare. Besides, we did not find any experimental study about TB-COVID coinfection. Considering the quality of current evidences, we think more studies are urgently needed to evaluate the prevalence, treatment, and long-term outcome of TB-COVID coinfection.

In terms of global prevalence status, 19 countries reported coinfected patients, including low TB prevalence countries like Belgium and Switzerland, as well as high TB prevalence countries like China and India. Therefore, we assumed the coinfection

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was common worldwide, which every country and region should beware of. The only large-scale study about the prevalence rate came from West Cape Province, South Africa. The rate was 0.04% on June 1, 2020 and 0.06% on June 9, 2020, increasing by 50% within only 8 days, not to mention the public sector tests were limited to patients >55 years of age or with comorbidity. From the perspective of hindsight, though the rate was low, considering it was at the beginning of the pandemic, the situation was still worrying. However, we cannot infer the global prevalence rate based on this very limited information, and we believe more studies are urgently needed, especially for countries with high COVID-19 cases and high TB prevalence.

Although case report is not a very sound study design for us to evaluate treatment and outcome information, we didn't find any other kinds of study design about this topic. The analysis of case reports showed the treatment for coinfection was of great heterogeneity, while we still identified 4 main paradigms: 1) simultaneous antiviral treatment and ATT, 2) only ATT, 3) only antiviral treatment, 4) no antiviral treatment and ATT. Some studies implied the doctor just followed the local clinical guidelines; however, very few studies mentioned regimen adjust due to characteristics of TB and COVID-19 coinfected patients. Therefore, we assumed the current treatment plans were more like a mix of the suggested TB regime and the COVID-19 regime. Besides, we did not detect any specific treatment protocol for patients with inactive TB, namely the previous TB or latent TB patients. Furthermore, HCQ, the most commonly used antiviral drug in these cases, was not taken by any patient from China's study.

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Traditional Chinese Medicine (TCM), which was widely used for COVID-19 patients in China[60], was also seldomly mentioned in these China's reports. The reason might be that these cases were reported at the beginning of the pandemic, and the doctors lacked the experience to deal with the situation. We tried multiple ways to explore the relationship between treatments and outcomes, but we did not find meaningful results. Therefore, we could not find evidence on which one is the best clinical practice due to a lack of sound investigation or experimental study, which suggested the urgency of further research.

The prolonged hospital stay for coinfected posed a significant challenge to the health care system, especially for those low-resources countries. Besides, the vaguely described health status at discharge and absence of long-term outcomes are rather alarming. Moreover, for discharged patients who were still on ATT, the adverse impact from pandemic to delivery of TB care services which had been confirmed in multiple countries[61-63], might impede their accessibility to services.

We tried to conduct MA on overall and in-hospital fatality rates, and the estimated results were 13.9% and 17.5%. According to Meyerowitz-Katz, the fatality rate of patients with single COVID-19 was 0.68% (95%CI: 0.053% ~ 0.82%)[64], which was relatively smaller. Therefore, we believe the coinfected patients were facing more challenges and calling for the need to refine medical equipment and interventions. In

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addition, significant heterogeneity was detected in both MAs on overall and in-hospital fatality rates. In contrast, no heterogeneity was detected in both sub-groups' MAs on in-hospital fatality rate. We did not detect any pattern about treatment in both high-income countries and LMICs, based on case reports. Considering patients in high-income countries are likely to receive care of better quality than patients in LMICs, we guess the different results of MAs' heterogeneity might hint that the quality of care is highly associated with outcomes. In other words, the patients in high-income countries shared a better condition than those in LMICs and were more likely to recover from TB and COVID-19 coinfection. It's necessary to point out that the results of MA were based on limited number of studies, of which defects in the quality was a frequent feature.

This systematic review has a few limitations. We did not include comorbidity as a keyword and MeSH term in the searching process, which might keep us from some relevant studies. All preprint databases and gray literature databases were not included in searching process. Besides, the diagnosis and testing criteria of COVID-19 and TB differs from countries and regions, and we were not able to transfer the data under a universal standard. Therefore, there might be some false-positive patients. Only 6 studies were included in MA and significant heterogeneity was detected among them. Therefore, the results should be interpreted with caution.

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In conclusion, the results of our systematic review show coinfection was not rare worldwide, and outcomes based on current evidences shared significant differences between high-income countries and LMICs. Current evidence is insufficient to evaluate the treatment and outcome, not to estimate the prevalence rate of TB-COVID coinfection globally. More study about this field is in urgent need, especially experimental study and cohort study. Given that some countries have higher TB and COVID-19 prevalence simultaneously, global efforts are needed to work collaboratively on TB-COVID coinfection, especially in response to those with limited medical resources.

Ethical Statement N/A

Data availability statement Data are from the published study. Extracted data are available on request from the corresponding author.

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designed this study; Jie Zhao was responsible for the literature search; Quan Wang and Shasha Guo conducted the study selection (review of titles and abstracts, review of full texts, and final decision), data abstraction, and quality assessment independently; Ning Xu, as a medical doctor, provided necessary information about treatment and drugs; Quan Wang finished the manuscript; Hui Li, Qiang Sun, Xiaolin Wei and Quanfang Dong refined the manuscript; Qiang Sun was responsible for this study. The corresponding author, Qiang Sun, attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

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Dissemination to participants and related patient and public communities It is not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

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Figure legends

Figure 1 A flow diagram outlines the study selection process.

Figure 2 Forest plots of fatality rate accorting to hospitalization, subgroups according to economic status of regions. A: overall fatality rate; B: in-hospital fatality rate; C: in-hospital fatality rate (high-income country subgroup); D: in-hospital fatality rate (low- and middle-income country subgroup)

Figure 3 Funnel plots of fatality rate accorting to hospitalization, subgroups according to economic status of regions. A: overall fatality rate; B: in-hospital fatality rate; C: in-hospital fatality rate (high-income country subgroup); D: in-hospital fatality rate (low- and middle-income country subgroup)

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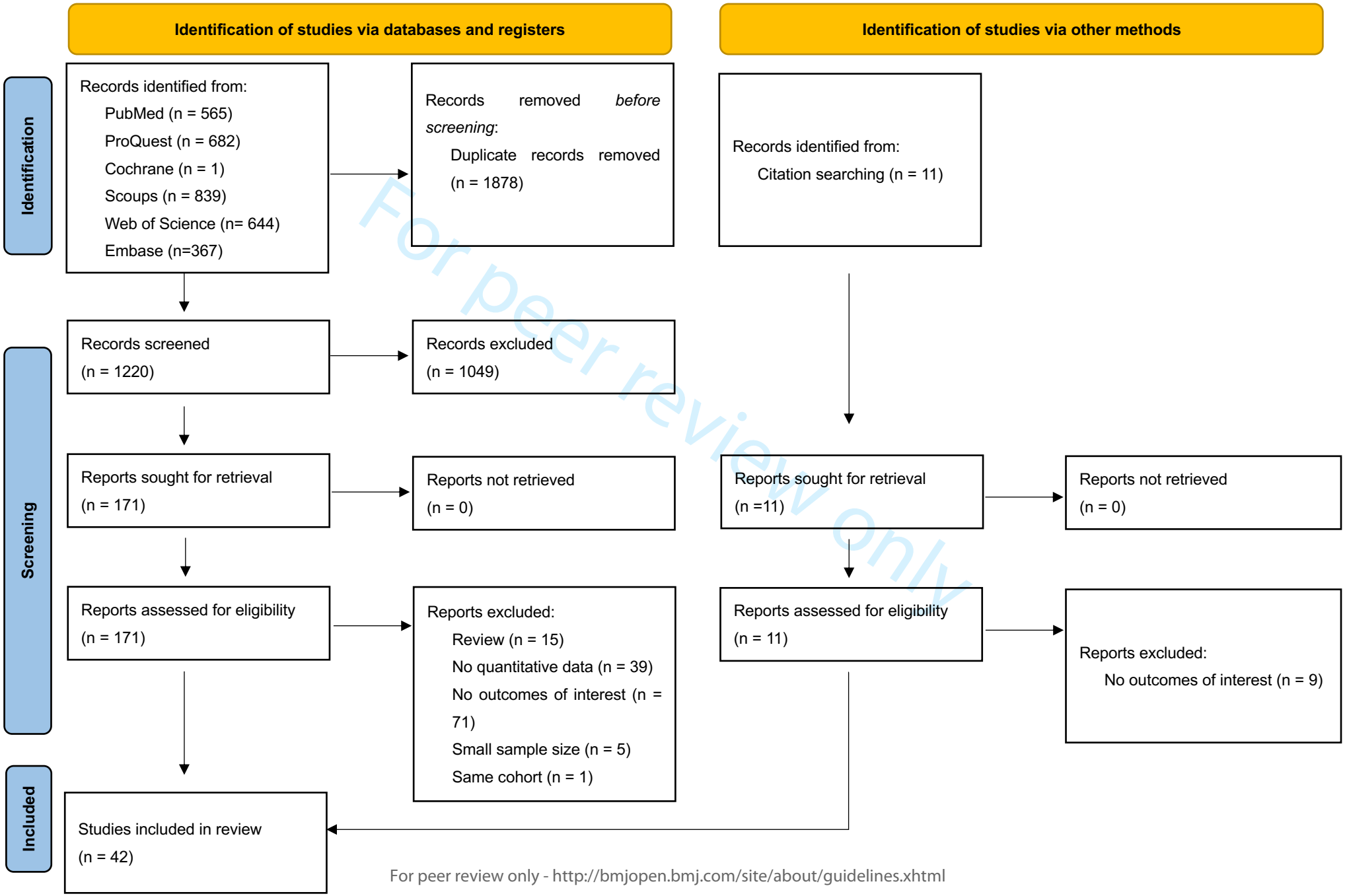
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Figure 1 Flow chart of literature search



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Figure 2 Forest plot of included studies

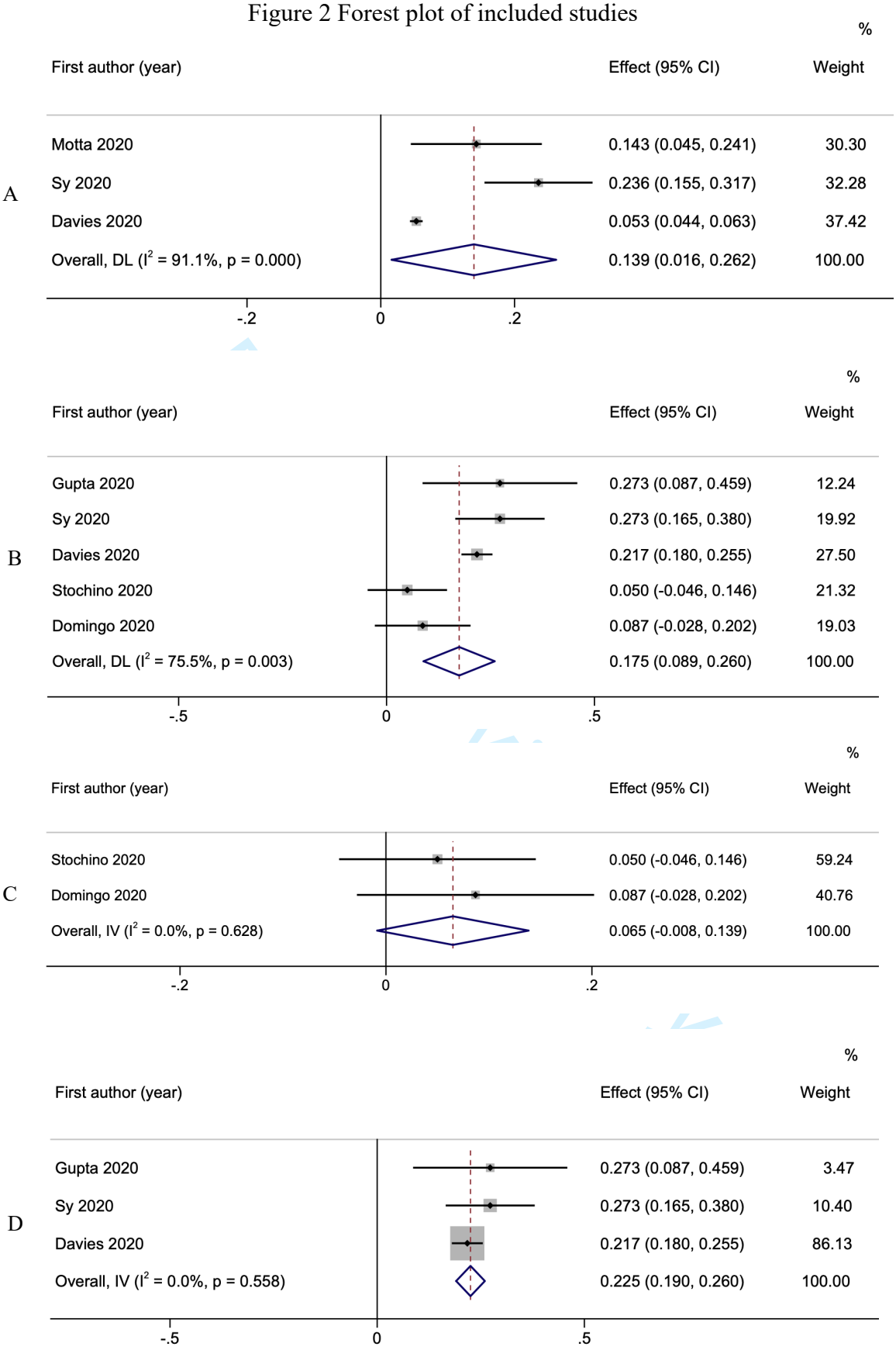
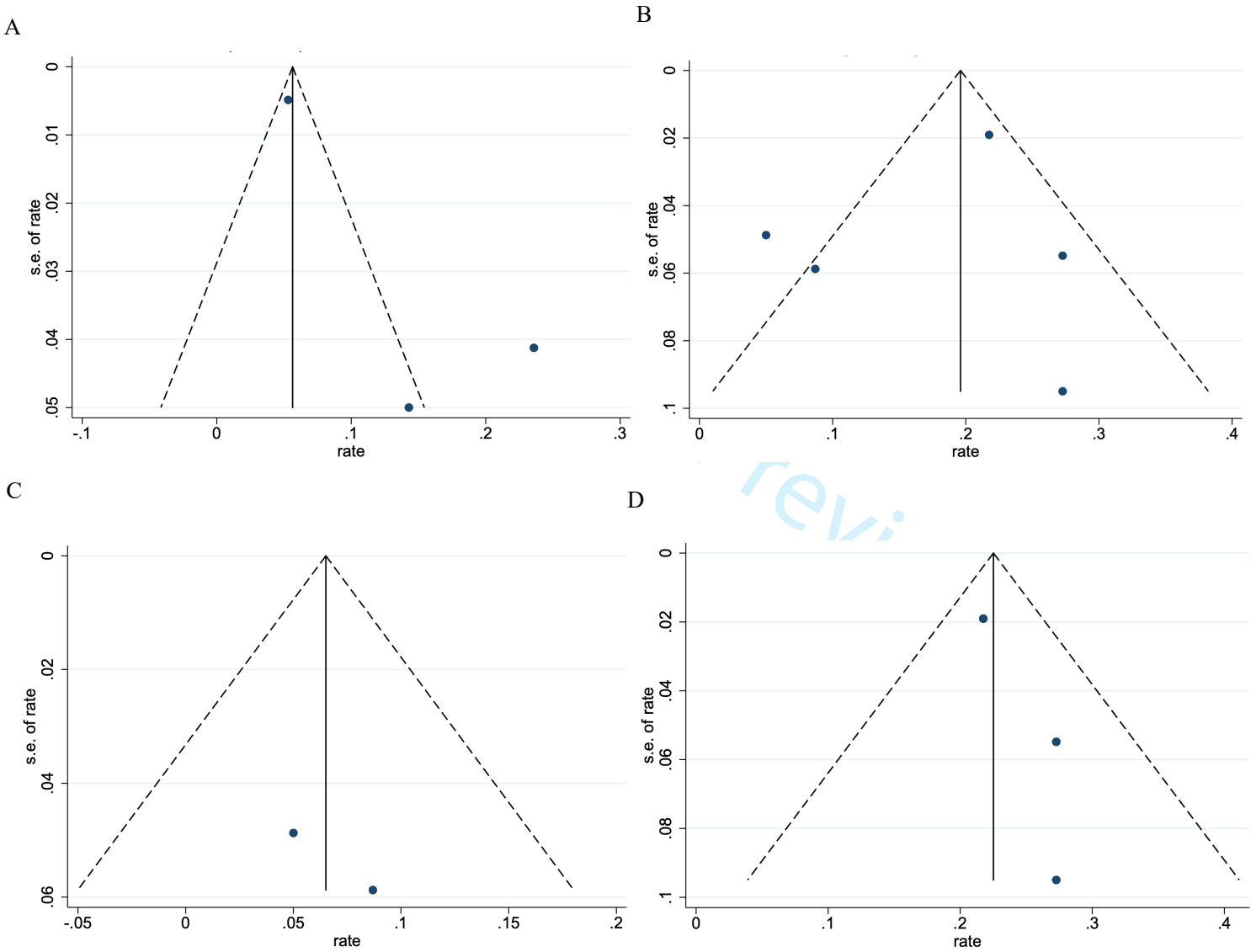


Figure 3 Funnel plot of included studies



Appendix

Table 1 The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P1-P2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P4-P5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P6-P7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	P7, appendix 2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P8-P9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the	P8-P9

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Section and Topic	Item #	Checklist item	Location where item is reported
		process.	
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P8-P9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P8-P9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P9
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	P8-P9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P9-P10
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P9-P10
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P9-P10
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P9-P10
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A

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Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P9-P10
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P10-P11
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	P11
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	P19 appendix 4
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	P10-P18, appendix 3
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P10-P20
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	P19-P20
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P19
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	P19
Certainty of	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P15-P20

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Section and Topic	Item #	Checklist item	Location where item is reported
evidence			
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P21
	23b	Discuss any limitations of the evidence included in the review.	P24
	23c	Discuss any limitations of the review processes used.	N/A
	23d	Discuss implications of the results for practice, policy, and future research.	P22-P25
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P6-P7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	P6-P7
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P25
Competing interests	26	Declare any competing interests of review authors.	P25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	P25

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Table 2 Search strategies to identify studies reporting the prevalence status, treatment and outcomes of tuberculosis and COVID-19

Database searched: PubMed

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	("covid-19"[MeSH Terms] OR "sars-cov-2"[MeSH Terms]) AND "tuberculosis"[MeSH Terms]	244
2	((covid-19[Title/Abstract]) OR (Sars-cov-2[Title/Abstract])) AND((TB[Title/Abstract]) OR (tuberculos*[Title/Abstract]))	523
3	#1 OR #2	565

Database searched: Web of Science

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	((TS=(covid-19) OR TS=(sars-cov-2) OR AB=(covid-19) OR AB=(sars-cov-2)) OR (TS=(covid-19)) OR (TS=(sars-cov-2)) OR (AB=(covid-19))) AND (TS=(tuberculosis)) OR (TS=(Mycobacterium tuberculosis)) OR (TS=(Mycobacterium tuberculosis)) OR (AB=(tuberculo*)) OR (AB=(TB)) OR (AB=(Mycobacterium tuberculosis)) OR (AB=(M.tuberculosis)) OR (AB=(tubercle bacillus)))	644

Database searched: ProQuest

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	su(COVID-19 OR SARS-COV-2) AND su(TB OR tuberculo* OR (Mycobacterium tuberculosis) OR (M.tuberculosis))	682

Database searched: Scopus

Date: Mars 19th, 2021

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Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	(TITLE-ABS-KEY (covid-19 OR sars-cov-2) AND TITLE-ABS-KEY (tb OR tuberculos* OR (mycobacterium AND tuberculosis) OR (m.tuberculosis)))	839

Database searched: Cochrane database

Date: Mars 19, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	MeSH descriptor: [COVID-19] explode all trees	257
2	MeSH descriptor: [SARS-COV-2] explode all trees	204
3	MeSH descriptor: [tuberculosis] explode all trees	2,029
4	(#1 OR #2) AND 3	1

Database searched: Embase

Date: Mars 19, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	('covid 19':ab,ti OR 'sars cov 2':ab,ti) AND [01-11-2019]/sd NOT [20-3-2021]/sd	81,923
2	(tuberculosis:ab,ti OR 'mycobacterium tuberculosis':ab,ti OR tuberculo*:ab,ti OR 'tb':ab,ti OR 'm.tuberculosis':ab,ti OR 'tubercle bacillus':ab,ti) AND [01-11-2019]/sd NOT [20-3-2021]/sd	19,510
3	#1 AND #2	367

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Table 3 Characteristics of reported TB and COVID-19 coinfection patients (n=55)

Basic information	Number (%)
Gender	
female	12 (21.8)
male	43 (78.2)
Age	
< 20	3 (5.5)
20-39	23 (41.8)
40-59	22 (40.0)
≥60	7 (12.7)
Comorbidity	
diabetes mellitus	8 (14.5)
hypertension	5 (9.1)
HIV	4 (7.3)
Type of TB	
active TB	50 (90.9)
post TB	4 (7.3)
latent TB	1 (1.8)
TB site	
pulmonary TB	47 (85.5)
central nervous system involvement	2 (3.6)
tuberculous meningitis	1 (1.8)
tuberculous pleuritis	1 (1.8)
peritoneal TB	1 (1.8)
not reported	6 (10.9)

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BCG Vaccination	
yes	2 (3.6)
not reported	53 (96.4)
Drug resistance	
no	15 (27.3)
yes	2 (3.6)
likely	1 (1.8)
not reported	37 (67.3)
Infection order	
TB before COVID-19	17
COVID-19 before TB	0
not reported	38
Symptomatic COVID-19	
yes	55 (100.0)
no	0 (0.0)

Table 4 Detailed basic information of included case reports (n=35)

Author (Year)	Country	Cases number	Age	Gender	Comorbidity	BCG Vaccination	TB site	Drug resistance	Active TB	TB before COVID-19	Symptomatic COVID-19

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Sarma 2020[1]	India	1	53	female	chronic kidney disease (CKD) stage 4, type 2 diabetes mellitus	/	pulmonary TB	likely	yes	yes	yes
Singh 2020[2]	India	1	76	female	/	/	pulmonary TB	/	yes	/	yes
Kumar 2020[3]	India	1	38	male	/	/	pulmonary TB	/	yes	/	yes
Ata 2020[4]	India	1	28	male	no HIV	/	pulmonary TB central nervous system involvement	/	yes	/	yes
Yadav 2020[5]	India	1	43	male	non-diabetes	/	pulmonary TB	no	yes	/	yes
Baskara 2021[6]	Indonesian	1	42	male	diabetes mellitus	/	pulmonary TB	no	yes	yes	yes
Wang 2020[7]	China	1	45	male	/	/	tuberculous meningitis	/	yes	/	yes
He 2020[8]	China	3	67	male	hypertension	/	/	/	no	yes	yes
			26	male	/	/	/	/	no	yes	yes
			76	male	exfoliative dermatitis, Arthrolithiasis	/	/	/	no	yes	yes
Cao 2020[9]	China	1	47	female	/	/	pulmonary TB	/	yes	yes	yes

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			50	male	anemia, hypoproteinemia	/	pulmonary TB	/	yes	yes	yes
Yao 2020[10]	China	3	44	male	diabetes, pleural effusion	/	pulmonary TB, tuberculous pleuritis	/	yes	/	yes
			57	male	diabetes	/	pulmonary TB	/	yes	yes	yes
			48	male	/	/	/	/	no	/	yes
			26	male	/	/	pulmonary TB	yes	yes	yes	yes
Liu 2020[11]	China	3	46	male	/	/	pulmonary TB, tuberculous pleuritis	/	no	yes	yes
Khayat 2021[12]	Saudi Arabia	1	40	female	no HIV and hepatitis	/	pulmonary TB	no	yes	/	yes
Faqihi 2020[13]	Saudi Arabia	1	60	male	hypertensive and diabetic	/	pulmonary TB	no	yes	/	yes
			34	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
Yousaf 2020[14]	Qatar	6	32	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			50	male	no HIV, hepatitis B, and hepatitis C	/	pulmonary TB	/	yes	/	yes

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					infection; have diabetes						
			35	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			27	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			35	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB, tuberculous pleuritis	/	yes	/	yes
AlKhateeb 2020[15]	Qatar	1	28	male	/	/	pulmonary TB	/	yes	/	yes
Tham 2020[16]	Singapore	4	32	male	/	/	pulmonary TB	/	yes	/	yes
			33	male	/	/	pulmonary TB	/	yes	/	yes
			22	male	/	/	pulmonary TB	/	yes	/	yes
			40	male	/	/	pulmonary TB	/	yes	/	yes
Cinar 2020[17]	Turkey	1	55	male	/	/	/	/	yes	/	yes
Sarinoglu 2020[18]	Turkey	2	77	female	diabetes mellitus, hypertension, chronic obstructive lung disease, chronic kidney failure	/	pulmonary TB	/	yes	/	yes
			39	female	/	/	pulmonary TB	/	yes	/	yes
Lopinto	France	1	58	male	pulmonary	/	pulmonary TB	/	yes	yes	yes

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2020[19]					embolism						
Musso 2021[20]	Italy	1	45	male	No HIV	/	pulmonary TB	no	yes	yes	yes
Luciani 2020[21]	Italy	1	32	female	No HIV	yes	pulmonary TB	/	yes	/	yes
Bouaré 2020[22]	Morocco	1	32	female	/	/	pulmonary TB	/	yes	/	yes
Essajee 2020[23]	South Africa	1	2	female	/	yes	pulmonary TB	no	yes	/	yes
Pillay 2021[24]	South Africa	1	44	female	HIV, pneumocystis iirovecii pneumonia, community-acquired pneumonia, pulmonary embolism	/	pulmonary TB	/	yes	/	yes
Agada 2021[25]	Nigeria	1	50	male	No HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	no	yes	/	yes
Gbenga 2020[26]	Nigeria	2	33	male	/	/	pulmonary TB	/	yes	/	yes
			30	male	/	/	pulmonary TB	/	yes	/	yes
Rivas 2020[27]	Panama	2	29	male	pleural empyema	/	pulmonary TB	no	yes	/	yes
			53	male	HIV	/	pulmonary TB	/	yes	yes	yes
Farias 2020[28]	Brazil	2	39	male	HIV, hepatitis B	/	pulmonary TB	no	yes	yes	yes
			43	male	HIV, anxiety	/	pulmonary TB	no	yes	/	yes

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Pinheiro 2020[29]	Brazil	1	68	male	disorder diabetes, hypertension, chronic liver disease secondary to schistosomiasis	/	pulmonary TB	no	yes	/	yes
Gerstein 2021[30]	USA	1	49	male	cirrhosis, portal hypertension	/	pulmonary TB, peritoneal TB	/	yes	/	yes
Cutler 2020[31]	USA	1	61	male	Parkinson's disease	/	pulmonary TB	/	yes	/	yes
Sasson 2020[32]	USA	1	44	male	no HIV; hypertension, type II diabetes mellitus, a previous stroke secondary to a left atrial thrombus	/	/	no	yes	/	yes
Freij 2020[33]	USA	1	5	female	no HIV	/	central nervous system involvement	/	yes	/	yes
Vanzetti 2020[34]	Argentina	3	33	female	/	/	pulmonary TB	no	yes	/	yes
			34	male	/	/	pulmonary TB	no	yes	yes	yes
			17	male	/	/	pulmonary TB	no	yes	/	yes
Vilbrun 2020[35]	Haiti	1	26	male	no HIV	/	pulmonary TB	yes	yes	yes	yes

/: No information provided

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Table 5 Detailed treatment information of included cases (n=55)

Author (Year)	Cases number	ICU	Treatment before hospitalization	Treatment in hospital	Treatment after discharge	Post-intervention clinical condition	Outcome	Days to die or discharge
Sarma 2020[1]	1	yes	/	**Supportive treatment:** oxygen support (intubated) **Antibiotic:** ampicillin, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** vitamin C, midazolam, fentanyl	/	/	resolved	/
Singh 2020[2]	1	no	/	**Supportive treatment:** / **Antibiotic:** ceftriaxone, azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** N-acetyl cysteine, methylprednisolone, vitamin C	/	/	on treatment	/

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Kumar 2020[3]	1	no	/	**Supportive treatment:** oxygen support, mechanical ventilation **Antibiotic:** ATT, other antibiotic (no detail) **Antiviral drug:** hydroxychloroquine sulphate **Other treatment:** /	/	/	died	/
Ata 2020[4]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** steroids	rifampicin, ethambutol, and pyrazinamide	/	resolved	28

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Yadav 2020[5]	1	no	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** / **Other treatment:** /	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, ethambutol, and pyrazinamide) **Antiviral drug:** / **Other treatment:** /	/	/	/	/
Baskara 2021[6]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), azithromycin, ceftriaxone, ceftazidime **Antiviral drug:** hydroxychloroquine, oseltamivir **Other treatment:** anti-coagulant drugs	treatment for diabetes mellitus	/	resolved	13

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Wang 2020[7]	1	no	/	**Supportive treatment:** / **Antibiotic:** rifampicin, amikacin, moxifloxacin **Antiviral drug:** arbidol **Other treatment:** tolvaptan, aspirin, mannitol, potassium supplement, sodium supplement	/	/	died	14
He 2020[8]	3	no	/	**Supportive treatment:** noninvasive ventilation support **Antibiotic:** antibiotic (no details and not ATT) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** methylprednisolone, probiotics, traditional Chinese medicine, immunoglobulin	/	/	resolved	37

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no	/	**Supportive treatment:** / **Antibiotic:** antibiotic (no details and not ATT) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** probiotics, traditional Chinese medicine	/	skin rashes with pruritus	ninth day after discharge, returned to the hospital because of positive recurrence of SARS-CoV-2 RNA	28
no	/	**Supportive treatment:** nasal cannula oxygen support **Antibiotic:** ATT, antibiotic (no details) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** methylprednisolone, glucocorticoid, probiotics, traditional Chinese medicine, immunoglobulin	/	/	resolved	24

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Cao 2020[9]	1	no	/	**Supportive treatment:** noninvasive ventilation support **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), moxifloxacin hydrochloride **Antiviral drug:** lopinavir and ritonavir, umifenovir **Other treatment:** polyene phosphati-dylcholine, reduced glutathione	/	Hepatic injury during the treatment	resolved	More than 14 days
Yao 2020[10]	3	no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** ATT **Antiviral drug:** / **Other treatment:** corticosteroid	/	/	died	22

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		no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** antiviral drugs (no details) **Other treatment:** acarbose, metformin	ATT	/	resolved	26
		no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** antibiotics (no details) **Antiviral drug:** antiviral drugs (no details) **Other treatment:** corticosteroids, immunoglobulin	/	/	resolved	27
Liu 2020[11]	3	no	/	**Supportive treatment:** extracorporeal membrane oxygenation, haemoperfusion **Antibiotic:** moxifloxacin, linezolid, no ATT **Antiviral drug:** arbidol **Other treatment:**	/	/	resolved	14

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		caspofungin, thymopentin					
no	/	**Supportive treatment:** nasal high-flow oxygen therapy **Antibiotic:** moxifloxacin, linezolid, cycloserine, clofazimine, pyrazinamide **Antiviral drug:** arbidol **Other treatment:** /	ATT	/	resolved	14
no	/	**Supportive treatment:** nasal high-flow oxygen therapy **Antibiotic:** moxifloxacin **Antiviral drug:** arbidol **Other treatment:** /	/	/	resolved	9

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Khayat 2021[12]	1	no	/	/	/	/	/	/
Faqihi 2020[13]	1	yes	/	**Supportive treatment:** respiratory support **Antibiotic:** moxifloxacin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir, ritonavir, ribavirin **Other treatment:** dexamethasone, prophylactic anticoagulation	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), isoniazid and rifampicin (later)	/	resolved	27
Yousaf 2020[14]	6	no	/	**Supportive treatment:** / **Antibiotic:** ceftriaxone, azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/

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AlKhateeb 2020[15]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details) **Antiviral drug:** / **Other treatment:** /	/	/	resolved	1 month
Tham 2020[16]	4	no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details) **Antiviral drug:** / **Other treatment:** /	/	/	/	/
		no	/		/	/	/	/
		no	/		/	/	/	/
		no	/		/	/	/	/
Cinar 2020[17]	1	yes	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), meropenem **Antiviral drug:** favipiravir **Other treatment:**	/	myelodysplastic syndrome (MDS/RAEB1 FAB subtype) during the treatment	/	/

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				convalescent plasma product, tocilizumab					
Sarinoglu 2020[18]	2	yes	hemodialysis	**Supportive treatment:** / **Antibiotic:** ATT (no details), piperacillin/tazobactam **Antiviral drug:** favipiravir, hydroxychloroquine **Other treatment:** /	hemodialysis	/	/	/	
		no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details), azithromycin, plaquanil, ciprofloxacine, tazocilli **Antiviral drug:** / **Other treatment:**	/	/	/	/	

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				/					
Lopinto 2020[19]	1	yes	/	/	/	/	resolved	14
Musso 2021[20]	1	no	/	**Supportive treatment:** chest tube drainage **Antibiotic:** amikacin, moxifloxacin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hidroxicloroquine **Other treatment:** corticosteroids	/	/	died	14

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Luciani 2020[21]	1	no	/	**Supportive treatment:** / **Antibiotic:** amoxicillin/clavulanic acid, clarithromycin, piperacillin/tazobactam, linezolid **Antiviral drug:** hydroxychloroquine, lopinavir, ritonavir **Other treatment:** paracetamol	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	/	resolved	> 12
Bouaré 2020[22]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), azithromycin **Antiviral drug:** chloroquine **Other treatment:** /	/	/	resolved	10

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Essajee 2020[23]	1	no	/	**Supportive treatment:** high flow nasal oxygen **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), **Antiviral drug:** / **Other treatment:** prednisone, aspirin, dexamethasone	/	/	on treatment	/
Pillay 2021[24]	1	no	/	**Supportive treatment:** rebreather mask **Antibiotic:** ATT, amoxicillin/clavulanic acid, azithromycin, trimethoprim, sulfamethoxazole **Antiviral drug:** / **Other treatment:** pyridoxine, prednisone, prophylactic clexane, dexamethasone, vitamin D, zinc, thiamine, ascorbic acid, therapeutic anticoagulation	/	/	on treatment	/

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Agada 2021[25]	1	no	/	**Supportive treatment:** supplemental oxygen **Antibiotic:** ceftriaxone, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** no **Other treatment:** dexamethasone, acetaminophen, pyridoxine phosphate, zinc-sulphate, Vitamin C	ATT	/	resolved	14
Gbenga 2020[26]	2	no	/	**Supportive treatment:** / **Antibiotic:** azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir, ritonavir **Other treatment:** Vitamin C, Zinc sulfate, oral prednisolone, tabs pyridoxine	/	/	resolved	17

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		no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir/ritonavir, **Other treatment:** vitamin C, zinc sulfate, tabs pyridoxine, IV hydrocortisone, clexane	/	/	died	6
Rivas 2020[27]	2	yes	/	**Supportive treatment:** invasive mechanical ventilatory support **Antibiotic:** ATT (isoniazid, rifampin, ethambutol, and pyrazinamide), levofloxacin, azithromycin, **Antiviral drug:** hydroxychloroquine, tenofovir, lamivudine, and dolutegravir **Other treatment:** low molecular weight heparin	/	nosocomial pneumonia	resolved	2 months

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		no	rifampicin and isoniazid	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ceftriaxone, azithromycin, **Antiviral drug:** hydroxychloroquine **Other treatment:** enoxapari	/	/	resolved	7
Farias 2020[28]	2	no	/	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ATT (isoniazid, rifampin, ethambutol, and pyrazinamide), ceftriaxone, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	21

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		no	/	**Supportive treatment:** / **Antibiotic:** azithromycin, ceftriaxone, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	7
Pinheiro 2020[29]	1	no	/	/	/	/	/	/
Gerstein 2021[30]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (levofloxacin, rifampicin, isoniazid, ethambutol) **Antiviral drug:** hydroxychloroquine, **Other treatment:** convalescent plasma	ATT (rifampicin, isoniazid)	/	resolved	42

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Cutler 2020[31]	1	no	/	**Supportive treatment:** nasal canula oxygen supplementation **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	/
Sasson 2020[32]	1	yes	intubated for hypoxia	**Supportive treatment:** intubated for hypoxia **Antibiotic:** azithromycin, ceftriaxone, cefepime, vancomycin, piperacillin/tazobactam, meropenem, caspofungin, metronidazole, voriconazole, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** dexamethasone, methylprednisolone, vitamin B6, metformin, alogliptin, long-acting insulin and rapid-acting insulin	/	/	resolved	> 35

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Freij 2020[33]	1	no	amoxicillin	**Supportive treatment:** / **Antibiotic:** azithromycin **Antiviral drug:** hydroxychloroquine, remdesivir **Other treatment:** dexamethasone, antihypertensive medications	/	/	died	26
		no	/	/	/	/	resolved	
		no	/	/	/	/	resolved	
Vanzetti 2020[34]	3	no	/	**Supportive treatment:** oxygen support **Antibiotic:** / **Antiviral drug:** / **Other treatment:** corticosteroids	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	/	resolved	longer hospital stay than average

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Vilbrun 2020[35]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (bedaquiline, levofloxacin, linezolid, clofazimine, pyrazinimide) **Antiviral drug:** / **Other treatment:** /	/	/	resolved	3 weeks from the diagnosis

/: No information provided

Table 6 Detailed treatment information of included retrospective cohort studies (n=7)

First author (year)	Country	Time	single center	Sample size	In-hospital	age	gender	Comorbidity	Died	Fatality rate (in-hospital or total)
Gupta 2020[36]	India	1 February 2020 to 14 June 2020	yes	22	yes	19-67	20 male (90.9%) 2 female (9.1%)	Hypertension 4 (18.2%) Diabetes 3 (13.6%) Seizure disorder 2 (9.1) Hypothyroidism 1 (4.5%)	6	27.3 (in-hospital)
Motta 2020[37]	Belgium, Brazil, France, Italy, Russia, Singapore,	12 March, 2020 to 5 May, 2020	no	49	no	27-70	/	/	7	14.3% (total)

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	Spain, and Switzerland									
Sy 2020[38]	Philippines	May 17, 2020 to June 15, 2020	no	106	no	/	73 male (68.9%) 33 female (31.1%)	Hypertension 22 (20.8) Diabetes 14 (13.2) Cancer 1 (0.9) Renal cancer 4 (3.8) Cardiac disease 8 (7.5) Asthma 4 (3.8) COPD 3 (2.8)	25	23.6% (total)
				66	yes	/		Hypertension 16 (24.2) Diabetes 10 (15.2) Cancer 1 (1.5) Renal cancer 4 (6.1) Cardiac disease 7 (10.6) Asthma 1 (1.5) COPD 1 (1.5)	18	27.3% (in hospital)
Stochino 2020[39]	Italy	/	yes	20	yes	27-47	12 male (60%) 8 female (40%)	/	1	5% (in-hospital)
Davies 2020[40]	South Africa	Till to 9 June, 2020	no	2128	no	20 or above	/	/	113	5.3% (total)
			no	469	yes	20 or	/	/	102	21.7%

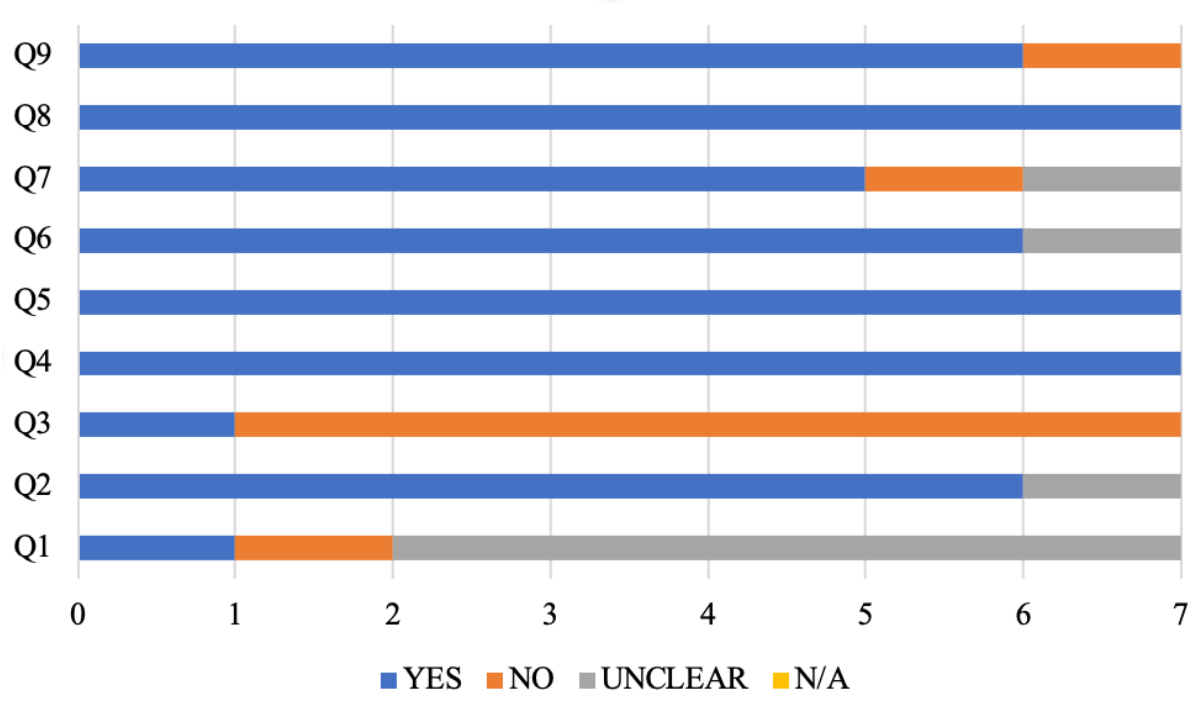
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						above				(in-hospital)
Domingo 2020[41]	Argentina	1 March to 30 June, 2020	yes	23	yes	5-82	18 male (78.3%) 5 female (21.7%)	Smoking and other addictions 16(69.6) HIV 4(17.4) Psychosis 1(4.3) Pulmonary thromboembolism 1(4.3) Arterial hypertension 1(4.3)	2	8.7% (in-hospital)
Gubkina 2020[42]	Russia	March to June, 2020	24	yes	/	/	/		0	0% (in-hospital)

/: No information provided

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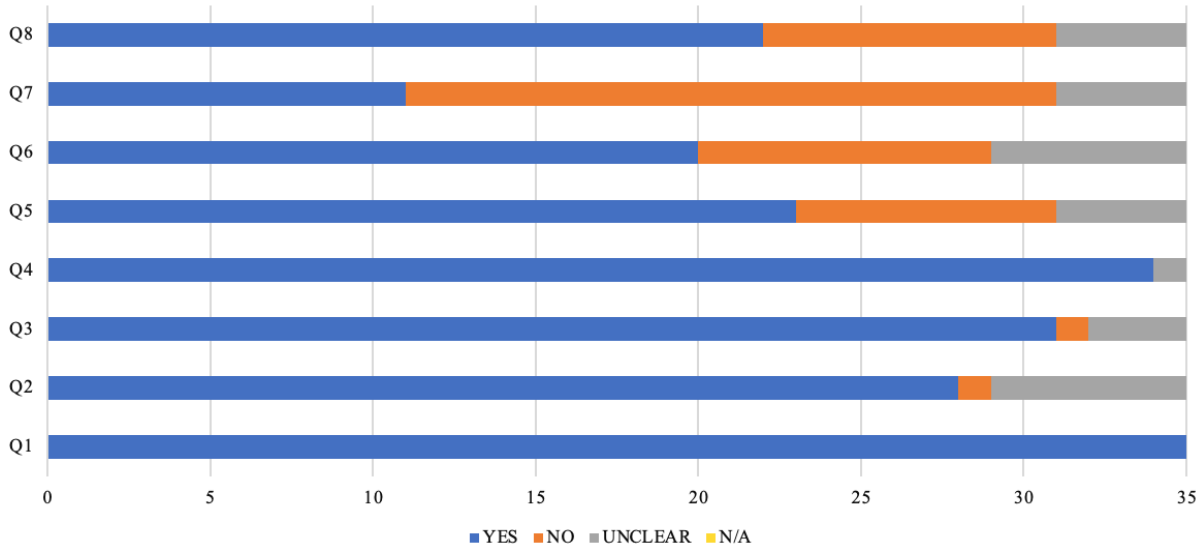
Figure 1 The results of quality assessment for prevalence study



Note: Four options for each question: yes, no, unclear and not appliable. Q1: Was the sample frame appropriate to address the target population? Q2: Were study participants sampled in an appropriate way? Q3: Was the sample size adequate? Q4: Were the study subjects and the setting described in detail? Q5: Was the data analysis conducted with sufficient coverage of the identified sample? Q6: Were valid methods used for the

identification of the condition? Q7: Was the condition measured in a standard, reliable way for all participants? Q8: Was there appropriate statistical analysis? Q9: Was the response rate adequate, and if not, was the low response rate managed appropriately?

Figure 2 The results of quality assessment for case report



Note: Four options for each question: yes, no, unclear and not appliable. Q1: Were patient's demographic characteristics clearly described? Q2: Was the patient's history clearly described and presented as a timeline? Q3: Was the current clinical condition of the patient on presentation clearly

described? Q4: Were diagnostic tests or assessment methods and the results clearly described? Q5: Was the intervention(s) or treatment procedure(s) clearly described? Q6: Was the post-intervention clinical condition clearly described? Q7: Were adverse events (harms) or unanticipated events identified and described? Q8: Does the case report provide takeaway lessons?

Table 7 Quality assessment of included case reports (n=35)

Study (Year) \ Results	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Agada 2021	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Yao 2020	Yes	Yes	Yes	Unclear	Unclear	No	No	No
Yadav 2020	Yes	Yes	Yes	Yes	No	No	No	No
Wang 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Vilbrun 2020	Yes	Yes	Yes	Yes	No	Unclear	No	Yes
Tham 2020	Yes	Yes	Yes	Yes	No	No	No	No
Singh 2020	Yes	Yes	Yes	Yes	Unclear	No	Yes	Yes
Sasson 2020	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear
Sarma 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Sarinoglu 2020	Yes	Yes	Yes	Yes	Unclear	Yes	No	Yes
Rivas 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Pinheiro 2020	Yes	Yes	Unclear	Yes	No	No	No	No
Pillay 2021	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Musso 2021	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Luciani 2020	Yes	Yes	Yes	Yes	Yes	No	No	Yes

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Lopinto 2020	Yes	Yes	Yes	Yes	No	Unclear	Yes	Yes
Liu 2020	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Kumar 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Khayat 2021	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Gbenga 2020	Yes	Yes	Yes	Yes	Unclear	Unclear	Unclear	No
Freij 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cutler 2020	Yes	Yes	Unclear	Yes	Yes	Yes	No	Yes
Cinar 2020	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes
Cao 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farias 2020	Yes	Yes	Yes	Yes	Yes	Unclear	No	Yes
Bouare 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Baskara 2021	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
AlKhateeb 2020	Yes	Yes	Yes	Yes	No	Yes	No	Unclear

Note: Four options for each question: yes, no, unclear and not appliable. Q1: Were patient's demographic characteristics clearly described? Q2: Was the patient's history clearly described and presented as a timeline? Q3: Was the current clinical condition of the patient on presentation clearly described? Q4: Were diagnostic tests or assessment methods and the results clearly described? Q5: Was the intervention(s) or treatment procedure(s) clearly described? Q6: Was the post-intervention clinical condition clearly described? Q7: Were adverse events (harms) or unanticipated events identified and described? Q8: Does the case report provide takeaway lessons?

Table 8 Quality assessment of included retrospective cohort studies (n=7)

Results / Study (Years)	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
Gupta 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Motta 2020	Unclear	Unclear	No	Yes	Yes	Yes	Yes	Yes	Yes

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Sy 2020	Yes	Yes	No	Yes	Yes	Unclear	Unclear	Yes	No
Stochino 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Davies 2020	Unclear	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Domingo 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Gubkina 2020	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes

Note: Four options for each question: yes, no, unclear and not appliable. Q1: Was the sample frame appropriate to address the target population? Q2: Were study participants sampled in an appropriate way? Q3: Was the sample size adequate? Q4: Were the study subjects and the setting described in detail? Q5: Was the data analysis conducted with sufficient coverage of the identified sample? Q6: Were valid methods used for the identification of the condition? Q7: Was the condition measured in a standard, reliable way for all participants? Q8: Was there appropriate statistical analysis? Q9: Was the response rate adequate, and if not, was the low response rate managed appropriately?

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The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis (From November 2019 to March 2021)

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The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis (From November 2019 to March 2021)

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The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis (From November 2019 to March 2021)

Abstract

Introduction The breakout of COVID-19 poses a significant threat to the TB patients. TB and COVID-19 (TB-COVID) coinfection means the disease caused by both mycobacterium tuberculosis and SARS-COV-2 infection. Currently, the prevalence status, treatment, and outcomes of the coinfection are poorly characterised. We aimed to systematically review the evidence on this topic and provide comprehensive information to guide the control and treatment of TB-COVID coinfection.

Methods An extensive screening was conducted using 6 electronic databases to search eligible studies from 1 November 2019 to 19 March 2021. Prevalence rate, treatment, and outcomes of TB-COVID coinfection were extracted. Random effects models were used to calculate mean fatality rates of coinfection with 95% confidence intervals (CI). The risks of bias were assessed with JBI Critical Appraisal Checklist for Study Reporting Prevalence Data and JBI Critical Appraisal Checklist for Case Report. A meta-analysis was conducted for subgroups on in-hospital fatality rate.

Results Forty-two studies were included into the analysis (35 case reports and 7 retrospective cohort studies). Nineteen countries reported coinfected patients, including high and low TB prevalence countries. The only study revealing prevalence rate came

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from West Cape Province, South Africa (people aged above 20, 0.04% till 1 June 2020 and 0.06% till 9 June 2020). The treatment regimes for coinfected patients were highly heterogeneous. The mean overall and in-hospital fatality rates of coinfection were 13.9% (95% CI: 1.6% to 26.2%) and 17.5% (95% CI: 8.9% to 26.0%). The mean in-hospital fatality rate for high-income country (Italy and Argentina) and low- and middle-income countries (LMICs) (India, Philippines, South Africa) were 6.5% (95% CI: -0.8% ~ 13.9%) and 22.5% (95% CI: 19.0% ~ 26.0%).

Conclusion TB-COVID coinfection is common globally, and the coinfected patients suffer from higher fatality risk than normal COVID-19 patients. Outcomes shared significant differences between high-income countries and LMICs.

Registration PROSPERO CRD42021253660

Keywords: COVID-19, tuberculosis, coinfection, prevalence, health outcome, treatment, systematic review

Strengths and limitations of this study

- This systematic review and meta-analysis provided comprehensive information on prevalence status, treatment plans and possible outcomes of TB-COVID coinfection.
- We collected evidence from case reports regarding information about patients' basic information, treatment regimens and outcomes.
- Sub-group analysis was conducted to explore the possible reason for different

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fatality rates in different areas.

- We did not include comorbidity as a keyword and MeSH term in the searching process, which might keep us from some relevant studies.

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1. Introduction

Individuals with chronic respiratory infections, including tuberculosis (TB), had been disproportionally hit by the coronavirus disease 2019 (COVID-19) pandemic[1]. The burden and impact of TB control and treatment are widely acknowledged with the ongoing pandemic. The number of people who fell ill with TB in 2019 is estimated to be 10 million[2]. The top seven countries account for two-thirds of the total global cases in 2018, while most high prevalence rate countries locate in South-East Asia and Africa[3]. Patients with TB have an increased risk of mortality during coinfection with COVID-19[4]. Recently, some researchers identified TB as a risk factor to severe COVID-19[5], and some studies suspected that the COVID-19 could reactivate or worsen TB[6 7]. From the perspective of pathology, the disorders induced by SARS-COS-2 and Mycobacterium tuberculosis to the immunomodulation tend to induce an unbalanced inflammatory response, promoting the progression and worsening of both diseases[8]. Besides, the use of immunosuppressive drugs on severe COVID-19 patients may increase the likelihood of active TB caused by reactivation or new infection[9]. On the other side, the COVID-19 outbreak has put on the TB patients and the entire health care system in an unprecedented crisis. Many experts have recognized the effects of the COVID-19 pandemic on TB treatment and control. As warned by Global Tuberculosis Network Group, COVID-19 has fueled TB infections and mortality[6]. In low- and middle-income countries (LMICs), the situation is even worse. As estimated by Kissler, the transmission of SARS-CoV-2 is likely to enter into regular circulation[10], which

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would be a deadly threat to people in LMICs, especially when most of them cannot stand the chance of getting COVID-19 vaccine[11]. Regrettably, people in these countries are also the main victims of tuberculosis[2].

Although cases of coinfection and studies about prevalence status have been reported from several countries[12], we did not find a standard definition of the TB and COVID-19 (TB-COVID) coinfection. Therefore, in this systematic, we preliminarily described the TB-COVID coinfection as the disease caused by both mycobacterium tuberculosis and SARS-COV-2 infection. However, much of the research up to now has been descriptive in nature and focused on the observation of several coinfected patients, which can only provide weak evidence to guide the treatment for TB-COVID coinfection. Furthermore, few scholar has conducted a systematic review in this field, and there is no general agreement about the prevalence, treatment and outcome of the coinfection worldwide.

Although a few health alerts and guidelines for people with increased risk for adverse outcomes in COVID-19 have been issued by World Health Organization (WHO) and the Center for Disease Control and Prevention (CDC)[13], there is a dearth of information regarding the impact of COVID-19 in patients with TB and the actual impact of tuberculosis on occurrence and clinical outcomes of COVID-19 is not clear. This study aims to systematically analyze the current evidence on global prevalence status,

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treatment protocols, and possible outcomes of TB and COVID-19 coinfection.

2. Methods

This systematic review protocol followed the International Prospective Register of Systematic Reviews (PROSPERO) guideline and has been registered in PROSPERO's database (registration number: CRD42021253660). The study followed the Preferred Reporting Items in Systematic Reviews and Meta-Analyses (PRISMA) guideline[14] (Appendix Table 1).

2.1 Eligibility Criteria of Included Studies

Studies on patients who had been confirmed with both TB and COVID-19 were included. We excluded studies that only mentioned the fatality and the number of coinfected patients less than 10 to avoid a large deviation in fatality rate, which would affect the results of Meta-Analysis (MA).

This systematic review included any kind of treatment for TB or COVID-19, including supportive therapy, directed therapy, antiretroviral treatment, ICU care, etc.

We sought prevalence rate data of any region and figured out all countries that reported coinfected patients. The primary outcome of interest was in-hospital survival or

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mortality, and secondary outcomes were disease progression, time to discharge from the ICU and hospital (for survivors), and time to death (for the dyed).

Case reports, cross-sectional studies, and cohort studies were included to explore the global prevalence status. In addition, case reports, cross-sectional studies, cohort studies, and quasi-experimental studies that revealed the treatment and outcome of TB-COVID coinfection were targeted. We excluded all reviews, editorials or commentaries, and clinical guidelines.

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2.2 Search Strategy

An extensive literature search was conducted using PubMed, Web of Science, Scopus, ProQuest, Embase, and Cochrane library databases. Manual reference screens from included studies were performed to ensure the inclusion of all relevant studies. The study's publication date was between November 1, 2019 and March 19, 2021. With the librarian's (JZ) help, we used building blocks searching strategy to ensure all the relevant studies were screened. The searching process utilized Medical Subject Heading (MeSH) terms and Tittle/Abstract words around two areas: 1) COVID-19, including COVID-19, SARS-COV-2, 2) tuberculosis, including tuberculosis, TB, Mycobacterium tuberculosis, M.tuberculosis (Appendix Table 2). The citation manager EndNote 20 (Thomson Corporation, Canada) removed duplicate studies, and we spent special efforts on the same patients or cohorts among different papers.

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2.3 Data Extraction

Two reviewers (QW and SG) independently conducted the study selection (review of titles and abstracts, review of full texts, and final decision), data abstraction, and quality assessment. Covidence (Covidence Company, Australia) was utilized to help us finish all the above processes. All non-English and non-Chinese articles were translated to English by Baidu Translation (Baidu Co., Ltd, China). We recorded exclusion reasons for all excluded studies in the full-text screen step during the review process. The conflicts between 2 reviewers were discussed and eventually resolved with another author's help (NX).

Since this systematic review got 2 research objectives, 2 categories of information were extracted: 1) data about the prevalence, 2) data about treatment and outcome. For the first category, we stressed locations (nation or province, especially for case reports) of studies and the following information: age, gender, sample size, time, prevalence rate, and fatality rate. As for the second category, treatment for TB and COVID-19 were recorded separately, as well as the primary and secondary outcomes (e.g., disease progression, time to discharge from the ICU and hospital, and time to death). Besides, we also stressed patients' basic information in both parts, like TB site, TB type, drug resistant status, comorbidity, BCG vaccination, etc.

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2.4 Quality Assessment

We used the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Study Reporting Prevalence Data for the prevalence part as the tool[15]. Nine questions in this checklist were set to assess the quality of studies, including sample frame, sample size, study setting, data analysis, validation, measurement, and response rate. All included studies were case reports for the treatment and outcome review, so we applied JBI Critical Appraisal Checklist for Case Report[16] as the tool which consists of 8 questions about sample, patients' condition, test and assessment, intervention, data analysis, validation, measurement, and response rate.

2.5 Data Synthesis

A narrative synthesis method was introduced to list the prevalence status of coinfection. Key findings such as the countries and regions of the case reports were extracted to probe the potential prevalence of TB-COVID coinfection. We condensed the information about treatment into 3 categories: treatment before hospitalization, treatment in hospital, and treatment after discharge. We also sorted out the treatment in the hospital into 4 kinds: supportive treatment, antibiotics, antiviral drugs, and other treatment. The narrative synthesis in this systematic review followed Cochrane's Consumers and Communication Review Group (CC&CRG) reviews protocol[17]. We conducted MA about the fatality rate of coinfection patients using the inverse variance statistical method to pool the data and estimate its 95% confidence interval (CI). Due

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to the highly heterogeneous treatment method for patients as well as patients themselves, we chose the random-effects model in MA, and the heterogeneity was evaluated with I^2 statistic. The funnel plot was created to test the bias.

2.6 Subgroup Set

Studies were grouped into the following subgroups for further analysis: high-income country and low- and middle-income country groups. We would like to explore the possible difference in fatality rates among different types of country.

2.7 Patient and Public Involvement

Patients were not involved in this systematic review.

3. Results

We searched 565 studies from PubMed, 682 from ProQuest, 839 from Scopus, 644 from Web of Science, 367 from Embase, and 1 from the Cochrane database. After removing 1878 duplicate studies, we got 1220 studies and imported them into Covidence. With the help of Covidence, we excluded 1049 studies in the title and abstract screen step. Next, we carefully reviewed the full text of the remaining 171 studies, and then we excluded 131 studies from them. We also found 11 possibly relevant studies during the full text reviewing process, 9 of which were excluded. The

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most common reasons for exclusion were no outcomes of interest (n = 80) and no quantitative data (n=39). The entire process is shown in Figure 1. Among the 42 included studies, 35 were case reports, and 7 were retrospective cohort studies. No experimental study was identified.

<Insert Figure 1 here>

3.1. Global Prevalence Status

Till to Mar. 19 2021, there were 35 case reports from 17 countries, including India[18-22], Indonesia[23], China[24-28], Saudi Arabia[29 30], Qatar[31 32], Singapore[33], Turkey[34 35], Haiti[36], Argentina[37], U.S.[38-41], Brazil[42 43], Panama[44], Nigeria[45 46], South Africa[47 48], Morocco[49], Italy[50 51], and France[52]. Totally, 55 cases were described in these 35 reports. China, India and U.S. were the top 3 countries with the most studies, and 29 (82.9%) studies were published in 2020. Table 1 shows the study characteristics of these case reports. Besides, we also identified coinfected patients from Belgium and Switzerland in Motta's paper[53].

Table 1 Study characteristics of case reports (n=35)

Study characteristic	Studies (%)	Cases (%)
Publication year		
2020	29 (82.9)	49 (89.1)
2021	6 (17.1)	6 (10.9)
Country		
India	5 (14.3)	5 (9.1)
Indonesian	1 (2.9)	1 (1.8)
China	5 (14.3)	11(20.0)
Saudi Arabia	2 (5.7)	2 (3.6)

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Qatar	2 (5.7)	7 (12.7)
Singapore	1 (2.9)	4 (7.3)
Turkey	2 (5.7)	3 (5.5)
Haiti	1 (2.9)	1 (1.8)
Argentina	1 (2.9)	3 (5.5)
The U.S.	4 (11.4)	4 (7.3)
Brazil	2 (5.7)	3 (5.5)
Panama	1 (2.9)	2 (3.6)
Nigeria	2 (5.7)	3 (5.5)
South Africa	2 (5.7)	2 (3.6)
Morocco	1 (2.9)	1 (1.8)
Italy	2 (5.7)	2 (3.6)
France	1 (2.9)	1 (1.8)

The only large-scale study about the prevalence rate came from West Cape Province, South Africa[54]. The author used the Western Cape Provincial Health Data Centre (WCPHDC) data, which included 3,460,932 individuals aged 20 or above, about half of the population in West Cape Province. Until June 1, 2020, when testing criteria changed with public sector tests being limited to patients >55 years of age or with comorbidity, 1,489 coinfected patients (including previous TB and current TB) were recorded, and the prevalence rate of coinfection was about 0.04%. Till June 9, 2020, 2128 coinfected patients (including previous TB and current TB) were recorded, and the prevalence rate was about 0.06%.

3.2. Treatment and Outcome

Till the end of search, no cohort study, cross-sectional study, or experimental study was identified to provide detailed information on TB and COVID-19 coinfection treatment.

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Presented the results of the included 35 studies which reported 55 coinfected patients in total. As for the basic information, 12 patients were female, and 47 patients were aged 20 to 60 years. Up to 50 cases were reported as active TB patients, 12 of which got TB before COVID-19, and we could not identify infection orders about the rest 38 cases. Besides, 4 cases were reported as post TB[24], and 1 was reported as latent TB[27]. Although most cases were diagnosed with pulmonary TB, there were still some more complex conditions, like central nervous system involvement[21 38], tuberculous meningitis[28], tuberculous pleuritis[26 27 31], and peritoneal TB[41]. With regard to drug resistance, 37 patients' information was missing, 15, 2 and 1 of the remaining 18 cases were reported as negative[22 23 29 30 37 39 42-44 46 48 51], positive[27 36], and likely positive[18]. All the cases were symptomatic COVID-19 patients. Check Appendix Table 3 for detailed information.

Only 5 patients received treatment before hospitalization. Among them, 3 received Anti-TB Treatment (ATT)[22 34 44], 1 received antibiotic (amoxicillin)[38], and 1 took hemodialysis due to chronic kidney failure[35]. During the hospitalization, 7 patients were transferred to the Intensive Care Unit (ICU)[18 30 34 35 39 44 52]. Up to 50 cases' detailed treatment information was reported. Various kinds of oxygen support were the most common supportive treatment (21 cases), and 1 patient in China received extracorporeal membrane oxygenation[27]. Except for several cases that did not provide any detailed ATT regime (12 cases) or patients who did not receive ATT (11 cases), 1 or several kinds of first-line Anti-TB drugs, including rifampicin, isoniazid, ethambutol, and

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pyrazinamide, were utilized in most remaining cases (27 cases). For the drug resistance TB patients, the doctors would adjust the regime for them[27 36]. Other kinds of antibiotics were also utilized, including azithromycin (21 cases), ceftriaxone (13 cases), moxifloxacin (7 cases), and linezolid (4 cases), etc. However, we could not identify the purpose of these extra antibiotics due to the lack of information. We guessed it could be 1 or more of the following situations: 1) supplement to ATT regime, 2) prevention against possible bacterial infections, and 3) treatment for the existing bacterial infection. Among these 50 cases, 13 cases did not mention any kind of antiviral drug and 2 cases did not provide antiviral drugs' names. Within the remaining cases, hydroxychloroquine (HCQ) was the most utilized antiviral drug (21 cases), as well as lopinavir (8 cases), ritonavir (8 cases), and arbidol (7 cases). We also found that all the 20 Chinese coinfected patients did not take HCQ completely but took arbidol most. Besides, drugs like steroids (methylprednisolone, dexamethasone, prednisone, etc.), anti-coagulant drugs, and vitamins (vitamin C, vitamin B6, etc.) were also prescribed for some patients. We identified information concerning treatment after discharge for 10 patients; 8 of them received ATT, and 2 received treatment for existing comorbidity. Although there was no doubt that the regimes for TB and COVID-19 coinfected patients were highly heterogeneous, we still could identify 4 main paradigms about treatment during hospitalization: 1) simultaneous antiviral treatment and ATT[19-21 23-26 28 30 31 34 35 39-41 43-45 49 51], 2) only ATT[22 26 32 33 35 36 46-48], 3) only antiviral treatment[18 24 26 27 38 44 50], 4) no antiviral treatment and ATT[37]. Check Appendix Table 4 and Appendix Table 5 for detailed information about the treatment.

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A minimal number of studies discussed about pros and cons of the treatment and the possible effects of the regimes. Faqihi held that the first-line Anti-TB drug with antiviral treatment for COVID-19 might theoretically increase the risk of side-effects[30], and Yousaf stressed the importance of adjustment to the management of coinfected patients[31]. The antagonism from lopinavir/ritonavir against rifampicin had been noticed[20 25], and Cao suggested umifenovir could be the substitute for lopinavir/ritonavir[25]. Kumar warned additive hepatotoxicity from remdisivir due to simultaneous use of anti-tubercular drugs[20]. Faqihi gave isoniazid along with pyridoxal phosphate to avoid peripheral neuropathy[30], and Farias did not start antiretroviral therapy at first for the 2 HIV, TB and COVID-19 coinfected patients to avoid complications with ATT[43]. Confronted with liver function impairment during the treatment, Musso replaced isoniazid and pyrazinamide with amikacin and moxifloxacin to modify the ATT regimen[51]. Liu, He, Cao, and Singh discussed the impact of corticosteroids on treating lung inflammation[19 24 25 27]. Liu's recommendation was not to administer corticosteroids for COVID-19 and TB coinfected patients. Besides, since COVID-19 and TB disease might both cause immunosuppression, Liu recommended using immunomodulatory therapies such as thymosin in patients with progressively lower lymphocyte counts[27].

About the outcome of 55 reported cases, 35 were resolved, 6 were deceased, 3 were

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still in treatment in hospital, 1 returned to hospitalization due to positive recurrence of SARS-CoV-2 RNA, and 10 were missing. For patients discharged from the hospital, the length of stay ranged from 7 days to 2 months. Besides, in Vanzetti's study, 3 patients were described they had a longer hospital stay than average[37]. For deceased patients, the length of stay ranged from 6 to 26 days. Unfortunately, we could only find minimal information about coinfected patients' physical condition when they got discharged. Most studies only indicated that the test results for SARS-CoV-2 were negative or patients got recovered. Although Agada described the patient with poor prognostic indices, detailed physical conditions were also missing[46]. He believed the treated or untreated mycobacterium TB was a risk factor for a worse prognosis of COVID-19 patients[24]. In addition, long-term outcome or follow-up outcome for the TB and COVID-19 coinfected patient was absent.

3.3. Fatality Rate of TB and COVID-19 Coinfection

We identified 7 retrospective cohort studies that revealed the fatality rate of TB and COVID-19 coinfection (Appendix Table 6 for detailed information). These studies came from India[55], Philippines[56], Italy[57], South Africa[54], Argentina[58], Russia[59] and 1 was based on data from multiple countries[53]. Motta's study provided 2 cohorts[53], 1 of which was same as Stochino's study[57]. Thusly, we removed the overlapped cohort. Except for Gubkina's study whose targeted population was children and no one deceased[59], the overall and in-hospital fatality rates fell within 5.3% ~ 23.6% and 5.0%

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~ 27.3%. Table 2 presents the results of the included 7 studies.

Table 2 Fatality rate of included studies (n=7)

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First author (year)	Country	Time span	Sample size	In-hospital	age	gender	Comorbidity	Died	Fatality rate (overall or in-hospital)
Gupta 2020	India	1 February 2020 to 14 June 2020	22	yes	19-67	20 male (90.9%) 2 female (9.1%)	Hypertension 4 (18.2%) Diabetes 3 (13.6%) Seizure disorder 2 (9.1%) Hypothyroidism 1 (4.5%)	6	27.3 (in-hospital)
Motta 2020	Belgium, Brazil, France, Italy, Russia, Singapore, Spain, and Switzerland	12 March 2020 to 5 May 2020	49	no	27-70	/	/	7	14.3% (overall)
Sy 2020	Philippines	17 May 2020 to 15 June 2020	106	no	/	73 male (68.9%) 33 female (31.1%)	Hypertension 22 (20.8%) Diabetes 14 (13.2%) Cancer 1 (0.9%) Renal cancer 4 (3.8%) Cardiac disease 8 (7.5%) Asthma 4 (3.8%) COPD 3 (2.8%)	25	23.6% (overall)
			66	yes	/		Hypertension 16 (24.2%) Diabetes 10 (15.2%) Cancer 1 (1.5%) Renal cancer 4 (6.1%) Cardiac disease 7 (10.6%) Asthma 1 (1.5%) COPD 1 (1.5%)	18	27.3% (in hospital)

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Stochino 2020	Italy	/	20	yes	27-47	12 male (60%) 8 female (40%)	/	1	5% (in-hospital)
Davies 2020	South Africa	Till 9 June 2020	2128	no	20 or above	/	/	113	5.3% (overall)
			469	yes	20 or above	/	/	102	21.7% (in-hospital)
Domingo 2020	Argentina	1 March to 30 June 2020	23	yes	5-82	18 male (78.3%) 5 female (21.7%)	Smoking and other addictions 16 (69.6%) HIV 4 (17.4%) Psychosis 1 (4.3%) Pulmonary thromboembolism 1 (4.3%) Arterial hypertension 1 (4.3%)	2	8.7% (in-hospital)
Gubkina 2020	Russia	March to June 2020	24	yes	/	/	/	0	0% (in-hospital)

/: No information provided

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3.4. Quality of Included Studies

We utilized JBI Critical Appraisal Checklist for Study Reporting Prevalence Data as the tool to assess 7 included studies that revealed prevalence rate or fatality rate. The grade (yes, no, unclear or N/A) was shown in Appendix Figure 1 and Appendix Table 7. The results suggested that insufficient sample size and inappropriate sample frame were the main risks of bias of these studies.

We utilized JBI Critical Appraisal Checklist for Case Report[16] as the tool to assess 35 included case reports, as shown in Appendix Figure 2 and Appendix Table 8. All the studies described patients' demographic characteristics clearly, and we could identify basic information about patients' history (28 studies), clinical condition (31 studies), diagnostic tests or assessment methods and the results (34 studies) for most studies. Whereas only 23, 20, and 22 studies provided clear messages about treatment procedures, post-intervention clinical conditions, and takeaway lessons, respectively. The most poorly reported part was adverse or unanticipated events, and only 11 studies gave a clear description, which could be a high risk of bias.

3.5. Meta-Analysis of Fatality Rate

We included 6 qualified studies (8 cohorts) to estimate the fatality rate of TB and COVID-19 coinfection, 3 studies for overall fatality rate[53 54 56] and 5 studies for in-hospital fatality rate[54-58]. Gubkina's study was excluded because its targeted population

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was children, which might lead high heterogeneity in the MA. Although Davies's study sample was people aged 20 or above, we still included it into the MA.

As shown in Figure 2A, the estimated overall fatality was 13.9% (95% CI: 1.6% ~ 26.2%). Significant heterogeneity was observed between the studies (I^2=91.07%, p-value for Q statistics <0.001). The estimated in-hospital fatality rate was 17.5% (95% CI: 8.9% to 26.0%), and heterogeneity was detected among these studies too (I^2=75.47%, p-value for Q statistics <0.01) (Figure 2B). We utilized funnel plots to explore the existence of publication bias, and the results were both positive (Figure 3A, 3B).

To further explore the source of heterogeneity, we grouped these studies into the high-income country and LMICs subgroups. However, due to the lack of enough data, we could only conduct subgroup MA on in-hospital fatality rate. The mean in-hospital fatality rate for high-income country (Italy and Argentina) and LMICs (India, Philippines, South Africa) were 6.5% (95% CI: -0.8% ~ 13.9%) and 22.5% (95% CI: 19.0% ~ 26.0%). No heterogeneity was observed between studies for both subgroups (Figure 2C, 2D). Similarly, 2 funnel plots were created to explore the existence of publication bias, and the results were both negative (Figure 3C, 3D).

<Insert Figure 2 and Figure 3 here>

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4. Discussion

This systematic review identified the evidence on the prevalence status, treatment and possible outcome of TB and COVID-19 coinfection. Although studies about coinfection were limited and quantitative data was even rarer, we still identified 42 studies, of which 35 were case reports and 7 were retrospective cohort studies.

Regarding the quality of included studies, we found that the sample sizes were insufficient, and the sampling frames were not specified, which had been the leading risk of bias for prevalence studies. In addition, the lack of reporting of adverse and unanticipated events, intervention and treatment procedures, and post-intervention conditions were noticeable defects of the case reports. Publication bias was found in the overall and the in-hospital fatality rate of the TB and COVID-19 coinfection, which suggested the current results stood on selected studies. As of the searching date, the number of relevant studies was limited, and cohort studies were even rarer. Besides, we did not find any experimental study about TB-COVID coinfection. Considering the quality of current evidence, we think more studies are urgently needed to evaluate the prevalence, treatment, and long-term outcome of TB-COVID coinfection.

In terms of global prevalence status, 19 countries reported coinfected patients, including low TB prevalence countries like Belgium and Switzerland, as well as high TB prevalence countries like China and India. Therefore, we assumed that the TB-

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COVID coinfection was common worldwide, which every country and region should beware of. The only large-scale study about the prevalence rate came from West Cape Province, South Africa. The rate was 0.04% on June 1, 2020, and 0.06% on June 9, 2020, increasing by 50% within only 8 days, not to mention that the public sector tests were limited to patients >55 years of age or with comorbidity. From the perspective of hindsight, though the rate was low, considering it was at the beginning of the pandemic, the situation was still worrying. However, we cannot infer the global prevalence rate based on this very limited information, and we believe more studies are urgently needed, especially for countries with high COVID-19 cases and high TB prevalence.

Although case report is not a very sound study design for us to evaluate treatment and outcome information, we didn't find any other kinds of study design on this topic. The analysis of case reports showed the treatment for coinfection was of great heterogeneity, while we still identified 4 main paradigms: 1) simultaneous antiviral treatment and ATT, 2) only ATT, 3) only antiviral treatment, 4) no antiviral treatment and ATT. Some studies implied the doctor just followed the local clinical guidelines; however, very few studies mentioned regimen adjustments due to characteristics of TB and COVID-19 coinfected patients. Therefore, we assumed the current treatment plans were more like a mix of the suggested TB regime and the COVID-19 regime. Besides, we did not detect any specific treatment protocol for patients with inactive TB, namely the previous TB or latent TB patients. Furthermore, HCQ, the most commonly used antiviral drug in these cases, was not taken by any patient in China's study. Traditional Chinese

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Medicine (TCM), which was widely used for COVID-19 patients in China[60], was also seldomly mentioned in these China's reports. The reason might be that these cases were reported at the beginning of the pandemic, and the doctors lacked the experience to deal with the situation. We tried multiple ways to explore the relationship between treatments and outcomes, but we did not find meaningful results. Therefore, we could not find evidence on which one is the best clinical practice due to a lack of sound investigation or experimental study, which suggested the urgency of further research.

The prolonged hospital stay for coinfected posed a significant challenge to the health care system, especially for those low-resource countries. Besides, the vaguely described health status at discharge and the absence of long-term outcomes are alarming. Moreover, for discharged patients who were still on ATT, the adverse impact of the pandemic on the delivery of TB care services which had been confirmed in multiple countries[61-63], might impede their accessibility to services.

We tried to conduct MA on overall and in-hospital fatality rates, and the estimated results were 13.9% and 17.5%. According to Meyerowitz-Katz, the fatality rate of patients with single COVID-19 was 0.68% (95%CI: 0.053% ~ 0.82%)[64], which was relatively smaller. Therefore, we believe the coinfected patients faced more challenges and called for the need to refine medical equipment and interventions. In addition, significant heterogeneity was detected in both MAs on overall and in-hospital fatality

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rates. In contrast, neither heterogeneity was detected in both sub-groups' MAs on the in-hospital fatality rate. Based on case reports, we did not detect any pattern about treatment in both high-income countries and LMICs. Considering that patients in high-income countries are likely to receive care of better quality than patients in LMICs, we guess the different results of MAs' heterogeneity might hint that the quality of care is highly associated with outcomes. In other words, the patients in high-income countries shared a better condition than those in LMICs and were more likely to recover from TB and COVID-19 coinfection. It's necessary to point out that the results of MA were based on a limited number of studies, of which defects in the quality were a frequent feature.

This systematic review has a few limitations. We did not include comorbidity as a keyword and MeSH term in the searching process, which might keep us from some relevant studies. All preprint and gray literature databases were not included in the search process. Besides, the diagnosis and testing criteria for COVID-19 and TB differ in countries and regions, and we were not able to transfer the data under a universal standard. Therefore, there might be some false-positive patients. Only 6 studies were included in MA, and significant heterogeneity was detected among them. Therefore, the results should be interpreted with caution.

In conclusion, the results of our systematic review show coinfection was not rare

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worldwide, and outcomes based on current evidence shared significant differences between high-income countries and LMICs. However, current evidence is insufficient to evaluate the treatment and outcome or estimate the prevalence rate of TB-COVID coinfection globally. More study about this field is urgently needed, especially experimental study and cohort study. Given that some countries have higher TB and COVID-19 prevalence simultaneously, global efforts are needed to work collaboratively on TB-COVID coinfection, especially in response to those with limited medical resources.

Ethical Statement N/A

Data availability statement Data are from the published study. Extracted data are available on request from the corresponding author.

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Contributor and Guarantor Information： Quan Wang put up with the idea and designed this study; Jie Zhao was responsible for the literature search; Quan Wang and

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Shasha Guo conducted the study selection (review of titles and abstracts, review of full texts, and final decision), data abstraction, and quality assessment independently; Ning Xu, as a medical doctor, provided necessary information about treatment and drugs; Quan Wang finished the manuscript; Hui Li, Qiang Sun, Xiaolin Wei and Quanfang Dong refined the manuscript; Qiang Sun was responsible for this study. Quan Wang and Shasha Guo have equal contributions to this study. The corresponding author, Qiang Sun, attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

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Figure legends

Figure 1 A flow diagram outlines the study selection process.

Figure 2 Forest plots of fatality rate accorting to hospitalization, subgroups according to economic status of regions. A: overall fatality rate; B: in-hospital fatality rate; C: in-hospital fatality rate (high-income country subgroup); D: in-hospital fatality rate (low- and middle-income country subgroup)

Figure 3 Funnel plots of fatality rate accorting to hospitalization, subgroups according to economic status of regions. A: overall fatality rate; B: in-hospital fatality rate; C: in-hospital fatality rate (high-income country subgroup); D: in-hospital fatality rate (low- and middle-income country subgroup)

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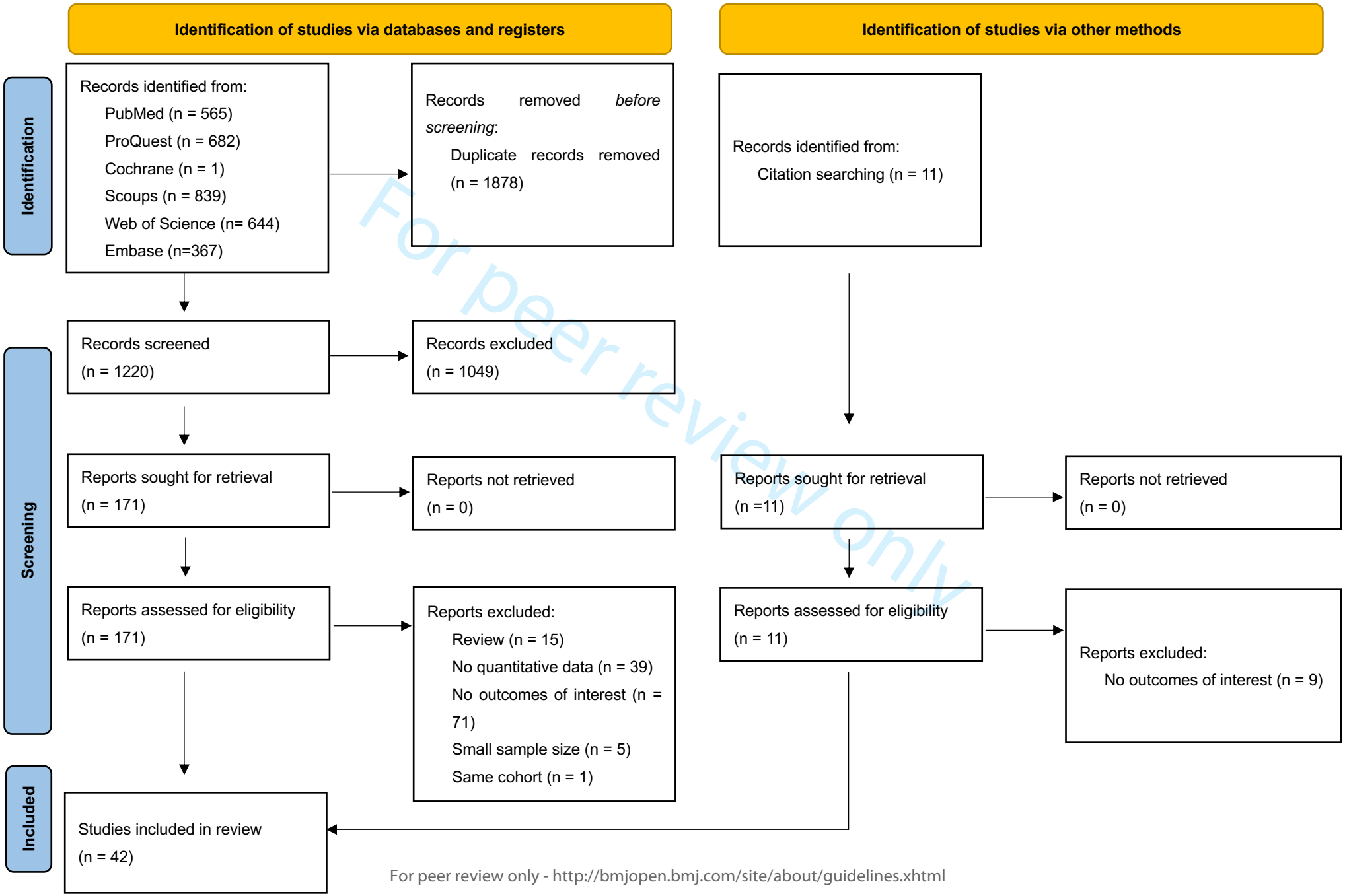
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Figure 1 Flow chart of literature search



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Figure 2 Forest plot of included studies

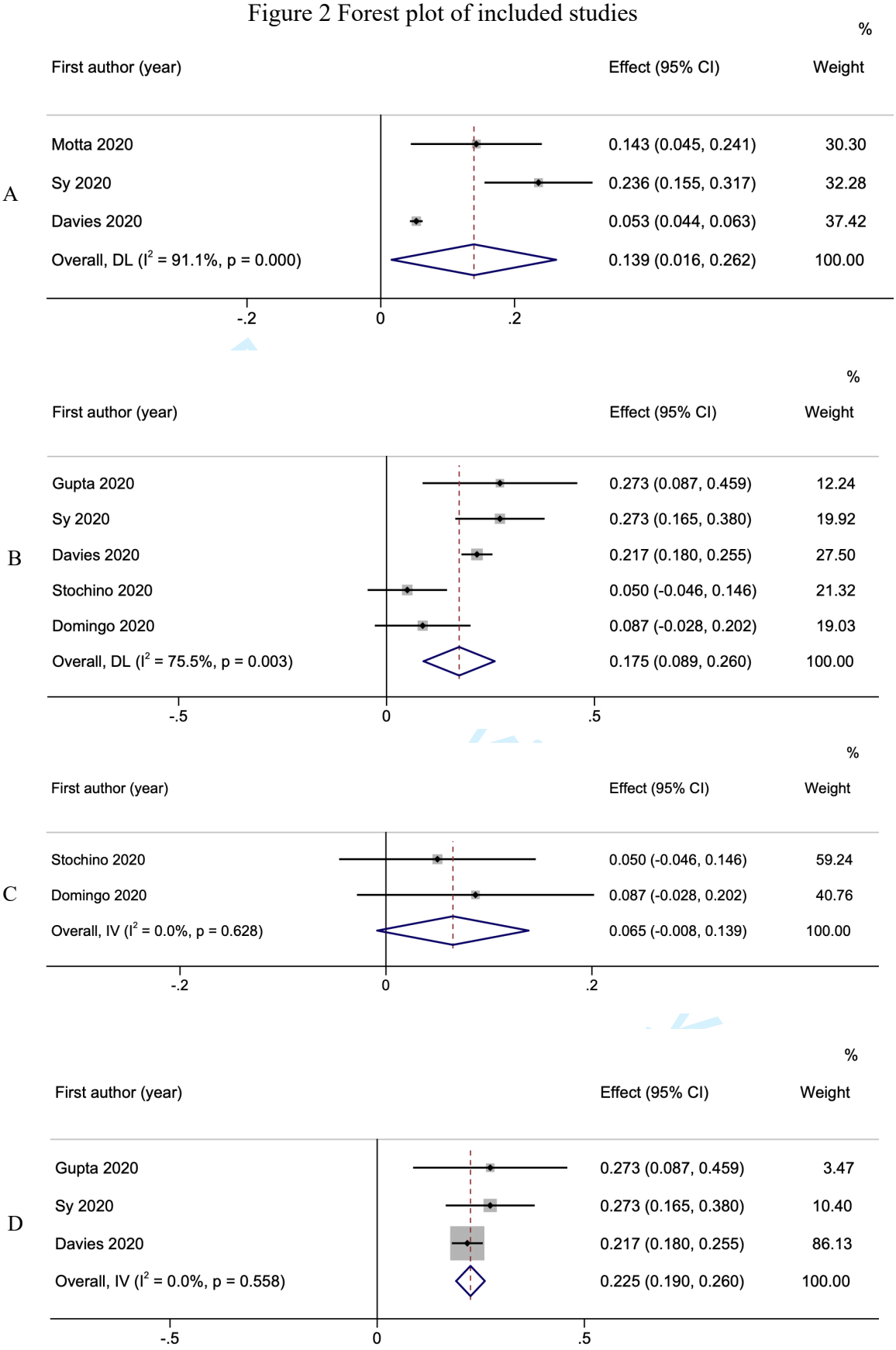
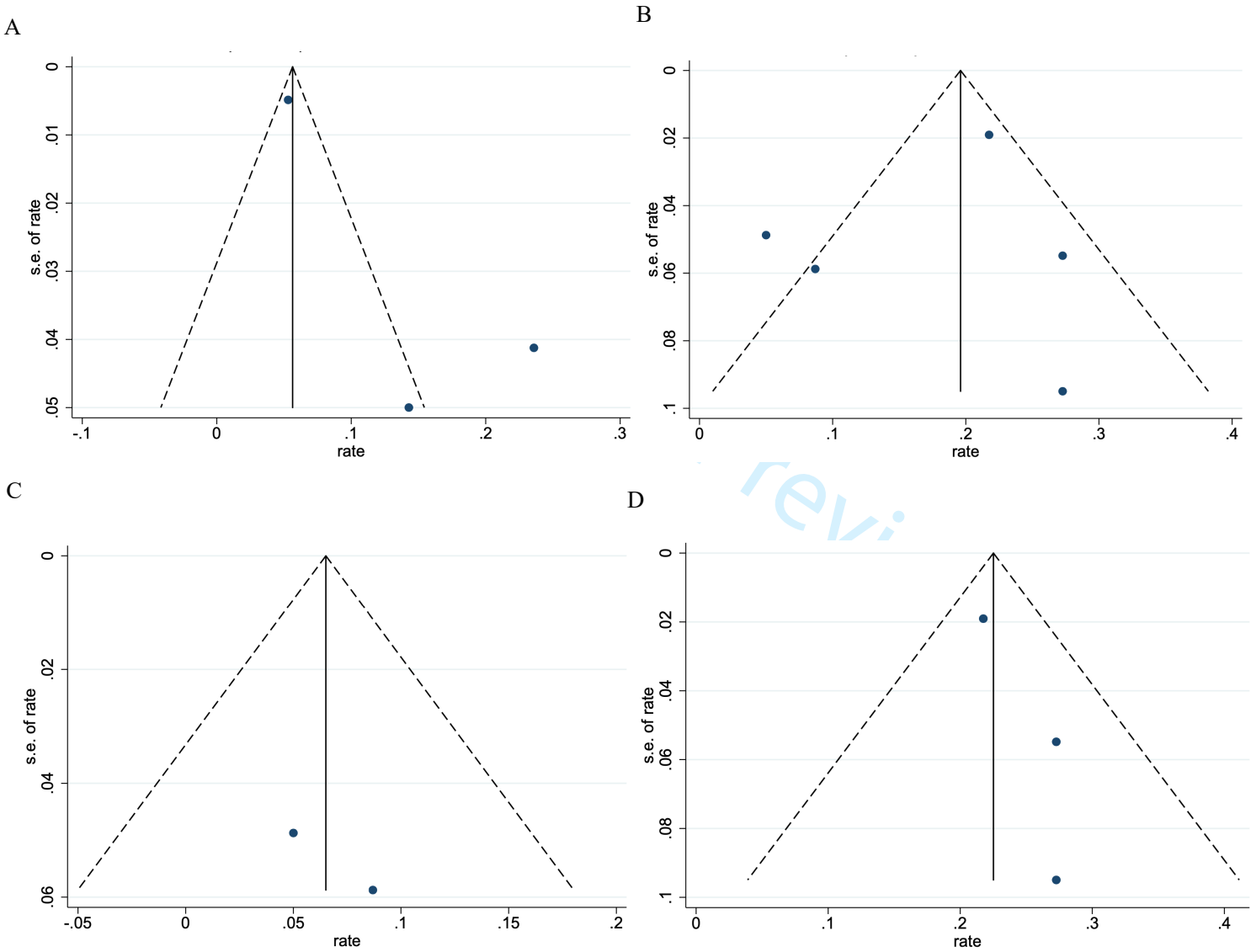


Figure 3 Funnel plot of included studies



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Appendix

Table 1 The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P1-P2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P4-P5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P6-P7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	P7, appendix 2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P8-P9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the	P8-P9

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Section and Topic	Item #	Checklist item	Location where item is reported
		process.	
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P8-P9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P8-P9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P9
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	P8-P9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P9-P10
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P9-P10
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P9-P10
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P9-P10
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A

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Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P9-P10
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P10-P11
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	P11
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	P19 appendix 4
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	P10-P18, appendix 3
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P10-P20
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	P19-P20
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P19
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	P19
Certainty of	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P15-P20

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Section and Topic	Item #	Checklist item	Location where item is reported
evidence			
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P21
	23b	Discuss any limitations of the evidence included in the review.	P24
	23c	Discuss any limitations of the review processes used.	N/A
	23d	Discuss implications of the results for practice, policy, and future research.	P22-P25
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P6-P7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	P6-P7
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P25
Competing interests	26	Declare any competing interests of review authors.	P25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	P25

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Table 2 Search strategies to identify studies reporting the prevalence status, treatment and outcomes of tuberculosis and COVID-19

Database searched: PubMed

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	("covid-19"[MeSH Terms] OR "sars-cov-2"[MeSH Terms]) AND "tuberculosis"[MeSH Terms]	244
2	((covid-19[Title/Abstract]) OR (Sars-cov-2[Title/Abstract])) AND((TB[Title/Abstract]) OR (tuberculos*[Title/Abstract]))	523
3	#1 OR #2	565

Database searched: Web of Science

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	((TS=(covid-19) OR TS=(sars-cov-2) OR AB=(covid-19) OR AB=(sars-cov-2)) OR (TS=(covid-19)) OR (TS=(sars-cov-2)) OR (AB=(covid-19))) AND (TS=(tuberculosis)) OR (TS=(Mycobacterium tuberculosis)) OR (TS=(Mycobacterium tuberculosis)) OR (AB=(tuberculo*)) OR (AB=(TB)) OR (AB=(Mycobacterium tuberculosis)) OR (AB=(M.tuberculosis)) OR (AB=(tubercle bacillus)))	644

Database searched: ProQuest

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	su(COVID-19 OR SARS-COV-2) AND su(TB OR tuberculo* OR (Mycobacterium tuberculosis) OR (M.tuberculosis))	682

Database searched: Scopus

Date: Mars 19th, 2021

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Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	(TITLE-ABS-KEY (covid-19 OR sars-cov-2) AND TITLE-ABS-KEY (tb OR tuberculos* OR (mycobacterium AND tuberculosis) OR (m.tuberculosis)))	839

Database searched: Cochrane database

Date: Mars 19, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	MeSH descriptor: [COVID-19] explode all trees	257
2	MeSH descriptor: [SARS-COV-2] explode all trees	204
3	MeSH descriptor: [tuberculosis] explode all trees	2,029
4	(#1 OR #2) AND 3	1

Database searched: Embase

Date: Mars 19, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	('covid 19':ab,ti OR 'sars cov 2':ab,ti) AND [01-11-2019]/sd NOT [20-3-2021]/sd	81,923
2	(tuberculosis:ab,ti OR 'mycobacterium tuberculosis':ab,ti OR tuberculo*:ab,ti OR 'tb':ab,ti OR 'm.tuberculosis':ab,ti OR 'tubercle bacillus':ab,ti) AND [01-11-2019]/sd NOT [20-3-2021]/sd	19,510
3	#1 AND #2	367

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Table 3 Characteristics of reported TB and COVID-19 coinfection patients (n=55)

Basic information	Number (%)
Gender	
female	12 (21.8)
male	43 (78.2)
Age	
< 20	3 (5.5)
20-39	23 (41.8)
40-59	22 (40.0)
≥60	7 (12.7)
Comorbidity	
diabetes mellitus	8 (14.5)
hypertension	5 (9.1)
HIV	4 (7.3)
Type of TB	
active TB	50 (90.9)
post TB	4 (7.3)
latent TB	1 (1.8)
TB site	
pulmonary TB	47 (85.5)
central nervous system involvement	2 (3.6)
tuberculous meningitis	1 (1.8)
tuberculous pleuritis	1 (1.8)
peritoneal TB	1 (1.8)
not reported	6 (10.9)

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BCG Vaccination	
yes	2 (3.6)
not reported	53 (96.4)
Drug resistance	
no	15 (27.3)
yes	2 (3.6)
likely	1 (1.8)
not reported	37 (67.3)
Infection order	
TB before COVID-19	17
COVID-19 before TB	0
not reported	38
Symptomatic COVID-19	
yes	55 (100.0)
no	0 (0.0)

Table 4 Detailed basic information of included case reports (n=35)

Author (Year)	Country	Cases number	Age	Gender	Comorbidity	BCG Vaccination	TB site	Drug resistance	Active TB	TB before COVID-19	Symptomatic COVID-19

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Sarma 2020[1]	India	1	53	female	chronic kidney disease (CKD) stage 4, type 2 diabetes mellitus	/	pulmonary TB	likely	yes	yes	yes
Singh 2020[2]	India	1	76	female	/	/	pulmonary TB	/	yes	/	yes
Kumar 2020[3]	India	1	38	male	/	/	pulmonary TB	/	yes	/	yes
Ata 2020[4]	India	1	28	male	no HIV	/	pulmonary TB central nervous system involvement	/	yes	/	yes
Yadav 2020[5]	India	1	43	male	non-diabetes	/	pulmonary TB	no	yes	/	yes
Baskara 2021[6]	Indonesian	1	42	male	diabetes mellitus	/	pulmonary TB	no	yes	yes	yes
Wang 2020[7]	China	1	45	male	/	/	tuberculous meningitis	/	yes	/	yes
He 2020[8]	China	3	67	male	hypertension	/	/	/	no	yes	yes
			26	male	/	/	/	/	no	yes	yes
			76	male	exfoliative dermatitis, Arthrolithiasis	/	/	/	no	yes	yes
Cao 2020[9]	China	1	47	female	/	/	pulmonary TB	/	yes	yes	yes

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			50	male	anemia, hypoproteinemia	/	pulmonary TB	/	yes	yes	yes
Yao 2020[10]	China	3	44	male	diabetes, pleural effusion	/	pulmonary TB, tuberculous pleuritis	/	yes	/	yes
			57	male	diabetes	/	pulmonary TB	/	yes	yes	yes
			48	male	/	/	/	/	no	/	yes
			26	male	/	/	pulmonary TB	yes	yes	yes	yes
Liu 2020[11]	China	3	46	male	/	/	pulmonary TB, tuberculous pleuritis	/	no	yes	yes
Khayat 2021[12]	Saudi Arabia	1	40	female	no HIV and hepatitis	/	pulmonary TB	no	yes	/	yes
Faqihi 2020[13]	Saudi Arabia	1	60	male	hypertensive and diabetic	/	pulmonary TB	no	yes	/	yes
			34	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
Yousaf 2020[14]	Qatar	6	32	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			50	male	no HIV, hepatitis B, and hepatitis C	/	pulmonary TB	/	yes	/	yes

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					infection; have diabetes						
			35	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			27	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			35	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB, tuberculous pleuritis	/	yes	/	yes
AlKhateeb 2020[15]	Qatar	1	28	male	/	/	pulmonary TB	/	yes	/	yes
Tham 2020[16]	Singapore	4	32	male	/	/	pulmonary TB	/	yes	/	yes
			33	male	/	/	pulmonary TB	/	yes	/	yes
			22	male	/	/	pulmonary TB	/	yes	/	yes
			40	male	/	/	pulmonary TB	/	yes	/	yes
Cinar 2020[17]	Turkey	1	55	male	/	/	/	/	yes	/	yes
Sarinoglu 2020[18]	Turkey	2	77	female	diabetes mellitus, hypertension, chronic obstructive lung disease, chronic kidney failure	/	pulmonary TB	/	yes	/	yes
			39	female	/	/	pulmonary TB	/	yes	/	yes
Lopinto	France	1	58	male	pulmonary	/	pulmonary TB	/	yes	yes	yes

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2020[19]					embolism						
Musso 2021[20]	Italy	1	45	male	No HIV	/	pulmonary TB	no	yes	yes	yes
Luciani 2020[21]	Italy	1	32	female	No HIV	yes	pulmonary TB	/	yes	/	yes
Bouaré 2020[22]	Morocco	1	32	female	/	/	pulmonary TB	/	yes	/	yes
Essajee 2020[23]	South Africa	1	2	female	/	yes	pulmonary TB	no	yes	/	yes
Pillay 2021[24]	South Africa	1	44	female	HIV, pneumocystis iirovecii pneumonia, community-acquired pneumonia, pulmonary embolism	/	pulmonary TB	/	yes	/	yes
Agada 2021[25]	Nigeria	1	50	male	No HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	no	yes	/	yes
Gbenga 2020[26]	Nigeria	2	33	male	/	/	pulmonary TB	/	yes	/	yes
			30	male	/	/	pulmonary TB	/	yes	/	yes
Rivas 2020[27]	Panama	2	29	male	pleural empyema	/	pulmonary TB	no	yes	/	yes
			53	male	HIV	/	pulmonary TB	/	yes	yes	yes
Farias 2020[28]	Brazil	2	39	male	HIV, hepatitis B	/	pulmonary TB	no	yes	yes	yes
			43	male	HIV, anxiety	/	pulmonary TB	no	yes	/	yes

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Pinheiro 2020[29]	Brazil	1	68	male	disorder diabetes, hypertension, chronic liver disease secondary to schistosomiasis	/	pulmonary TB	no	yes	/	yes
Gerstein 2021[30]	USA	1	49	male	cirrhosis, portal hypertension	/	pulmonary TB, peritoneal TB	/	yes	/	yes
Cutler 2020[31]	USA	1	61	male	Parkinson's disease	/	pulmonary TB	/	yes	/	yes
Sasson 2020[32]	USA	1	44	male	no HIV; hypertension, type II diabetes mellitus, a previous stroke secondary to a left atrial thrombus	/	/	no	yes	/	yes
Freij 2020[33]	USA	1	5	female	no HIV	/	central nervous system involvement	/	yes	/	yes
Vanzetti 2020[34]	Argentina	3	33	female	/	/	pulmonary TB	no	yes	/	yes
			34	male	/	/	pulmonary TB	no	yes	yes	yes
			17	male	/	/	pulmonary TB	no	yes	/	yes
Vilbrun 2020[35]	Haiti	1	26	male	no HIV	/	pulmonary TB	yes	yes	yes	yes

/: No information provided

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Table 5 Detailed treatment information of included cases (n=55)

Author (Year)	Cases number	ICU	Treatment before hospitalization	Treatment in hospital	Treatment after discharge	Post-intervention clinical condition	Outcome	Days to die or discharge
Sarma 2020[1]	1	yes	/	**Supportive treatment:** oxygen support (intubated) **Antibiotic:** ampicillin, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** vitamin C, midazolam, fentanyl	/	/	resolved	/
Singh 2020[2]	1	no	/	**Supportive treatment:** / **Antibiotic:** ceftriaxone, azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** N-acetyl cysteine, methylprednisolone, vitamin C	/	/	on treatment	/

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Kumar 2020[3]	1	no	/	**Supportive treatment:** oxygen support, mechanical ventilation **Antibiotic:** ATT, other antibiotic (no detail) **Antiviral drug:** hydroxychloroquine sulphate **Other treatment:** /	/	/	died	/
Ata 2020[4]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** steroids	rifampicin, ethambutol, and pyrazinamide	/	resolved	28

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Yadav 2020[5]	1	no	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** / **Other treatment:** /	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, ethambutol, and pyrazinamide) **Antiviral drug:** / **Other treatment:** /	/	/	/	/
Baskara 2021[6]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), azithromycin, ceftriaxone, ceftazidime **Antiviral drug:** hydroxychloroquine, oseltamivir **Other treatment:** anti-coagulant drugs	treatment for diabetes mellitus	/	resolved	13

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Wang 2020[7]	1	no	/	**Supportive treatment:** / **Antibiotic:** rifampicin, amikacin, moxifloxacin **Antiviral drug:** arbidol **Other treatment:** tolvaptan, aspirin, mannitol, potassium supplement, sodium supplement	/	/	died	14
He 2020[8]	3	no	/	**Supportive treatment:** noninvasive ventilation support **Antibiotic:** antibiotic (no details and not ATT) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** methylprednisolone, probiotics, traditional Chinese medicine, immunoglobulin	/	/	resolved	37

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no	/	**Supportive treatment:** / **Antibiotic:** antibiotic (no details and not ATT) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** probiotics, traditional Chinese medicine	/	skin rashes with pruritus	ninth day after discharge, returned to the hospital because of positive recurrence of SARS-CoV-2 RNA	28
no	/	**Supportive treatment:** nasal cannula oxygen support **Antibiotic:** ATT, antibiotic (no details) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** methylprednisolone, glucocorticoid, probiotics, traditional Chinese medicine, immunoglobulin	/	/	resolved	24

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Cao 2020[9]	1	no	/	**Supportive treatment:** noninvasive ventilation support **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), moxifloxacin hydrochloride **Antiviral drug:** lopinavir and ritonavir, umifenovir **Other treatment:** polyene phosphati-dylcholine, reduced glutathione	/	Hepatic injury during the treatment	resolved	More than 14 days
Yao 2020[10]	3	no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** ATT **Antiviral drug:** / **Other treatment:** corticosteroid	/	/	died	22

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		no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** antiviral drugs (no details) **Other treatment:** acarbose, metformin	ATT	/	resolved	26
		no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** antibiotics (no details) **Antiviral drug:** antiviral drugs (no details) **Other treatment:** corticosteroids, immunoglobulin	/	/	resolved	27
Liu 2020[11]	3	no	/	**Supportive treatment:** extracorporeal membrane oxygenation, haemoperfusion **Antibiotic:** moxifloxacin, linezolid, no ATT **Antiviral drug:** arbidol **Other treatment:**	/	/	resolved	14

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		caspofungin, thymopentin					
no	/	**Supportive treatment:** nasal high-flow oxygen therapy **Antibiotic:** moxifloxacin, linezolid, cycloserine, clofazimine, pyrazinamide **Antiviral drug:** arbidol **Other treatment:** /	ATT	/	resolved	14
no	/	**Supportive treatment:** nasal high-flow oxygen therapy **Antibiotic:** moxifloxacin **Antiviral drug:** arbidol **Other treatment:** /	/	/	resolved	9

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Khayat 2021[12]	1	no	/	/	/	/	/	/
Faqihi 2020[13]	1	yes	/	**Supportive treatment:** respiratory support **Antibiotic:** moxifloxacin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir, ritonavir, ribavirin **Other treatment:** dexamethasone, prophylactic anticoagulation	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), isoniazid and rifampicin (later)	/	resolved	27
Yousaf 2020[14]	6	no	/	**Supportive treatment:** / **Antibiotic:** ceftriaxone, azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/

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AlKhateeb 2020[15]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details) **Antiviral drug:** / **Other treatment:** /	/	/	resolved	1 month
Tham 2020[16]	4	no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details) **Antiviral drug:** / **Other treatment:** /	/	/	/	/
		no	/		/	/	/	/
		no	/		/	/	/	/
		no	/		/	/	/	/
Cinar 2020[17]	1	yes	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), meropenem **Antiviral drug:** favipiravir **Other treatment:**	/	myelodysplastic syndrome (MDS/RAEB1 FAB subtype) during the treatment	/	/

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				convalescent plasma product, tocilizumab					
Sarinoglu 2020[18]	2	yes	hemodialysis	**Supportive treatment:** / **Antibiotic:** ATT (no details), piperacillin/tazobactam **Antiviral drug:** favipiravir, hydroxychloroquine **Other treatment:** /	hemodialysis	/	/	/	
		no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details), azithromycin, plaquanil, ciprofloxacine, tazocilli **Antiviral drug:** / **Other treatment:**	/	/	/	/	

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				/				
Lopinto 2020[19]	1	yes	/	/	/	/	resolved	14
Musso 2021[20]	1	no	/	**Supportive treatment:** chest tube drainage **Antibiotic:** amikacin, moxifloxacin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hidroxicloroquine **Other treatment:** corticosteroids	/	/	died	14

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Luciani 2020[21]	1	no	/	**Supportive treatment:** / **Antibiotic:** amoxicillin/clavulanic acid, clarithromycin, piperacillin/tazobactam, linezolid **Antiviral drug:** hydroxychloroquine, lopinavir, ritonavir **Other treatment:** paracetamol	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	/	resolved	> 12
Bouaré 2020[22]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), azithromycin **Antiviral drug:** chloroquine **Other treatment:** /	/	/	resolved	10

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Essajee 2020[23]	1	no	/	**Supportive treatment:** high flow nasal oxygen **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), **Antiviral drug:** / **Other treatment:** prednisone, aspirin, dexamethasone	/	/	on treatment	/
Pillay 2021[24]	1	no	/	**Supportive treatment:** rebreather mask **Antibiotic:** ATT, amoxicillin/clavulanic acid, azithromycin, trimethoprim, sulfamethoxazole **Antiviral drug:** / **Other treatment:** pyridoxine, prednisone, prophylactic clexane, dexamethasone, vitamin D, zinc, thiamine, ascorbic acid, therapeutic anticoagulation	/	/	on treatment	/

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Agada 2021[25]	1	no	/	**Supportive treatment:** supplemental oxygen **Antibiotic:** ceftriaxone, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** no **Other treatment:** dexamethasone, acetaminophen, pyridoxine phosphate, zinc-sulphate, Vitamin C	ATT	/	resolved	14
Gbenga 2020[26]	2	no	/	**Supportive treatment:** / **Antibiotic:** azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir, ritonavir **Other treatment:** Vitamin C, Zinc sulfate, oral prednisolone, tabs pyridoxine	/	/	resolved	17

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		no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir/ritonavir, **Other treatment:** vitamin C, zinc sulfate, tabs pyridoxine, IV hydrocortisone, clexane	/	/	died	6
Rivas 2020[27]	2	yes	/	**Supportive treatment:** invasive mechanical ventilatory support **Antibiotic:** ATT (isoniazid, rifampin, ethambutol, and pyrazinamide), levofloxacin, azithromycin, **Antiviral drug:** hydroxychloroquine, tenofovir, lamivudine, and dolutegravir **Other treatment:** low molecular weight heparin	/	nosocomial pneumonia	resolved	2 months

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		no	rifampicin and isoniazid	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ceftriaxone, azithromycin, **Antiviral drug:** hydroxychloroquine **Other treatment:** enoxapari	/	/	resolved	7
Farias 2020[28]	2	no	/	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ATT (isoniazid, rifampin, ethambutol, and pyrazinamide), ceftriaxone, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	21

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		no	/	**Supportive treatment:** / **Antibiotic:** azithromycin, ceftriaxone, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	7
Pinheiro 2020[29]	1	no	/	/	/	/	/	/
Gerstein 2021[30]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (levofloxacin, rifampicin, isoniazid, ethambutol) **Antiviral drug:** hydroxychloroquine, **Other treatment:** convalescent plasma	ATT (rifampicin, isoniazid)	/	resolved	42

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Cutler 2020[31]	1	no	/	**Supportive treatment:** nasal canula oxygen supplementation **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	/
Sasson 2020[32]	1	yes	intubated for hypoxia	**Supportive treatment:** intubated for hypoxia **Antibiotic:** azithromycin, ceftriaxone, cefepime, vancomycin, piperacillin/tazobactam, meropenem, caspofungin, metronidazole, voriconazole, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** dexamethasone, methylprednisolone, vitamin B6, metformin, alogliptin, long-acting insulin and rapid-acting insulin	/	/	resolved	> 35

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Freij 2020[33]	1	no	amoxicillin	**Supportive treatment:** / **Antibiotic:** azithromycin **Antiviral drug:** hydroxychloroquine, remdesivir **Other treatment:** dexamethasone, antihypertensive medications	/	/	died	26
		no	/	/	/	/	resolved	
		no	/	/	/	/	resolved	
Vanzetti 2020[34]	3	no	/	**Supportive treatment:** oxygen support **Antibiotic:** / **Antiviral drug:** / **Other treatment:** corticosteroids	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	/	resolved	longer hospital stay than average

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Vilbrun 2020[35]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (bedaquiline, levofloxacin, linezolid, clofazimine, pyrazinimide) **Antiviral drug:** / **Other treatment:** /	/	/	/	resolved	3 weeks from the diagnosis

/: No information provided

Table 6 Detailed treatment information of included retrospective cohort studies (n=7)

First author (year)	Country	Time	single center	Sample size	In-hospital	age	gender	Comorbidity	Died	Fatality rate (in-hospital or total)
Gupta 2020[36]	India	1 February 2020 to 14 June 2020	yes	22	yes	19-67	20 male (90.9%) 2 female (9.1%)	Hypertension 4 (18.2%) Diabetes 3 (13.6%) Seizure disorder 2 (9.1) Hypothyroidism 1 (4.5%)	6	27.3 (in-hospital)
Motta 2020[37]	Belgium, Brazil, France, Italy, Russia, Singapore,	12 March, 2020 to 5 May, 2020	no	49	no	27-70	/	/	7	14.3% (total)

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	Spain, and Switzerland									
Sy 2020[38]	Philippines	May 17, 2020 to June 15, 2020	no	106	no	/	73 male (68.9%) 33 female (31.1%)	Hypertension 22 (20.8) Diabetes 14 (13.2) Cancer 1 (0.9) Renal cancer 4 (3.8) Cardiac disease 8 (7.5) Asthma 4 (3.8) COPD 3 (2.8)	25	23.6% (total)
				66	yes	/		Hypertension 16 (24.2) Diabetes 10 (15.2) Cancer 1 (1.5) Renal cancer 4 (6.1) Cardiac disease 7 (10.6) Asthma 1 (1.5) COPD 1 (1.5)	18	27.3% (in hospital)
Stochino 2020[39]	Italy	/	yes	20	yes	27-47	12 male (60%) 8 female (40%)	/	1	5% (in-hospital)
Davies 2020[40]	South Africa	Till to 9 June, 2020	no	2128	no	20 or above	/	/	113	5.3% (total)
			no	469	yes	20 or	/	/	102	21.7%

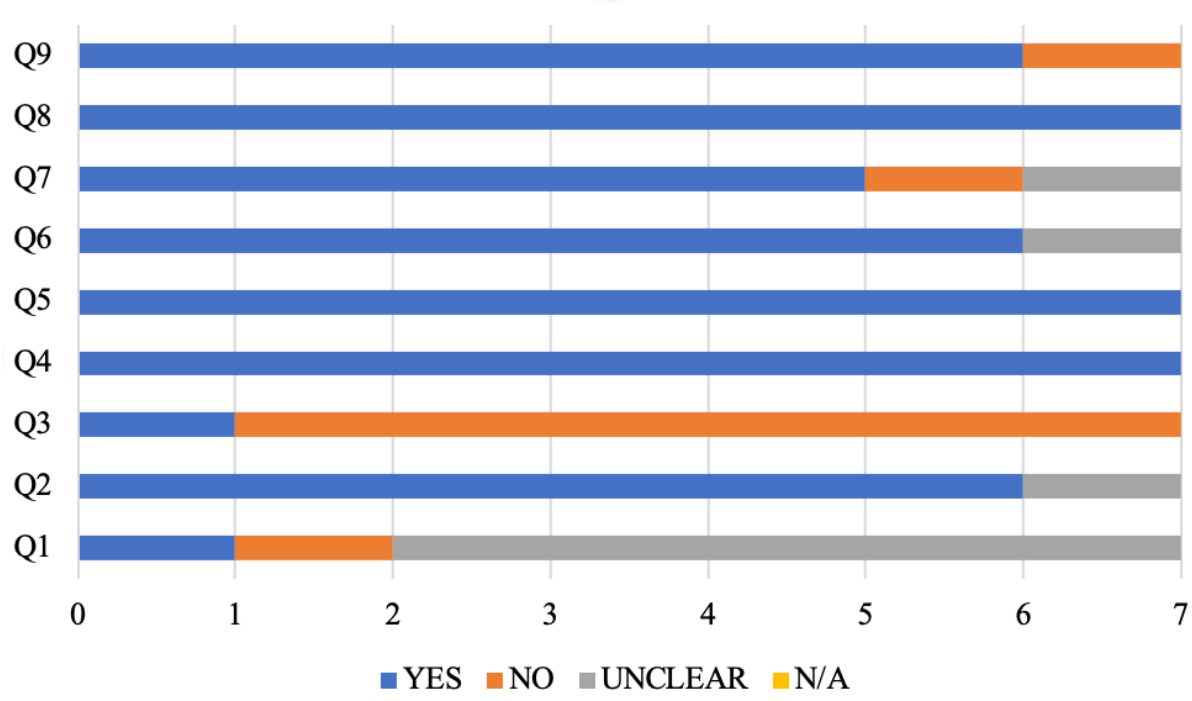
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						above				(in-hospital)
Domingo 2020[41]	Argentina	1 March to 30 June, 2020	yes	23	yes	5-82	18 male (78.3%) 5 female (21.7%)	Smoking and other addictions 16(69.6) HIV 4(17.4) Psychosis 1(4.3) Pulmonary thromboembolism 1(4.3) Arterial hypertension 1(4.3)	2	8.7% (in-hospital)
Gubkina 2020[42]	Russia	March to June, 2020	24	yes	/	/	/		0	0% (in-hospital)

/: No information provided

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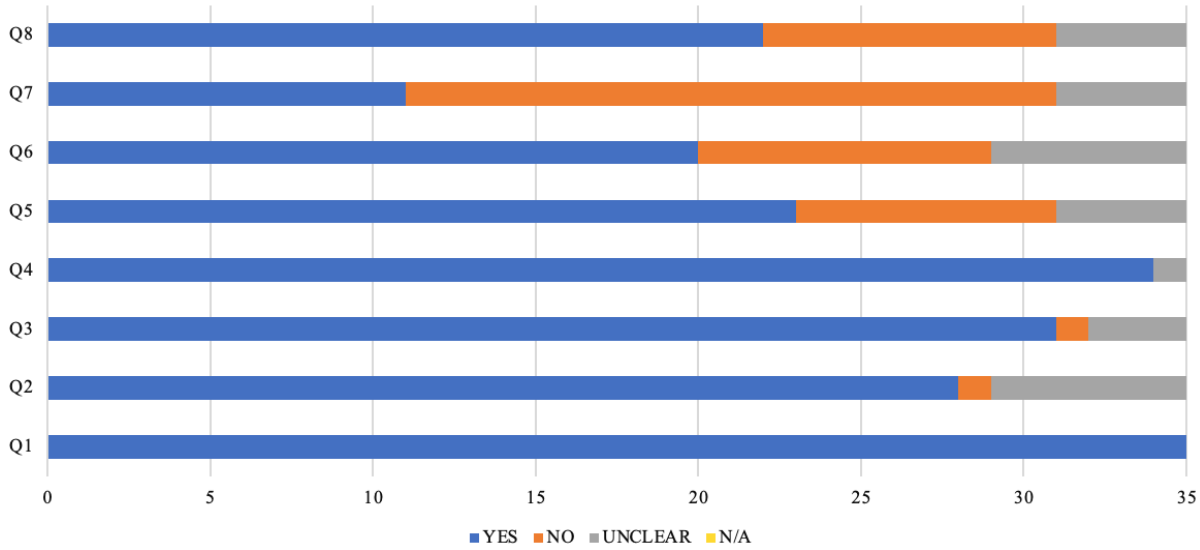
Figure 1 The results of quality assessment for prevalence study



Note: Four options for each question: yes, no, unclear and not appliable. Q1: Was the sample frame appropriate to address the target population? Q2: Were study participants sampled in an appropriate way? Q3: Was the sample size adequate? Q4: Were the study subjects and the setting described in detail? Q5: Was the data analysis conducted with sufficient coverage of the identified sample? Q6: Were valid methods used for the

identification of the condition? Q7: Was the condition measured in a standard, reliable way for all participants? Q8: Was there appropriate statistical analysis? Q9: Was the response rate adequate, and if not, was the low response rate managed appropriately?

Figure 2 The results of quality assessment for case report



Note: Four options for each question: yes, no, unclear and not appliable. Q1: Were patient's demographic characteristics clearly described? Q2: Was the patient's history clearly described and presented as a timeline? Q3: Was the current clinical condition of the patient on presentation clearly

described? Q4: Were diagnostic tests or assessment methods and the results clearly described? Q5: Was the intervention(s) or treatment procedure(s) clearly described? Q6: Was the post-intervention clinical condition clearly described? Q7: Were adverse events (harms) or unanticipated events identified and described? Q8: Does the case report provide takeaway lessons?

Table 7 Quality assessment of included case reports (n=35)

Study (Year) \ Results	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Agada 2021	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Yao 2020	Yes	Yes	Yes	Unclear	Unclear	No	No	No
Yadav 2020	Yes	Yes	Yes	Yes	No	No	No	No
Wang 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Vilbrun 2020	Yes	Yes	Yes	Yes	No	Unclear	No	Yes
Tham 2020	Yes	Yes	Yes	Yes	No	No	No	No
Singh 2020	Yes	Yes	Yes	Yes	Unclear	No	Yes	Yes
Sasson 2020	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear
Sarma 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Sarinoglu 2020	Yes	Yes	Yes	Yes	Unclear	Yes	No	Yes
Rivas 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Pinheiro 2020	Yes	Yes	Unclear	Yes	No	No	No	No
Pillay 2021	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Musso 2021	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Luciani 2020	Yes	Yes	Yes	Yes	Yes	No	No	Yes

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Lopinto 2020	Yes	Yes	Yes	Yes	No	Unclear	Yes	Yes
Liu 2020	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Kumar 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Khayat 2021	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Gbenga 2020	Yes	Yes	Yes	Yes	Unclear	Unclear	Unclear	No
Freij 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cutler 2020	Yes	Yes	Unclear	Yes	Yes	Yes	No	Yes
Cinar 2020	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes
Cao 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farias 2020	Yes	Yes	Yes	Yes	Yes	Unclear	No	Yes
Bouare 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Baskara 2021	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
AlKhateeb 2020	Yes	Yes	Yes	Yes	No	Yes	No	Unclear

Note: Four options for each question: yes, no, unclear and not appliable. Q1: Were patient's demographic characteristics clearly described? Q2: Was the patient's history clearly described and presented as a timeline? Q3: Was the current clinical condition of the patient on presentation clearly described? Q4: Were diagnostic tests or assessment methods and the results clearly described? Q5: Was the intervention(s) or treatment procedure(s) clearly described? Q6: Was the post-intervention clinical condition clearly described? Q7: Were adverse events (harms) or unanticipated events identified and described? Q8: Does the case report provide takeaway lessons?

Table 8 Quality assessment of included retrospective cohort studies (n=7)

Results / Study (Years)	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
Gupta 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Motta 2020	Unclear	Unclear	No	Yes	Yes	Yes	Yes	Yes	Yes

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Sy 2020	Yes	Yes	No	Yes	Yes	Unclear	Unclear	Yes	No
Stochino 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Davies 2020	Unclear	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Domingo 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Gubkina 2020	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes

Note: Four options for each question: yes, no, unclear and not appliable. Q1: Was the sample frame appropriate to address the target population? Q2: Were study participants sampled in an appropriate way? Q3: Was the sample size adequate? Q4: Were the study subjects and the setting described in detail? Q5: Was the data analysis conducted with sufficient coverage of the identified sample? Q6: Were valid methods used for the identification of the condition? Q7: Was the condition measured in a standard, reliable way for all participants? Q8: Was there appropriate statistical analysis? Q9: Was the response rate adequate, and if not, was the low response rate managed appropriately?

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